

1.1 Vector Algebra

1.1.1 Vector Operations

If you walk 4 miles due north and then 3 miles due east (Fig. 1.1), you will have gone a total of 7 miles, but you're *not* 7 miles from where you set out – only 5. We need an arithmetic to describe quantities like this, which evidently do not add in the ordinary way. The reason they don't, of course, is that **displacements** (straight line segments going from one point to another) have *direction* as well as *magnitude* (length), and it is essential to take both into account when you combine them. Such objects are called **vectors**: velocity, acceleration, force, and momentum are other examples. By contrast, quantities that have magnitude but no direction are called **scalars**: examples include mass, charge, density, and temperature.

I shall use **boldface** (**A**, **B**, and so on) for vectors and ordinary type for scalars. The magnitude of a vector **A** is written $|\mathbf{A}|$ or, more simply, A . In diagrams, vectors are denoted by arrows: the length of the arrow is proportional to the magnitude of the vector, and the arrowhead indicates its direction. *Minus A* ($-\mathbf{A}$) is a vector with the same magnitude as **A** but the opposite direction (Fig. 1.2). Note that vectors have magnitude and direction but *not location*: a displacement of 4 miles due north from Washington is represented by the same vector as a displacement 4 miles north from Baltimore (neglecting, of course, the curvature of the Earth). On a diagram, therefore, you can slide the arrow around at will, as long as you don't change its length or direction.

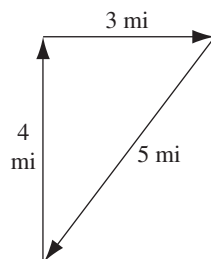


Fig. 1.1

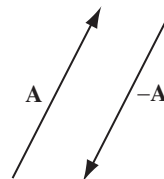


Fig. 1.2

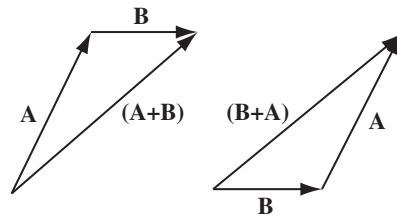


Fig. 1.3

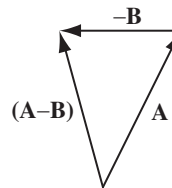


Fig. 1.4

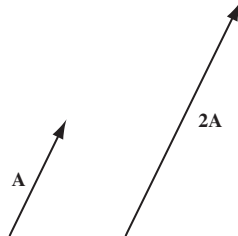


Fig. 1.5

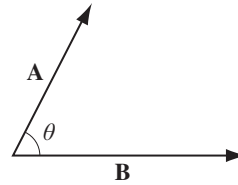


Fig. 1.6

We define four vector operations: addition and three kinds of multiplication.

(i) Addition of two vectors. Place the tail of $\mathbf{B}$ at the head of $\mathbf{A}$; the sum, $\mathbf{A} + \mathbf{B}$, is the vector from the tail of $\mathbf{A}$ to the head of $\mathbf{B}$ (Fig. 1.3). (This rule generalizes the obvious procedure for combining two displacements.) Addition is *commutative*:

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A};$$

3 miles east followed by 4 miles north gets you to the same place as 4 miles north followed by 3 miles east. Addition is also *associative*:

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C}).$$

To subtract a vector, add its opposite (Fig. 1.4):

$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B}).$$

(ii) Multiplication by a scalar. Multiplication of a vector by a positive scalar a multiplies the *magnitude* but leaves the direction unchanged (Fig. 1.5). (If a is negative, the direction is reversed.) Scalar multiplication is *distributive*:

$$a(\mathbf{A} + \mathbf{B}) = a\mathbf{A} + a\mathbf{B}.$$

(iii) Dot product of two vectors. The dot product of two vectors is defined by

$$\mathbf{A} \cdot \mathbf{B} \equiv AB \cos \theta, \quad (1.1)$$

where θ is the angle they form when placed tail to tail (Fig. 1.6). Note that $\mathbf{A} \cdot \mathbf{B}$ is itself a *scalar* (hence the alternative name **scalar product**). The dot product is *commutative*,

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A},$$

and *distributive*,

$$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}. \quad (1.2)$$

Geometrically, $\mathbf{A} \cdot \mathbf{B}$ is the product of A times the projection of $\mathbf{B}$ along $\mathbf{A}$ (or the product of B times the projection of $\mathbf{A}$ along $\mathbf{B}$). If the two vectors are parallel, then $\mathbf{A} \cdot \mathbf{B} = AB$. In particular, for any vector $\mathbf{A}$,

$$\mathbf{A} \cdot \mathbf{A} = A^2. \quad (1.3)$$

If $\mathbf{A}$ and $\mathbf{B}$ are perpendicular, then $\mathbf{A} \cdot \mathbf{B} = 0$.

Example 1.1. Let $\mathbf{C} = \mathbf{A} - \mathbf{B}$ (Fig. 1.7), and calculate the dot product of $\mathbf{C}$ with itself.

Solution.

$$\mathbf{C} \cdot \mathbf{C} = (\mathbf{A} - \mathbf{B}) \cdot (\mathbf{A} - \mathbf{B}) = \mathbf{A} \cdot \mathbf{A} - \mathbf{A} \cdot \mathbf{B} - \mathbf{B} \cdot \mathbf{A} + \mathbf{B} \cdot \mathbf{B},$$

or

$$C^2 = A^2 + B^2 - 2AB \cos \theta.$$

This is the **law of cosines**.

(iv) **Cross product of two vectors.** The cross product of two vectors is defined by

$$\mathbf{A} \times \mathbf{B} \equiv AB \sin \theta \hat{\mathbf{n}}, \quad (1.4)$$

where $\hat{\mathbf{n}}$ is a **unit vector** (vector of magnitude 1) pointing perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$. (I shall use a hat (^) to denote unit vectors.) Of course, there are *two* directions perpendicular to any plane: “in” and “out.” The ambiguity is resolved by the **right-hand rule**: let your fingers point in the direction of the first vector and curl around (via the smaller angle) toward the second; then your thumb indicates the direction of $\hat{\mathbf{n}}$. (In Fig. 1.8, $\mathbf{A} \times \mathbf{B}$ points *into* the page; $\mathbf{B} \times \mathbf{A}$ points *out* of the page.) Note that $\mathbf{A} \times \mathbf{B}$ is itself a *vector* (hence the alternative name **vector product**). The cross product is *distributive*,

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) + (\mathbf{A} \times \mathbf{C}), \quad (1.5)$$

but *not commutative*. In fact,

$$(\mathbf{B} \times \mathbf{A}) = -(\mathbf{A} \times \mathbf{B}). \quad (1.6)$$

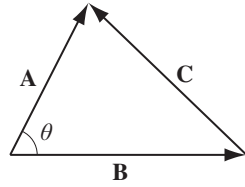


Fig. 1.7

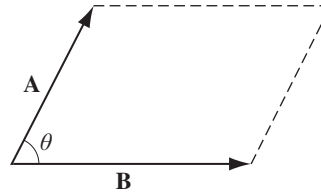


Fig. 1.8

Geometrically, $|\mathbf{A} \times \mathbf{B}|$ is the area of the parallelogram generated by $\mathbf{A}$ and $\mathbf{B}$ (Fig. 1.8). If two vectors are parallel, their cross product is zero. In particular,

$$\mathbf{A} \times \mathbf{A} = \mathbf{0}$$

for any vector $\mathbf{A}$. (Here $\mathbf{0}$ is the **zero vector**, with magnitude 0.)

Problem 1.1. Using the definitions in Eqs. 1.1 and 1.4, and appropriate diagrams, show that the dot product and cross product are distributive:

- (a) when the three vectors are coplanar;
- (b) in the general case.

! **Problem 1.2.** Is the cross product associative?

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} \stackrel{?}{=} \mathbf{A} \times (\mathbf{B} \times \mathbf{C}).$$

If so, *prove* it; if not, provide a counterexample (the simpler the better).

1.1.2 Vector Algebra: Component Form

In the previous section, I defined the four vector operations (addition, scalar multiplication, dot product, and cross product) in “abstract” form – that is, without reference to any particular coordinate system. In practice, it is often easier to set up Cartesian coordinates x , y , z and work with vector **components**. Let $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ be unit vectors parallel to the x -, y -, and z -axes, respectively (Fig. 1.9a). An arbitrary vector $\mathbf{A}$ can be expressed in terms of these **basis vectors** (Fig. 1.9b):

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}.$$

The coefficients A_x , A_y , and A_z are the “components” of $\mathbf{A}$; geometrically, they are the projections of $\mathbf{A}$ along the three coordinate axes ($A_x = \mathbf{A} \cdot \hat{\mathbf{x}}$, $A_y = \mathbf{A} \cdot \hat{\mathbf{y}}$, $A_z = \mathbf{A} \cdot \hat{\mathbf{z}}$). We can now reformulate each of the four vector operations as a rule for manipulating components:

$$\begin{aligned} \mathbf{A} + \mathbf{B} &= (A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}) + (B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}}) \\ &= (A_x + B_x) \hat{\mathbf{x}} + (A_y + B_y) \hat{\mathbf{y}} + (A_z + B_z) \hat{\mathbf{z}}. \end{aligned} \quad (1.7)$$

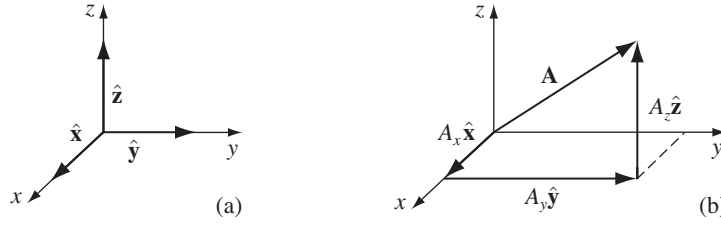


Fig. 1.9

Rule (i) To add vectors, add like components:

$$a\mathbf{A} = (aA_x)\hat{\mathbf{x}} + (aA_y)\hat{\mathbf{y}} + (aA_z)\hat{\mathbf{z}}. \quad (1.8)$$

Rule (ii) To multiply by a scalar, multiply each component.

Because $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ are mutually perpendicular unit vectors,

$$\hat{\mathbf{x}} \cdot \hat{\mathbf{x}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1, \quad \hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{x}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{z}} = 0. \quad (1.9)$$

Accordingly,

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= (A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}}) \cdot (B_x\hat{\mathbf{x}} + B_y\hat{\mathbf{y}} + B_z\hat{\mathbf{z}}) \\ &= A_xB_x + A_yB_y + A_zB_z. \end{aligned} \quad (1.10)$$

Rule (iii) To calculate the dot product, multiply like components, and add. In particular,

$$\mathbf{A} \cdot \mathbf{A} = A_x^2 + A_y^2 + A_z^2,$$

so

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}. \quad (1.11)$$

(This is, if you like, the three-dimensional generalization of the Pythagorean theorem.)

Similarly,¹

$$\begin{aligned} \hat{\mathbf{x}} \times \hat{\mathbf{x}} &= \hat{\mathbf{y}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{z}} = \mathbf{0}, \\ \hat{\mathbf{x}} \times \hat{\mathbf{y}} &= -\hat{\mathbf{y}} \times \hat{\mathbf{x}} = \hat{\mathbf{z}}, \\ \hat{\mathbf{y}} \times \hat{\mathbf{z}} &= -\hat{\mathbf{z}} \times \hat{\mathbf{y}} = \hat{\mathbf{x}}, \\ \hat{\mathbf{z}} \times \hat{\mathbf{x}} &= -\hat{\mathbf{x}} \times \hat{\mathbf{z}} = \hat{\mathbf{y}}. \end{aligned} \quad (1.12)$$

Therefore,

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= (A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}}) \times (B_x\hat{\mathbf{x}} + B_y\hat{\mathbf{y}} + B_z\hat{\mathbf{z}}) \\ &= (A_yB_z - A_zB_y)\hat{\mathbf{x}} + (A_zB_x - A_xB_z)\hat{\mathbf{y}} + (A_xB_y - A_yB_x)\hat{\mathbf{z}}. \end{aligned} \quad (1.13)$$

¹ These signs pertain to a *right-handed* coordinate system (x -axis out of the page, y -axis to the right, z -axis up, or any rotated version thereof). In a *left-handed* system (z -axis down), the signs would be reversed: $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = -\hat{\mathbf{z}}$, and so on. We shall use right-handed systems exclusively. Always check that you haven't inadvertently adopted a left-handed system.

This cumbersome expression can be written more neatly as a determinant:

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}. \quad (1.14)$$

Rule (iv) To calculate the cross product, form the determinant whose first row is $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$, whose second row is $\mathbf{A}$ (in component form), and whose third row is $\mathbf{B}$.

Example 1.2. Find the angle between the face diagonals of a cube.

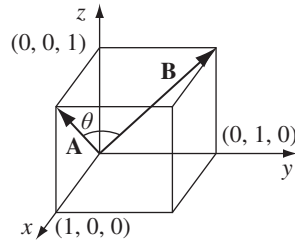


Fig. 1.10

Solution. We might as well use a cube of side 1, and place it as shown in Fig. 1.10, with one corner at the origin. The face diagonals $\mathbf{A}$ and $\mathbf{B}$ are

$$\mathbf{A} = 1\hat{\mathbf{x}} + 0\hat{\mathbf{y}} + 1\hat{\mathbf{z}}, \quad \mathbf{B} = 0\hat{\mathbf{x}} + 1\hat{\mathbf{y}} + 1\hat{\mathbf{z}}.$$

So, in component form,

$$\mathbf{A} \cdot \mathbf{B} = 1 \cdot 0 + 0 \cdot 1 + 1 \cdot 1 = 1.$$

On the other hand, in “abstract” form,

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta = \sqrt{2}\sqrt{2} \cos \theta = 2 \cos \theta.$$

Therefore,

$$\cos \theta = 1/2, \quad \text{or} \quad \theta = 60^\circ.$$

Of course, you can get the answer more easily by drawing in a diagonal across the top of the cube, completing the equilateral triangle. But in cases where the geometry is not so simple, this device of comparing the abstract and component forms of the dot product can be a very efficient means of finding angles.

Problem 1.3. Find the angle between the body diagonals of a cube.

Problem 1.4. Use the cross product to find the components of the unit vector $\hat{\mathbf{n}}$ perpendicular to the shaded plane in Fig. 1.11.

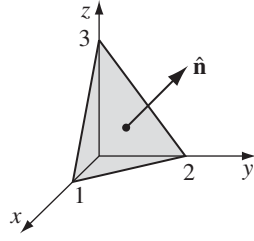


Fig. 1.11

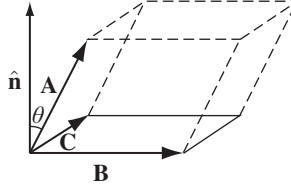


Fig. 1.12

1.1.3 Triple Products

Since the cross product of two vectors is itself a vector, it can be dotted or crossed with a third vector to form a *triple* product.

(i) Scalar triple product: $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$. Geometrically, $|\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})|$ is the volume of the parallelepiped generated by $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, since $|\mathbf{B} \times \mathbf{C}|$ is the area of the base, and $|\mathbf{A} \cos \theta|$ is the altitude (Fig. 1.12). Evidently,

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}), \quad (1.15)$$

for they all correspond to the same figure. Note that “alphabetical” order is preserved – in view of Eq. 1.6, the “nonalphabetical” triple products

$$\mathbf{A} \cdot (\mathbf{C} \times \mathbf{B}) = \mathbf{B} \cdot (\mathbf{A} \times \mathbf{C}) = \mathbf{C} \cdot (\mathbf{B} \times \mathbf{A}),$$

have the opposite sign. In component form,

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}. \quad (1.16)$$

Note that the dot and cross can be interchanged:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$$

(this follows immediately from Eq. 1.15); however, the placement of the parentheses is critical: $(\mathbf{A} \cdot \mathbf{B}) \times \mathbf{C}$ is a meaningless expression – you can’t make a cross product from a *scalar* and a vector.

(ii) Vector triple product: $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$. The vector triple product can be simplified by the so-called **BAC–CAB rule**:

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}). \quad (1.17)$$

Notice that

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = -\mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = -\mathbf{A}(\mathbf{B} \cdot \mathbf{C}) + \mathbf{B}(\mathbf{A} \cdot \mathbf{C})$$

is an entirely different vector (cross products are not associative). All *higher* vector products can be similarly reduced, by repeated application

of Eq. 1.17, so it is never necessary for an expression to contain more than one cross product in any term. For instance,

$$\begin{aligned}(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) &= (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C}), \\ \mathbf{A} \times [\mathbf{B} \times (\mathbf{C} \times \mathbf{D})] &= \mathbf{B}[\mathbf{A} \cdot (\mathbf{C} \times \mathbf{D})] - (\mathbf{A} \cdot \mathbf{B})(\mathbf{C} \times \mathbf{D}).\end{aligned}\quad (1.18)$$

Problem 1.5. Prove the **BAC–CAB** rule by writing out both sides in component form.

Problem 1.6. Prove that

$$[\mathbf{A} \times (\mathbf{B} \times \mathbf{C})] + [\mathbf{B} \times (\mathbf{C} \times \mathbf{A})] + [\mathbf{C} \times (\mathbf{A} \times \mathbf{B})] = \mathbf{0}.$$

Under what conditions does $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$?

1.1.4 Position, Displacement, and Separation Vectors

The location of a point in three dimensions can be described by listing its Cartesian coordinates (x, y, z) . The vector to that point from the origin ($\mathcal{O}$) is called the **position vector** (Fig. 1.13):

$$\mathbf{r} \equiv x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}. \quad (1.19)$$

I will reserve the letter $\mathbf{r}$ for this purpose, throughout the book. Its magnitude,

$$r = \sqrt{x^2 + y^2 + z^2}, \quad (1.20)$$

is the distance from the origin, and

$$\hat{\mathbf{r}} = \frac{\mathbf{r}}{r} = \frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}}{\sqrt{x^2 + y^2 + z^2}} \quad (1.21)$$

is a unit vector pointing radially outward. The **infinitesimal displacement vector**, from (x, y, z) to $(x + dx, y + dy, z + dz)$, is

$$d\mathbf{l} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}. \quad (1.22)$$

(We could call this $d\mathbf{r}$, since that's what it is, but it's useful to have a special notation for infinitesimal displacements.)

In electrodynamics, one frequently encounters problems involving *two* points – typically, a **source point**, $\mathbf{r}'$, where an electric charge is located, and a **field point**, $\mathbf{r}$, at which you are calculating the electric or magnetic field (Fig. 1.14). It pays to adopt right from the start some shorthand notation for the **separation vector** from the source point to the field point. I shall use for this purpose the cursive letter $\mathbf{r}$:

$$\mathbf{r} \equiv \mathbf{r} - \mathbf{r}'. \quad (1.23)$$

Its magnitude is

$$r = |\mathbf{r} - \mathbf{r}'|, \quad (1.24)$$

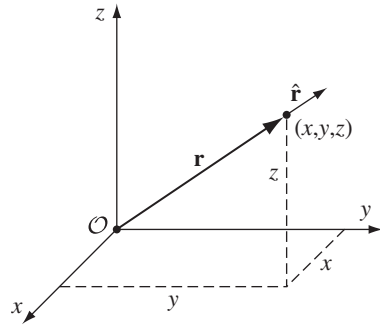


Fig. 1.13

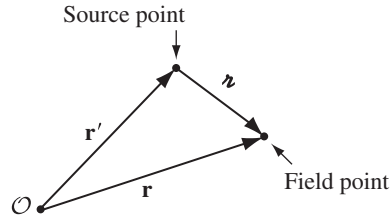


Fig. 1.14

and a unit vector in the direction from $\mathbf{r}'$ to $\mathbf{r}$ is

$$\hat{\mathbf{z}} = \frac{\mathbf{z}}{z} = \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|}. \quad (1.25)$$

In Cartesian coordinates,

$$\mathbf{z} = (x - x')\hat{\mathbf{x}} + (y - y')\hat{\mathbf{y}} + (z - z')\hat{\mathbf{z}}, \quad (1.26)$$

$$z = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}, \quad (1.27)$$

$$\hat{\mathbf{z}} = \frac{(x - x')\hat{\mathbf{x}} + (y - y')\hat{\mathbf{y}} + (z - z')\hat{\mathbf{z}}}{\sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}} \quad (1.28)$$

(from which you will appreciate the economy of the script- z notation).

Problem 1.7. Find the separation vector $\mathbf{z}$ from the source point $(2, 8, 7)$ to the field point $(4, 6, 8)$. Determine its magnitude (z), and construct the unit vector $\hat{\mathbf{z}}$.

1.1.5 How Vectors Transform

The² definition of a vector as “a quantity with a magnitude and direction” is not altogether satisfactory: What precisely does “direction” *mean*? This may seem a pedantic question, but we shall soon encounter a species of derivative that *looks* rather like a vector, and we’ll want to know for sure whether it *is* one.

You might be inclined to say that a vector is anything that has three components that combine properly under addition. Well, how about this: We have a barrel of fruit that contains N_x pears, N_y apples, and N_z bananas. Is $\mathbf{N} = N_x\hat{\mathbf{x}} + N_y\hat{\mathbf{y}} + N_z\hat{\mathbf{z}}$ a vector? It has three components, and when you add another barrel with M_x pears, M_y apples, and M_z bananas the result is

² This section can be skipped without loss of continuity.

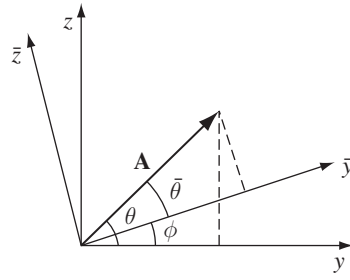


Fig. 1.15

$(N_x + M_x)$ pears, $(N_y + M_y)$ apples, $(N_z + M_z)$ bananas. So it does *add* like a vector. Yet it's obviously *not* a vector, in the physicist's sense of the word, because it doesn't really have a *direction*. What exactly is wrong with it?

The answer is that **N** *does not transform properly when you change coordinates*. The coordinate frame we use to describe positions in space is of course entirely arbitrary, but there is a specific geometrical transformation law for converting vector components from one frame to another. Suppose, for instance, the $\bar{x}, \bar{y}, \bar{z}$ system is rotated by angle ϕ , relative to x, y, z , about the common $x = \bar{x}$ -axes. From Fig. 1.15,

$$A_y = A \cos \theta, \quad A_z = A \sin \theta,$$

while

$$\begin{aligned} \bar{A}_y &= A \cos \bar{\theta} = A \cos(\theta - \phi) = A(\cos \theta \cos \phi + \sin \theta \sin \phi) \\ &= \cos \phi A_y + \sin \phi A_z, \\ \bar{A}_z &= A \sin \bar{\theta} = A \sin(\theta - \phi) = A(\sin \theta \cos \phi - \cos \theta \sin \phi) \\ &= -\sin \phi A_y + \cos \phi A_z. \end{aligned}$$

We might express this conclusion in matrix notation:

$$\begin{pmatrix} \bar{A}_y \\ \bar{A}_z \end{pmatrix} = \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} A_y \\ A_z \end{pmatrix}. \quad (1.29)$$

More generally, for rotation about an *arbitrary* axis in three dimensions, the transformation law takes the form

$$\begin{pmatrix} \bar{A}_x \\ \bar{A}_y \\ \bar{A}_z \end{pmatrix} = \begin{pmatrix} R_{xx} & R_{xy} & R_{xz} \\ R_{yx} & R_{yy} & R_{yz} \\ R_{zx} & R_{zy} & R_{zz} \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix}, \quad (1.30)$$

or, more compactly,

$$\bar{A}_i = \sum_{j=1}^3 R_{ij} A_j, \quad (1.31)$$

where the index 1 stands for x , 2 for y , and 3 for z . The elements of the matrix R can be ascertained, for a given rotation, by the same sort of trigonometric arguments as we used for a rotation about the x -axis.

Now, *do* the components of $\mathbf{N}$ transform in this way? Of *course* not – it doesn't matter what coordinates you use to represent positions in space; there are still just as many apples in the barrel. You can't convert a pear into a banana by choosing a different set of axes, but you *can* turn A_x into $\bar{A}_y$. Formally, then, a *vector* is a set of three components that transforms the same way as (x, y, z) under a rotation of the coordinate axes. As always, displacement is the *archetype* for the behavior of all vectors.³

By the way, a (second-rank) **tensor** is a quantity with *nine* components, $T_{xx}, T_{xy}, T_{xz}, T_{yx}, \dots, T_{zz}$, which transform with *two* factors of R :

$$\begin{aligned}\bar{T}_{xx} = & R_{xx}(R_{xx}T_{xx} + R_{xy}T_{xy} + R_{xz}T_{xz}) \\ & + R_{xy}(R_{xx}T_{yx} + R_{xy}T_{yy} + R_{xz}T_{yz}) \\ & + R_{xz}(R_{xx}T_{zx} + R_{xy}T_{zy} + R_{xz}T_{zz}), \dots\end{aligned}$$

or, more compactly,

$$\bar{T}_{ij} = \sum_{k=1}^3 \sum_{l=1}^3 R_{ik} R_{jl} T_{kl}. \quad (1.32)$$

In general, an n th-rank tensor has n indices and 3^n components, and transforms with n factors of R . In this hierarchy, a vector is a tensor of rank 1, and a scalar is a tensor of rank zero.⁴

Problem 1.8.

- Prove that the two-dimensional rotation matrix (Eq. 1.29) preserves dot products. (That is, show that $\bar{A}_y \bar{B}_y + \bar{A}_z \bar{B}_z = A_y B_y + A_z B_z$.)
- What constraints must the elements (R_{ij}) of the three-dimensional rotation matrix (Eq. 1.30) satisfy in order to preserve the length of $\mathbf{A}$ (for all vectors $\mathbf{A}$)?

Problem 1.9. Find the transformation matrix R that describes a rotation by 120° about an axis from the origin through the point $(1, 1, 1)$. The rotation is clockwise as you look down the axis toward the origin.

Problem 1.10. Vectors (and scalars, and tensors) are defined by their behavior under a *rotation* of the coordinate axes. But it is sometimes of interest to enquire how they might transform (using displacements as a model) under other coordinate changes – translations, say, or inversions.

³ If you're a mathematician you might want to contemplate generalized vector spaces in which the "axes" have nothing to do with direction and the basis vectors are no longer $\hat{x}$, $\hat{y}$, and $\hat{z}$ (indeed, there may be more than three dimensions). This is the subject of **linear algebra**. But for our purposes all vectors live in ordinary 3-space (or, in Chapter 12, in 4-dimensional space-time).

⁴ A scalar does not change when you change coordinates. In particular, the individual *components* of a vector are *not* scalars, but the magnitude *is*.

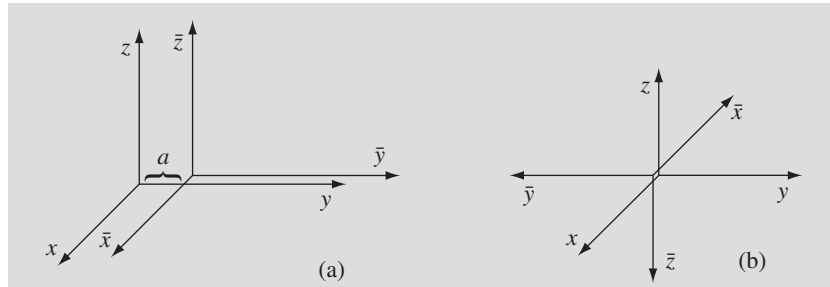


Fig. 1.16

- (a) How do the components of a vector⁵ transform under a **translation** of coordinates ($\bar{x} = x, \bar{y} = y - a, \bar{z} = z$, Fig. 1.16a)?
- (b) How do the components of a displacement transform under an **inversion** of coordinates ($\bar{x} = -x, \bar{y} = -y, \bar{z} = -z$, Fig. 1.16b)?
- (c) How do the components of a cross product (Eq. 1.13) transform under inversion? (The cross product of two vectors is called a **pseudovector** because of this “anomalous” behavior under inversions. It’s still a *vector* – that is determined by its behavior under *rotations*; where the distinction is at issue I’ll call a vector with the “normal” behavior under inversions an **ordinary vector**.) Is the cross product of two pseudovectors an ordinary vector, or a pseudovector? Name two pseudovector quantities in classical mechanics.
- (d) How does the scalar triple product of three vectors transform under inversions? (Such an object is called a **pseudoscalar**.)

1.2 Differential Calculus

1.2.1 “Ordinary” Derivatives

Suppose we have a function of one variable: $f(x)$. *Question:* What does the derivative, df/dx , do for us? *Answer:* It tells us how rapidly the function $f(x)$ varies when we change the argument x by a tiny amount, dx :

$$df = \left(\frac{df}{dx} \right) dx. \quad (1.33)$$

In words, if we increment x by an infinitesimal amount dx , then f changes by an amount df ; the derivative is the proportionality factor. For example,

⁵ *Beware:* The vector $\mathbf{r}$ (Eq. 1.19) goes from a specific point in space (the origin, $\mathcal{O}$) to the point $P = (x, y, z)$. Under a translation the *new* origin ($\bar{\mathcal{O}}$) is at a different location, and the arrow from $\bar{\mathcal{O}}$ to P is a completely different vector. The original vector $\mathbf{r}$ still goes from $\mathcal{O}$ to P , regardless of the coordinates used to label these points.

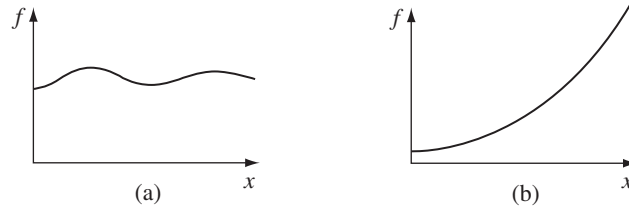


Fig. 1.17

in Fig. 1.17a, the function varies slowly with x , and the derivative is correspondingly small. In Fig. 1.17b, f increases rapidly with x , and the derivative is large, as you move away from $x = 0$.

Geometrical Interpretation: The derivative df/dx is the *slope* of the graph of f versus x .

1.2.2 Gradient

Suppose now that we have a function of *three* variables – say, the temperature $T(x, y, z)$ in this room. (Start out in one corner, and set up a system of axes; then for each point (x, y, z) in the room, T gives the temperature at that location.) We want to generalize the notion of “derivative” to functions like T , which depend not on *one* but on *three* variables.

A derivative is supposed to tell us how fast the function varies if we move a little distance. But this time the situation is more complicated, because it depends on what *direction* we move: if we go straight up, then the temperature will probably increase fairly rapidly, but if we move horizontally, it may not change much at all. In fact, the question “How fast does T vary?” has an infinite number of answers, one for each direction we might choose to explore.

Fortunately, the problem is not as bad as it looks. A theorem on partial derivatives states that

$$dT = \left(\frac{\partial T}{\partial x}\right) dx + \left(\frac{\partial T}{\partial y}\right) dy + \left(\frac{\partial T}{\partial z}\right) dz. \quad (1.34)$$

This tells us how T changes when we alter all three variables by the infinitesimal amounts dx , dy , dz . Notice that we do *not* require an infinite number of derivatives – *three* will suffice: the *partial* derivatives along each of the three coordinate directions.

Equation 1.34 is reminiscent of a dot product:

$$\begin{aligned} dT &= \left(\frac{\partial T}{\partial x} \hat{\mathbf{x}} + \frac{\partial T}{\partial y} \hat{\mathbf{y}} + \frac{\partial T}{\partial z} \hat{\mathbf{z}}\right) \cdot (dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}) \\ &= (\nabla T) \cdot (d\mathbf{l}), \end{aligned} \quad (1.35)$$

where

$$\nabla T \equiv \frac{\partial T}{\partial x} \hat{\mathbf{x}} + \frac{\partial T}{\partial y} \hat{\mathbf{y}} + \frac{\partial T}{\partial z} \hat{\mathbf{z}} \quad (1.36)$$

is called the **gradient** of T . Note that ∇T is a *vector* quantity, with three components; it is the generalized derivative we have been looking for. Equation 1.35 is the three-dimensional version of Eq. 1.33.

Geometrical Interpretation of the Gradient: Like any vector, the gradient has *magnitude* and *direction*. To determine its geometrical meaning, let's rewrite the dot product (Eq. 1.35) using Eq. 1.1:

$$dT = \nabla T \cdot d\mathbf{l} = |\nabla T| |d\mathbf{l}| \cos \theta, \quad (1.37)$$

where θ is the angle between ∇T and $d\mathbf{l}$. Now, if we *fix* the *magnitude* $|d\mathbf{l}|$ and search around in various *directions* (that is, vary θ), the *maximum* change in T evidently occurs when $\theta = 0$ (for then $\cos \theta = 1$). That is, for a fixed distance $|d\mathbf{l}|$, dT is greatest when I move in the *same direction* as ∇T . Thus,

the gradient ∇T points in the direction of maximum increase of the function T .

Moreover,

the magnitude $|\nabla T|$ gives the slope (rate of increase) along this maximal direction.

Imagine you are standing on a hillside. Look all around you, and find the direction of steepest ascent. That is the *direction* of the gradient. Now measure the *slope* in that direction (rise over run). That is the *magnitude* of the gradient. (Here the function we're talking about is the height of the hill, and the coordinates it depends on are positions – latitude and longitude, say. This function depends on only *two* variables, not *three*, but the geometrical meaning of the gradient is easier to grasp in two dimensions.) Notice from Eq. 1.37 that the direction of maximum *descent* is opposite to the direction of maximum *ascent*, while at right angles ($\theta = 90^\circ$) the slope is zero (the gradient is perpendicular to the contour lines). You can conceive of surfaces that do not have these properties, but they always have “kinks” in them, and correspond to nondifferentiable functions.

What would it mean for the gradient to vanish? If $\nabla T = \mathbf{0}$ at (x, y, z) , then $dT = 0$ for small displacements about the point (x, y, z) . This is, then, a **stationary point** of the function $T(x, y, z)$. It could be a maximum (a summit), a minimum (a valley), a saddle point (a pass), or a “shoulder.” This is analogous to the situation for functions of *one* variable, where a vanishing derivative signals a maximum, a minimum, or an inflection. In particular, if you want to locate the extrema of a function of three variables, set its gradient equal to zero.

Example 1.3. Find the gradient of $r = \sqrt{x^2 + y^2 + z^2}$ (the magnitude of the position vector).

Solution.

$$\begin{aligned}\nabla r &= \frac{\partial r}{\partial x} \hat{\mathbf{x}} + \frac{\partial r}{\partial y} \hat{\mathbf{y}} + \frac{\partial r}{\partial z} \hat{\mathbf{z}} \\ &= \frac{1}{2} \frac{2x}{\sqrt{x^2 + y^2 + z^2}} \hat{\mathbf{x}} + \frac{1}{2} \frac{2y}{\sqrt{x^2 + y^2 + z^2}} \hat{\mathbf{y}} + \frac{1}{2} \frac{2z}{\sqrt{x^2 + y^2 + z^2}} \hat{\mathbf{z}} \\ &= \frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}}{\sqrt{x^2 + y^2 + z^2}} = \frac{\mathbf{r}}{r} = \hat{\mathbf{r}}.\end{aligned}$$

Does this make sense? Well, it says that the distance from the origin increases most rapidly in the radial direction, and that its *rate* of increase in that direction is 1 ... just what you'd expect.

Problem 1.11. Find the gradients of the following functions:

- (a) $f(x, y, z) = x^2 + y^3 + z^4$.
- (b) $f(x, y, z) = x^2 y^3 z^4$.
- (c) $f(x, y, z) = e^x \sin(y) \ln(z)$.

Problem 1.12. The height of a certain hill (in feet) is given by

$$h(x, y) = 10(2xy - 3x^2 - 4y^2 - 18x + 28y + 12),$$

where y is the distance (in miles) north, x the distance east, of South Hadley.

- (a) Where is the top of the hill located?
- (b) How high is the hill?
- (c) How steep is the slope (in feet per mile) at a point 1 mile north and 1 mile east of South Hadley? In what direction is the slope steepest, at that point?

- **Problem 1.13.** Let $\mathbf{r}$ be the separation vector from a fixed point (x', y', z') to the point (x, y, z) , and let r be its length. Show that

- (a) $\nabla(r^2) = 2\mathbf{r}$;
- (b) $\nabla(1/r) = -\mathbf{r}/r^2$.
- (c) What is the *general* formula for $\nabla(r^n)$?

! **Problem 1.14.** Suppose f is a function of two variables (y and z) only. Show that the gradient $\nabla f = (\partial f/\partial y)\hat{y} + (\partial f/\partial z)\hat{z}$ transforms as a vector under rotations, Eq. 1.29. [Hint: $(\partial f/\partial \bar{y}) = (\partial f/\partial y)(\partial y/\partial \bar{y}) + (\partial f/\partial z)(\partial z/\partial \bar{y})$, and the analogous formula for $\partial f/\partial \bar{z}$. We know that $\bar{y} = y \cos \phi + z \sin \phi$ and $\bar{z} = -y \sin \phi + z \cos \phi$; “solve” these equations for y and z (as functions of $\bar{y}$ and $\bar{z}$), and compute the needed derivatives $\partial y/\partial \bar{y}$, $\partial z/\partial \bar{y}$, etc.]

1.2.3 The Del Operator

The gradient has the formal appearance of a vector, ∇ , “multiplying” a scalar T :

$$\nabla T = \left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right) T. \quad (1.38)$$

(For once, I write the unit vectors to the *left*, just so no one will think this means $\partial \hat{x}/\partial x$, and so on – which would be zero, since $\hat{x}$ is constant.) The term in parentheses is called **del**:

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}. \quad (1.39)$$

Of course, del is *not* a vector, in the usual sense. Indeed, it doesn’t mean much until we give it a function to act on. Furthermore, it does not “multiply” T ; rather, it is an instruction to *differentiate* what follows. To be precise, then, we say that ∇ is a **vector operator** that *acts upon* T , not a vector that multiplies T .

With this qualification, though, ∇ mimics the behavior of an ordinary vector in virtually every way; almost anything that can be done with other vectors can also be done with ∇ , if we merely translate “multiply” as “act upon.” So by all means take the vector appearance of ∇ seriously: it is a marvelous piece of notational simplification, as you will appreciate if you ever consult Maxwell’s original work on electromagnetism, written without the benefit of ∇ .

Now, an ordinary vector $\mathbf{A}$ can multiply in three ways:

- (1) by a scalar a : $\mathbf{A}a$;
- (2) by a vector $\mathbf{B}$ via the dot product: $\mathbf{A} \cdot \mathbf{B}$;
- (3) by a vector $\mathbf{B}$ via the cross product: $\mathbf{A} \times \mathbf{B}$.

Correspondingly, there are three ways the operator ∇ can act:

- (1) on a scalar function T : ∇T (the gradient);
- (2) on a vector function $\mathbf{v}$ via the dot product: $\nabla \cdot \mathbf{v}$ (the **divergence**);
- (3) on a vector function $\mathbf{v}$ via the cross product: $\nabla \times \mathbf{v}$ (the **curl**).

I have already discussed the gradient. In the following sections we examine the other two vector derivatives: divergence and curl.

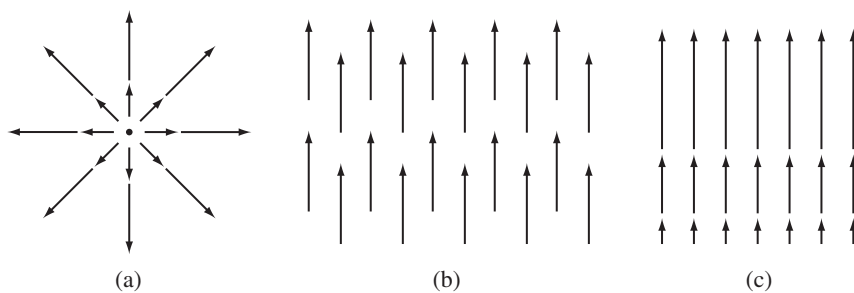


Fig. 1.18

1.2.4 The Divergence

From the definition of ∇ we construct the divergence:

$$\begin{aligned}\nabla \cdot \mathbf{v} &= \left(\hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \right) \cdot (v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} + v_z \hat{\mathbf{z}}) \\ &= \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}.\end{aligned}\quad (1.40)$$

Observe that the divergence of a vector function⁶ $\mathbf{v}$ is itself a *scalar*, $\nabla \cdot \mathbf{v}$.

Geometrical Interpretation: The name **divergence** is well chosen, for $\nabla \cdot \mathbf{v}$ is a measure of how much the vector $\mathbf{v}$ spreads out (diverges) from the point in question. For example, the vector function in Fig. 1.18a has a large (positive) divergence (if the arrows pointed *in*, it would be a *negative* divergence), the function in Fig. 1.18b has zero divergence, and the function in Fig. 1.18c again has a positive divergence. (Please understand that $\mathbf{v}$ here is a *function* – there’s a vector associated with *every point* in space. In the diagrams, of course, I can only draw the arrows at a few representative locations.)

Imagine standing at the edge of a pond. Sprinkle some sawdust or pine needles on the surface. If the material spreads out, then you dropped it at a point of positive divergence; if it collects together, you dropped it at a point of negative divergence. (The vector function $\mathbf{v}$ in this model is the velocity of the water at the surface – this is a *two-dimensional* example, but it helps give one a “feel” for what the divergence means. A point of positive divergence is a source, or “faucet”; a point of negative divergence is a sink, or “drain.”)

Example 1.4. Suppose the functions in Fig. 1.18 are $\mathbf{v}_a = \mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$, $\mathbf{v}_b = \hat{\mathbf{z}}$, and $\mathbf{v}_c = z\hat{\mathbf{z}}$. Calculate their divergences.

Solution.

$$\nabla \cdot \mathbf{v}_a = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3;$$

⁶ A vector function $\mathbf{v}(x, y, z) = v_x(x, y, z)\hat{\mathbf{x}} + v_y(x, y, z)\hat{\mathbf{y}} + v_z(x, y, z)\hat{\mathbf{z}}$ is really *three* functions – one for each component. There’s no such thing as the divergence of a scalar.

as anticipated, this function has a positive divergence.

$$\nabla \cdot \mathbf{v}_b = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(1) = 0 + 0 + 0 = 0,$$

as expected.

$$\nabla \cdot \mathbf{v}_c = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(z) = 0 + 0 + 1 = 1.$$

Problem 1.15. Calculate the divergence of the following vector functions:

(a) $\mathbf{v}_a = x^2\hat{\mathbf{x}} + 3xz^2\hat{\mathbf{y}} - 2xz\hat{\mathbf{z}}.$

(b) $\mathbf{v}_b = xy\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + 3zx\hat{\mathbf{z}}.$

(c) $\mathbf{v}_c = y^2\hat{\mathbf{x}} + (2xy + z^2)\hat{\mathbf{y}} + 2yz\hat{\mathbf{z}}.$

- **Problem 1.16.** Sketch the vector function

$$\mathbf{v} = \frac{\hat{\mathbf{r}}}{r^2},$$

and compute its divergence. The answer may surprise you ... how do you account for it?

- ! **Problem 1.17.** In two dimensions, show that the divergence transforms as a scalar under rotations. [Hint: Use Eq. 1.29 to determine $\bar{v}_y$ and $\bar{v}_z$, and the method of Prob. 1.14 to calculate the derivatives. Your aim is to show that $\partial\bar{v}_y/\partial\bar{y} + \partial\bar{v}_z/\partial\bar{z} = \partial v_y/\partial y + \partial v_z/\partial z$.]

1.2.5 The Curl

From the definition of ∇ we construct the curl:

$$\begin{aligned} \nabla \times \mathbf{v} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_x & v_y & v_z \end{vmatrix} \\ &= \hat{\mathbf{x}}\left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right) + \hat{\mathbf{y}}\left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right) + \hat{\mathbf{z}}\left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right). \end{aligned} \quad (1.41)$$

Notice that the curl of a vector function⁷ $\mathbf{v}$ is, like any cross product, a *vector*.

Geometrical Interpretation: The name **curl** is also well chosen, for $\nabla \times \mathbf{v}$ is a measure of how much the vector $\mathbf{v}$ swirls around the point in question. Thus the three functions in Fig. 1.18 all have zero curl (as you can easily check, using the expressions in Ex. 1.4), whereas the functions in Fig. 1.19 have a substantial curl, pointing in the z direction, as the natural right-hand rule would suggest. Imagine (again) you are standing at the edge of a pond.

⁷ There's no such thing as the curl of a scalar.

Float a small paddlewheel (a cork with toothpicks pointing out radially would do); if it starts to rotate, then you placed it at a point of nonzero *curl*. A whirlpool would be a region of large curl.

Example 1.5. Suppose the function sketched in Fig. 1.19a is $\mathbf{v}_a = -y\hat{\mathbf{x}} + x\hat{\mathbf{y}}$, and that in Fig. 1.19b is $\mathbf{v}_b = x\hat{\mathbf{y}}$. Calculate their curls.

Solution.

$$\nabla \times \mathbf{v}_a = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -y & x & 0 \end{vmatrix} = 2\hat{\mathbf{z}}$$

and

$$\nabla \times \mathbf{v}_b = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 0 & x & 0 \end{vmatrix} = \hat{\mathbf{z}}.$$

As expected, these curls point in the $+z$ direction. (Incidentally, they both have zero divergence, as you might guess from the pictures: nothing is “spreading out” ... it just “swirls around.”)

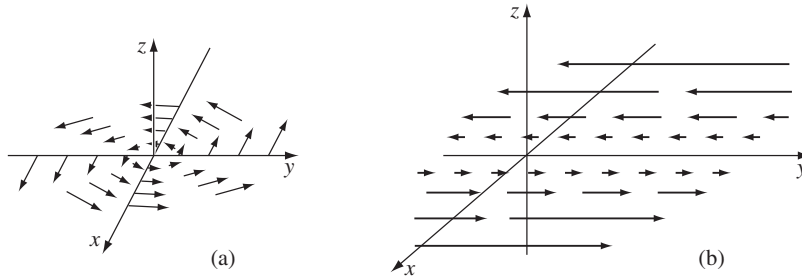


Fig. 1.19

Problem 1.18. Calculate the curls of the vector functions in Prob. 1.15.

Problem 1.19. Draw a circle in the xy plane. At a few representative points draw the unit vector $\mathbf{v}$ tangent to the circle, pointing in the clockwise direction. By comparing adjacent vectors, determine the *signs* of $\partial v_x/\partial y$ and $\partial v_y/\partial x$. According to Eq. 1.41 then, what is the direction of $\nabla \times \mathbf{v}$?

Problem 1.20. Construct a vector function that has zero divergence and zero curl everywhere. (A *constant* will do the job, of course, but make it something a little more interesting than that!)

1.2.6 Product Rules

The calculation of ordinary derivatives is facilitated by a number of rules, such as the sum rule:

$$\frac{d}{dx}(f + g) = \frac{df}{dx} + \frac{dg}{dx},$$

the rule for multiplying by a constant (c):

$$\frac{d}{dx}(cf) = c \frac{df}{dx},$$

the product rule:

$$\frac{d}{dx}(fg) = f \frac{dg}{dx} + g \frac{df}{dx},$$

and the quotient rule:

$$\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{g \frac{df}{dx} - f \frac{dg}{dx}}{g^2}.$$

Similar relations hold for the vector derivatives. Thus,

$$\nabla(f + g) = \nabla f + \nabla g, \quad \nabla \cdot (\mathbf{A} + \mathbf{B}) = (\nabla \cdot \mathbf{A}) + (\nabla \cdot \mathbf{B}),$$

$$\nabla \times (\mathbf{A} + \mathbf{B}) = (\nabla \times \mathbf{A}) + (\nabla \times \mathbf{B}),$$

and

$$\nabla(cf) = c\nabla f, \quad \nabla \cdot (c\mathbf{A}) = c(\nabla \cdot \mathbf{A}), \quad \nabla \times (c\mathbf{A}) = c(\nabla \times \mathbf{A}),$$

as you can check for yourself. The product rules are not quite so simple. There are two ways to construct a scalar as the product of two functions:

fg (product of two scalar functions),

$\mathbf{A} \cdot \mathbf{B}$ (dot product of two vector functions),

and two ways to make a vector:

$f\mathbf{A}$ (scalar times vector),

$\mathbf{A} \times \mathbf{B}$ (cross product of two vectors).

Accordingly, there are *six* product rules, two for gradients:

$$(i) \quad \nabla(fg) = f\nabla g + g\nabla f;$$

$$(ii) \quad \nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A};$$

two for divergences:

$$(iii) \quad \nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f);$$

$$(iv) \quad \nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B});$$

and two for curls:

$$(v) \quad \nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f);$$

$$(vi) \quad \nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}).$$

You will be using these product rules so frequently that I have put them inside the front cover for easy reference. The proofs come straight from the product rule for ordinary derivatives. For instance,

$$\begin{aligned} \nabla \cdot (f\mathbf{A}) &= \frac{\partial}{\partial x}(fA_x) + \frac{\partial}{\partial y}(fA_y) + \frac{\partial}{\partial z}(fA_z) \\ &= \left(\frac{\partial f}{\partial x}A_x + f\frac{\partial A_x}{\partial x}\right) + \left(\frac{\partial f}{\partial y}A_y + f\frac{\partial A_y}{\partial y}\right) + \left(\frac{\partial f}{\partial z}A_z + f\frac{\partial A_z}{\partial z}\right) \\ &= (\nabla f) \cdot \mathbf{A} + f(\nabla \cdot \mathbf{A}). \end{aligned}$$

It is also possible to formulate three quotient rules:

$$\begin{aligned} \nabla\left(\frac{f}{g}\right) &= \frac{g\nabla f - f\nabla g}{g^2}, \\ \nabla \cdot \left(\frac{\mathbf{A}}{g}\right) &= \frac{g(\nabla \cdot \mathbf{A}) - \mathbf{A} \cdot (\nabla g)}{g^2}, \\ \nabla \times \left(\frac{\mathbf{A}}{g}\right) &= \frac{g(\nabla \times \mathbf{A}) + \mathbf{A} \times (\nabla g)}{g^2}. \end{aligned}$$

However, since these can be obtained quickly from the corresponding product rules, there is no point in listing them separately.

Problem 1.21. Prove product rules (i), (iv), and (v).

Problem 1.22.

- If $\mathbf{A}$ and $\mathbf{B}$ are two vector functions, what does the expression $(\mathbf{A} \cdot \nabla)\mathbf{B}$ mean? (That is, what are its x , y , and z components, in terms of the Cartesian components of $\mathbf{A}$, $\mathbf{B}$, and ∇ ?)
- Compute $(\hat{\mathbf{r}} \cdot \nabla)\hat{\mathbf{r}}$, where $\hat{\mathbf{r}}$ is the unit vector defined in Eq. 1.21.
- For the functions in Prob. 1.15, evaluate $(\mathbf{v}_a \cdot \nabla)\mathbf{v}_b$.

Problem 1.23. (For masochists only.) Prove product rules (ii) and (vi). Refer to Prob. 1.22 for the definition of $(\mathbf{A} \cdot \nabla)\mathbf{B}$.

Problem 1.24. Derive the three quotient rules.

Problem 1.25.

(a) Check product rule (iv) (by calculating each term separately) for the functions

$$\mathbf{A} = x\hat{\mathbf{x}} + 2y\hat{\mathbf{y}} + 3z\hat{\mathbf{z}}, \quad \mathbf{B} = 3y\hat{\mathbf{x}} - 2x\hat{\mathbf{y}}.$$

(b) Do the same for product rule (ii).

(c) Do the same for rule (vi).

1.2.7 Second Derivatives

Gradient, divergence, and curl are the only first derivatives we can make with ∇ ; by applying ∇ *twice*, we can construct five species of *second* derivatives. The gradient ∇T is a *vector*, so we can take the *divergence* and *curl* of it:

(a) Divergence of gradient: $\nabla \cdot (\nabla T)$.

(b) Curl of gradient: $\nabla \times (\nabla T)$.

The divergence $\nabla \cdot \mathbf{v}$ is a *scalar* – all we can do is take its *gradient*:

(c) Gradient of divergence: $\nabla(\nabla \cdot \mathbf{v})$.

The curl $\nabla \times \mathbf{v}$ is a *vector*, so we can take its *divergence* and *curl*:

(d) Divergence of curl: $\nabla \cdot (\nabla \times \mathbf{v})$.

(e) Curl of curl: $\nabla \times (\nabla \times \mathbf{v})$.

This exhausts the possibilities, and in fact not all of them give us anything new. Let's consider them one at a time:

$$\begin{aligned} (1) \quad \nabla \cdot (\nabla T) &= \left(\hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \right) \cdot \left(\frac{\partial T}{\partial x} \hat{\mathbf{x}} + \frac{\partial T}{\partial y} \hat{\mathbf{y}} + \frac{\partial T}{\partial z} \hat{\mathbf{z}} \right) \\ &= \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}. \end{aligned} \quad (1.42)$$

This object, which we write as $\nabla^2 T$ for short, is called the **Laplacian** of T ; we shall be studying it in great detail later on. Notice that the Laplacian of a *scalar* T is a *scalar*. Occasionally, we shall speak of the Laplacian of a *vector*, $\nabla^2 \mathbf{v}$. By this we mean a *vector* quantity whose x -component is the Laplacian of v_x , and so on:⁸

$$\nabla^2 \mathbf{v} \equiv (\nabla^2 v_x) \hat{\mathbf{x}} + (\nabla^2 v_y) \hat{\mathbf{y}} + (\nabla^2 v_z) \hat{\mathbf{z}}. \quad (1.43)$$

This is nothing more than a convenient extension of the meaning of ∇^2 .

(2) The curl of a gradient is always zero:

$$\nabla \times (\nabla T) = \mathbf{0}. \quad (1.44)$$

This is an important fact, which we shall use repeatedly; you can easily prove it from the definition of ∇ , Eq. 1.39. *Beware*: You might think

⁸ In curvilinear coordinates, where the unit vectors themselves depend on position, they too must be differentiated (see Section 1.4.1).

Eq. 1.44 is “obviously” true – isn’t it just $(\nabla \times \nabla)T$, and isn’t the cross product of *any* vector (in this case, ∇) with itself always zero? This reasoning is suggestive, but not quite conclusive, since ∇ is an *operator* and does not “multiply” in the usual way. The proof of Eq. 1.44, in fact, hinges on the equality of cross derivatives:

$$\frac{\partial}{\partial x} \left(\frac{\partial T}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\partial T}{\partial x} \right). \quad (1.45)$$

If you think I’m being fussy, test your intuition on this one:

$$(\nabla T) \times (\nabla S).$$

Is *that* always zero? (It *would* be, of course, if you replaced the ∇ s by an ordinary vector.)

- (3) $\nabla(\nabla \cdot \mathbf{v})$ seldom occurs in physical applications, and it has not been given any special name of its own – it’s just the **gradient of the divergence**. Notice that $\nabla(\nabla \cdot \mathbf{v})$ is *not* the same as the Laplacian of a vector: $\nabla^2 \mathbf{v} = (\nabla \cdot \nabla) \mathbf{v} \neq \nabla(\nabla \cdot \mathbf{v})$.
- (4) The divergence of a curl, like the curl of a gradient, is always zero:

$$\nabla \cdot (\nabla \times \mathbf{v}) = 0. \quad (1.46)$$

You can prove this for yourself. (Again, there is a fraudulent shortcut proof, using the vector identity $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$.)

- (5) As you can check from the definition of ∇ ,

$$\nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{v}) - \nabla^2 \mathbf{v}. \quad (1.47)$$

So curl-of-curl gives nothing new; the first term is just number (3), and the second is the Laplacian (of a vector). (In fact, Eq. 1.47 is often used to *define* the Laplacian of a vector, in preference to Eq. 1.43, which makes explicit reference to Cartesian coordinates.)

Really, then, there are just two kinds of second derivatives: the Laplacian (which is of fundamental importance) and the gradient-of-divergence (which we seldom encounter). I could go through a similar exercise to work out *third* derivatives, but fortunately second derivatives suffice for most physical applications.

A final word on vector differential calculus: It *all* flows from the operator ∇ , and from taking seriously its vectorial character. Even if you remembered *only* the definition of ∇ , you could easily reconstruct all the rest.

Problem 1.26. Calculate the Laplacian of the following functions:

- (a) $T_a = x^2 + 2xy + 3z + 4$.
 (b) $T_b = \sin x \sin y \sin z$.
 (c) $T_c = e^{-5x} \sin 4y \cos 3z$.
 (d) $\mathbf{v} = x^2 \hat{\mathbf{x}} + 3xz^2 \hat{\mathbf{y}} - 2xz \hat{\mathbf{z}}$.

Problem 1.27. Prove that the divergence of a curl is always zero. Check it for function $\mathbf{v}_a$ in Prob. 1.15.

Problem 1.28. Prove that the curl of a gradient is always zero. Check it for function (b) in Prob. 1.11.

1.3 Integral Calculus

1.3.1 Line, Surface, and Volume Integrals

In electrodynamics, we encounter several different kinds of integrals, among which the most important are **line** (or **path**) **integrals**, **surface integrals** (or **flux**), and **volume integrals**.

(a) **Line Integrals.** A line integral is an expression of the form

$$\int_a^b \mathbf{v} \cdot d\mathbf{l}, \quad (1.48)$$

where $\mathbf{v}$ is a vector function, $d\mathbf{l}$ is the infinitesimal displacement vector (Eq. 1.22), and the integral is to be carried out along a prescribed path from point $\mathbf{a}$ to point $\mathbf{b}$ (Fig. 1.20). If the path in question forms a closed loop (that is, if $\mathbf{b} = \mathbf{a}$), I shall put a circle on the integral sign:

$$\oint \mathbf{v} \cdot d\mathbf{l}. \quad (1.49)$$

At each point on the path, we take the dot product of $\mathbf{v}$ (evaluated at that point) with the displacement $d\mathbf{l}$ to the next point along the path. To a physicist, the most familiar example of a line integral is the work done by a force $\mathbf{F}$: $W = \int \mathbf{F} \cdot d\mathbf{l}$.

Ordinarily, the value of a line integral depends critically on the path taken from $\mathbf{a}$ to $\mathbf{b}$, but there is an important special class of vector functions for which the line integral is *independent* of path and is determined entirely by the end points. It will be our business in due course to characterize this special class of vectors. (A *force* that has this property is said to be **conservative**.)

Example 1.6. Calculate the line integral of the function $\mathbf{v} = y^2\hat{\mathbf{x}} + 2x(y+1)\hat{\mathbf{y}}$ from the point $\mathbf{a} = (1, 1, 0)$ to the point $\mathbf{b} = (2, 2, 0)$, along the paths (1) and (2) in Fig. 1.21. What is $\oint \mathbf{v} \cdot d\mathbf{l}$ for the loop that goes from $\mathbf{a}$ to $\mathbf{b}$ along (1) and returns to $\mathbf{a}$ along (2)?

Solution. As always, $d\mathbf{l} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}$. Path (1) consists of two parts. Along the “horizontal” segment, $dy = dz = 0$, so

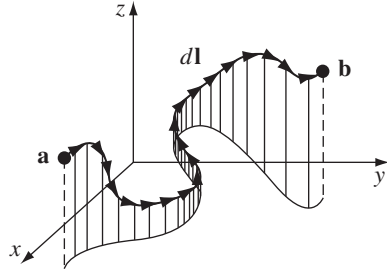


Fig. 1.20

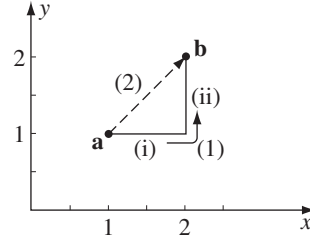


Fig. 1.21

(i) $d\mathbf{l} = dx \hat{\mathbf{x}}$, $y = 1$, $\mathbf{v} \cdot d\mathbf{l} = y^2 dx = dx$, so $\int \mathbf{v} \cdot d\mathbf{l} = \int_1^2 dx = 1$.

On the “vertical” stretch, $dx = dz = 0$, so

(ii) $d\mathbf{l} = dy \hat{\mathbf{y}}$, $x = 2$, $\mathbf{v} \cdot d\mathbf{l} = 2x(y+1) dy = 4(y+1) dy$, so

$$\int \mathbf{v} \cdot d\mathbf{l} = 4 \int_1^2 (y+1) dy = 10.$$

By path (1), then,

$$\int_a^b \mathbf{v} \cdot d\mathbf{l} = 1 + 10 = 11.$$

Meanwhile, on path (2) $x = y$, $dx = dy$, and $dz = 0$, so $d\mathbf{l} = dx \hat{\mathbf{x}} + dx \hat{\mathbf{y}}$, $\mathbf{v} \cdot d\mathbf{l} = x^2 dx + 2x(x+1) dx = (3x^2 + 2x) dx$, and

$$\int_a^b \mathbf{v} \cdot d\mathbf{l} = \int_1^2 (3x^2 + 2x) dx = (x^3 + x^2)|_1^2 = 10.$$

(The strategy here is to get everything in terms of one variable; I could just as well have eliminated x in favor of y .)

For the loop that goes *out* (1) and *back* (2), then,

$$\oint \mathbf{v} \cdot d\mathbf{l} = 11 - 10 = 1.$$

(b) **Surface Integrals.** A surface integral is an expression of the form

$$\int_S \mathbf{v} \cdot d\mathbf{a}, \quad (1.50)$$

where $\mathbf{v}$ is again some vector function, and the integral is over a specified surface S . Here $d\mathbf{a}$ is an infinitesimal patch of area, with direction perpendicular to the surface (Fig. 1.22). There are, of course, *two* directions perpendicular to any surface, so the *sign* of a surface integral is intrinsically ambiguous. If the surface is *closed* (forming a “balloon”), in which case I shall again put a circle on the integral sign

$$\oint \mathbf{v} \cdot d\mathbf{a},$$

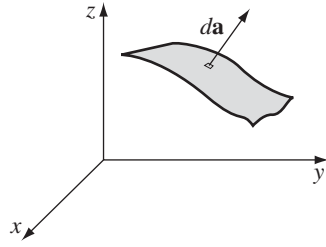


Fig. 1.22

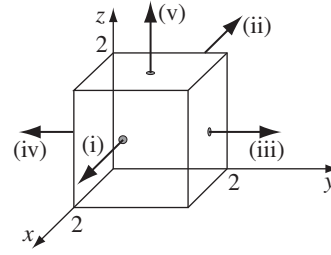


Fig. 1.23

then tradition dictates that “outward” is positive, but for open surfaces it’s arbitrary. If $\mathbf{v}$ describes the flow of a fluid (mass per unit area per unit time), then $\int \mathbf{v} \cdot d\mathbf{a}$ represents the total mass per unit time passing through the surface – hence the alternative name, “flux.”

Ordinarily, the value of a surface integral depends on the particular surface chosen, but there is a special class of vector functions for which it is *independent* of the surface and is determined entirely by the boundary line. An important task for us will be to characterize this special class of functions.

Example 1.7. Calculate the surface integral of $\mathbf{v} = 2xz\hat{\mathbf{x}} + (x+2)\hat{\mathbf{y}} + y(z^2 - 3)\hat{\mathbf{z}}$ over five sides (excluding the bottom) of the cubical box (side length 2) in Fig. 1.23. Let “upward and outward” be the positive direction, as indicated by the arrows.

Solution. Take the sides one at a time:

(i) $x = 2$, $d\mathbf{a} = dy dz \hat{\mathbf{x}}$, $\mathbf{v} \cdot d\mathbf{a} = 2xz dy dz = 4z dy dz$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = 4 \int_0^2 dy \int_0^2 z dz = 16.$$

(ii) $x = 0$, $d\mathbf{a} = -dy dz \hat{\mathbf{x}}$, $\mathbf{v} \cdot d\mathbf{a} = -2xz dy dz = 0$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = 0.$$

(iii) $y = 2$, $d\mathbf{a} = dx dz \hat{\mathbf{y}}$, $\mathbf{v} \cdot d\mathbf{a} = (x+2) dx dz$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = \int_0^2 (x+2) dx \int_0^2 dz = 12.$$

(iv) $y = 0$, $d\mathbf{a} = -dx dz \hat{\mathbf{y}}$, $\mathbf{v} \cdot d\mathbf{a} = -(x+2) dx dz$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = - \int_0^2 (x+2) dx \int_0^2 dz = -12.$$

(v) $z = 2$, $d\mathbf{a} = dx dy \hat{\mathbf{z}}$, $\mathbf{v} \cdot d\mathbf{a} = y(z^2 - 3) dx dy = y dx dy$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = \int_0^2 dx \int_0^2 y dy = 4.$$

The *total* flux is

$$\int_{\text{surface}} \mathbf{v} \cdot d\mathbf{a} = 16 + 0 + 12 - 12 + 4 = 20.$$

(c) **Volume Integrals.** A volume integral is an expression of the form

$$\int_V T d\tau, \quad (1.51)$$

where T is a scalar function and $d\tau$ is an infinitesimal volume element. In Cartesian coordinates,

$$d\tau = dx dy dz. \quad (1.52)$$

For example, if T is the density of a substance (which might vary from point to point), then the volume integral would give the total mass. Occasionally, we shall encounter volume integrals of *vector* functions:

$$\int \mathbf{v} d\tau = \int (v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} + v_z \hat{\mathbf{z}}) d\tau = \hat{\mathbf{x}} \int v_x d\tau + \hat{\mathbf{y}} \int v_y d\tau + \hat{\mathbf{z}} \int v_z d\tau; \quad (1.53)$$

because the unit vectors ($\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$) are constants, they come outside the integral.

Example 1.8. Calculate the volume integral of $T = xyz^2$ over the prism in Fig. 1.24.

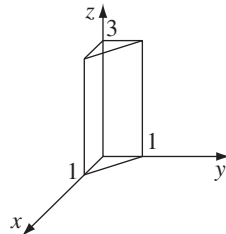


Fig. 1.24

Solution. You can do the three integrals in any order. Let's do x first: it runs from 0 to $(1 - y)$, then y (it goes from 0 to 1), and finally z (0 to 3):

$$\begin{aligned} \int T d\tau &= \int_0^3 z^2 \left\{ \int_0^1 y \left[\int_0^{1-y} x dx \right] dy \right\} dz \\ &= \frac{1}{2} \int_0^3 z^2 dz \int_0^1 (1-y)^2 y dy = \frac{1}{2} (9) \left(\frac{1}{12} \right) = \frac{3}{8}. \end{aligned}$$

Problem 1.29. Calculate the line integral of the function $\mathbf{v} = x^2\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + y^2\hat{\mathbf{z}}$ from the origin to the point (1,1,1) by three different routes:

- (a) $(0, 0, 0) \rightarrow (1, 0, 0) \rightarrow (1, 1, 0) \rightarrow (1, 1, 1)$.
- (b) $(0, 0, 0) \rightarrow (0, 0, 1) \rightarrow (0, 1, 1) \rightarrow (1, 1, 1)$.
- (c) The direct straight line.
- (d) What is the line integral around the closed loop that goes *out* along path (a) and *back* along path (b)?

Problem 1.30. Calculate the surface integral of the function in Ex. 1.7, over the *bottom* of the box. For consistency, let “upward” be the positive direction. Does the surface integral depend only on the boundary line for this function? What is the total flux over the *closed* surface of the box (including the bottom)? [Note: For the *closed* surface, the positive direction is “outward,” and hence “down,” for the bottom face.]

Problem 1.31. Calculate the volume integral of the function $T = z^2$ over the tetrahedron with corners at $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$.

1.3.2 The Fundamental Theorem of Calculus

Suppose $f(x)$ is a function of one variable. The **fundamental theorem of calculus** says that

$$\int_a^b \left(\frac{df}{dx} \right) dx = f(b) - f(a). \quad (1.54)$$

In case this doesn’t look familiar, I’ll write it another way:

$$\int_a^b F(x) dx = f(b) - f(a),$$

where $df/dx = F(x)$. The fundamental theorem tells you how to integrate $F(x)$: you think up a function $f(x)$ whose *derivative* is equal to F .

Geometrical Interpretation: According to Eq. 1.33, $df = (df/dx) dx$ is the infinitesimal change in f when you go from (x) to $(x+dx)$. The fundamental theorem (Eq. 1.54) says that if you chop the interval from a to b (Fig. 1.25) into many tiny pieces, dx , and add up the increments df from each little piece, the result is (not surprisingly) equal to the total change in f : $f(b) - f(a)$. In other words, there are two ways to determine the total change in the function: *either* subtract the values at the ends *or* go step by step, adding up all the tiny increments as you go. You’ll get the same answer either way.

Notice the structure of the fundamental theorem: the *integral* of a *derivative* over some *region* is given by the *value of the function* at the end points (*boundaries*). In vector calculus there are three species of derivative (gradient, divergence, and curl), and each has its own “fundamental theorem,” with essentially the same format. I don’t plan to prove these theorems

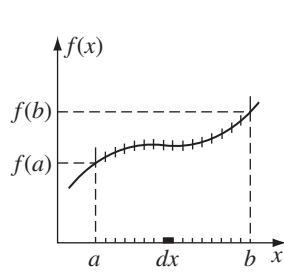


Fig. 1.25

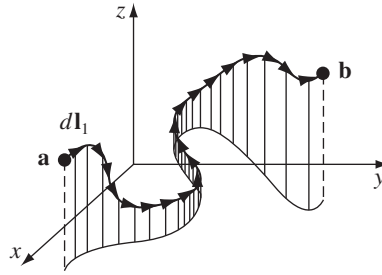


Fig. 1.26

here; rather, I will explain what they *mean*, and try to make them *plausible*. Proofs are given in Appendix A.

1.3.3 The Fundamental Theorem for Gradients

Suppose we have a scalar function of three variables $T(x, y, z)$. Starting at point **a**, we move a small distance $d\mathbf{l}_1$ (Fig. 1.26). According to Eq. 1.37, the function T will change by an amount

$$dT = (\nabla T) \cdot d\mathbf{l}_1.$$

Now we move a little farther, by an additional small displacement $d\mathbf{l}_2$; the incremental change in T will be $(\nabla T) \cdot d\mathbf{l}_2$. In this manner, proceeding by infinitesimal steps, we make the journey to point **b**. At each step we compute the gradient of T (at that point) and dot it into the displacement $d\mathbf{l}$... this gives us the change in T . Evidently, the *total* change in T in going from **a** to **b** (along the path selected) is

$$\int_a^b (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a}). \quad (1.55)$$

This is the **fundamental theorem for gradients**; like the “ordinary” fundamental theorem, it says that the integral (here a *line* integral) of a derivative (here the *gradient*) is given by the value of the function at the boundaries (**a** and **b**).

Geometrical Interpretation: Suppose you wanted to determine the height of the Eiffel Tower. You could climb the stairs, using a ruler to measure the rise at each step, and adding them all up (that’s the left-hand side of Eq. 1.55), or you could place altimeters at the top and the bottom, and subtract the two readings (that’s the right-hand side); you should get the same answer either way (that’s the fundamental theorem).

Incidentally, as we found in Ex. 1.6, line integrals ordinarily depend on the *path* taken from **a** to **b**. But the *right-hand* side of Eq. 1.55 makes no reference to the path – only to the end points! Evidently, *gradients* have the special property that their line integrals are path independent:

Corollary 1. The integral $\int_a^b (\nabla T) \cdot d\mathbf{l}$ is independent of the path taken from $\mathbf{a}$ to $\mathbf{b}$.

Corollary 2. The integral $\oint (\nabla T) \cdot d\mathbf{l} = 0$, since the beginning and end points are identical, and hence $T(\mathbf{b}) - T(\mathbf{a}) = 0$.

Example 1.9. Let $T = xy^2$, and take point $\mathbf{a}$ to be the origin $(0, 0, 0)$ and $\mathbf{b}$ the point $(2, 1, 0)$. Check the fundamental theorem for gradients.

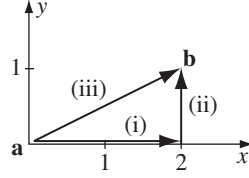


Fig. 1.27

Solution. Although the integral is independent of path, we must *pick* a specific path in order to evaluate it. Let's go out along the x -axis (step i) and then up (step ii) (Fig. 1.27). As always, $d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}$; $\nabla T = y^2 \hat{\mathbf{x}} + 2xy \hat{\mathbf{y}}$.

(i) $y = 0$, $d\mathbf{l} = dx \hat{\mathbf{x}}$, $\nabla T \cdot d\mathbf{l} = y^2 dx = 0$, so

$$\int_{(i)} \nabla T \cdot d\mathbf{l} = 0.$$

(ii) $x = 2$, $d\mathbf{l} = dy \hat{\mathbf{y}}$, $\nabla T \cdot d\mathbf{l} = 2xy dy = 4y dy$, so

$$\int_{(ii)} \nabla T \cdot d\mathbf{l} = \int_0^1 4y dy = 2y^2 \Big|_0^1 = 2.$$

The total line integral is 2. Is this consistent with the fundamental theorem? Yes: $T(\mathbf{b}) - T(\mathbf{a}) = 2 - 0 = 2$.

Now, just to convince you that the answer is independent of path, let me calculate the same integral along path (iii) (the straight line from $\mathbf{a}$ to $\mathbf{b}$):

(iii) $y = \frac{1}{2}x$, $dy = \frac{1}{2}dx$, $\nabla T \cdot d\mathbf{l} = y^2 dx + 2xy dy = \frac{3}{4}x^2 dx$, so

$$\int_{(iii)} \nabla T \cdot d\mathbf{l} = \int_0^2 \frac{3}{4}x^2 dx = \frac{1}{4}x^3 \Big|_0^2 = 2. \quad \checkmark$$

Problem 1.32. Check the fundamental theorem for gradients, using $T = x^2 + 4xy + 2yz^3$, the points $\mathbf{a} = (0, 0, 0)$, $\mathbf{b} = (1, 1, 1)$, and the three paths in Fig. 1.28:

(a) $(0, 0, 0) \rightarrow (1, 0, 0) \rightarrow (1, 1, 0) \rightarrow (1, 1, 1)$;

(b) $(0, 0, 0) \rightarrow (0, 0, 1) \rightarrow (0, 1, 1) \rightarrow (1, 1, 1)$;

(c) the parabolic path $z = x^2, y = x$.

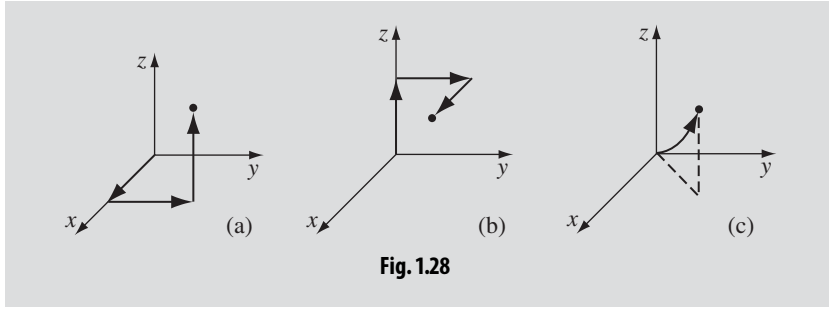


Fig. 1.28

1.3.4 The Fundamental Theorem for Divergences

The fundamental theorem for divergences states that

$$\int_{\mathcal{V}} (\nabla \cdot \mathbf{v}) d\tau = \oint_{\mathcal{S}} \mathbf{v} \cdot d\mathbf{a}. \quad (1.56)$$

In honor, I suppose, of its great importance, this theorem has at least three different names: **Gauss's theorem**, **Green's theorem**, or simply the **divergence theorem**. Like the other “fundamental theorems,” it says that the *integral* of a *derivative* (in this case the *divergence*) over a *region* (in this case a *volume*, $\mathcal{V}$) is equal to the value of the function at the *boundary* (in this case the *surface* $\mathcal{S}$ that bounds the volume). Notice that the boundary term is itself an integral (specifically, a surface integral). This is reasonable: the “boundary” of a *line* is just two end points, but the boundary of a *volume* is a (closed) surface.

Geometrical Interpretation: If $\mathbf{v}$ represents the flow of an incompressible fluid, then the *flux* of $\mathbf{v}$ (the right-hand side of Eq. 1.56) is the total amount of stuff passing out through the surface, per unit time. Now, the divergence measures the “spreading out” of the vectors from a point – a place of high divergence is like a “faucet,” pouring out liquid. If we have a bunch of faucets in a region completely filled with incompressible fluid, an equal amount of liquid will be forced out through the boundaries of the region. In fact, there are *two* ways we could determine how much is being produced: (a) we could count up all the faucets, recording how much each puts out, or (b) we could go around the boundary, measuring the outflow at each point, and add it all up. You get the same answer either way:

$$\int (\text{faucets within the volume}) = \oint (\text{flow out through the surface}).$$

This, in essence, is what the divergence theorem says.

Example 1.10. Check the divergence theorem using the function

$$\mathbf{v} = y^2 \hat{\mathbf{x}} + (2xy + z^2) \hat{\mathbf{y}} + (2yz) \hat{\mathbf{z}}$$

and a unit cube at the origin (Fig. 1.29).

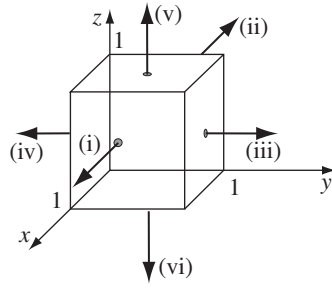


Fig. 1.29

Solution. In this case,

$$\nabla \cdot \mathbf{v} = 2(x + y)$$

and

$$\int_V 2(x + y) d\tau = 2 \int_0^1 \int_0^1 \int_0^1 (x + y) dx dy dz,$$

$$\int_0^1 (x + y) dx = \frac{1}{2} + y, \quad \int_0^1 \left(\frac{1}{2} + y\right) dy = 1, \quad \int_0^1 1 dz = 1.$$

Thus,

$$\int_V \nabla \cdot \mathbf{v} d\tau = 2.$$

So much for the left-hand side of the divergence theorem. To evaluate the surface integral we must consider separately the six faces of the cube:

$$\begin{aligned} \text{(i)} \quad \int \mathbf{v} \cdot d\mathbf{a} &= \int_0^1 \int_0^1 y^2 dy dz = \frac{1}{3}. \\ \text{(ii)} \quad \int \mathbf{v} \cdot d\mathbf{a} &= - \int_0^1 \int_0^1 y^2 dy dz = -\frac{1}{3}. \\ \text{(iii)} \quad \int \mathbf{v} \cdot d\mathbf{a} &= \int_0^1 \int_0^1 (2x + z^2) dx dz = \frac{4}{3}. \\ \text{(iv)} \quad \int \mathbf{v} \cdot d\mathbf{a} &= - \int_0^1 \int_0^1 z^2 dx dz = -\frac{1}{3}. \\ \text{(v)} \quad \int \mathbf{v} \cdot d\mathbf{a} &= \int_0^1 \int_0^1 2y dx dy = 1. \\ \text{(vi)} \quad \int \mathbf{v} \cdot d\mathbf{a} &= - \int_0^1 \int_0^1 0 dx dy = 0. \end{aligned}$$

So the total flux is

$$\oint_S \mathbf{v} \cdot d\mathbf{a} = \frac{1}{3} - \frac{1}{3} + \frac{4}{3} - \frac{1}{3} + 1 + 0 = 2,$$

as expected.

Problem 1.33. Test the divergence theorem for the function $\mathbf{v} = (xy)\hat{\mathbf{x}} + (2yz)\hat{\mathbf{y}} + (3zx)\hat{\mathbf{z}}$. Take as your volume the cube shown in Fig. 1.30, with sides of length 2.

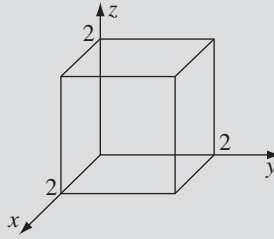


Fig. 1.30

1.3.5 The Fundamental Theorem for Curls

The fundamental theorem for curls, which goes by the special name of **Stokes' theorem**, states that

$$\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint_P \mathbf{v} \cdot d\mathbf{l}. \quad (1.57)$$

As always, the *integral* of a *derivative* (here, the *curl*) over a *region* (here, a patch of *surface*, S) is equal to the value of the function at the *boundary* (here, the perimeter of the patch, P). As in the case of the divergence theorem, the boundary term is itself an integral – specifically, a closed line integral.

Geometrical Interpretation: Recall that the curl measures the “twist” of the vectors $\mathbf{v}$; a region of high curl is a whirlpool – if you put a tiny paddle wheel there, it will rotate. Now, the integral of the curl over some surface (or, more precisely, the *flux* of the curl *through* that surface) represents the “total amount of swirl,” and we can determine that just as well by going around the edge and finding how much the flow is following the boundary (Fig. 1.31). Indeed, $\oint \mathbf{v} \cdot d\mathbf{l}$ is sometimes called the **circulation** of $\mathbf{v}$.

You may have noticed an apparent ambiguity in Stokes' theorem: concerning the boundary line integral, which *way* are we supposed to go around (clockwise or counterclockwise)? If we go the “wrong” way, we'll pick up an overall sign error. The answer is that it doesn't matter which way you go as long as you are consistent, for there is a compensating sign ambiguity in the surface integral: Which way does $d\mathbf{a}$ point? For a *closed* surface (as in the divergence theorem), $d\mathbf{a}$ points in the direction of the *outward* normal; but for an *open* surface, which way is “out”? Consistency in Stokes' theorem (as in all such matters) is given by the right-hand rule: if your fingers point in the direction of the line integral, then your thumb fixes the direction of $d\mathbf{a}$ (Fig. 1.32).

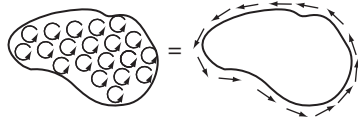


Fig. 1.31

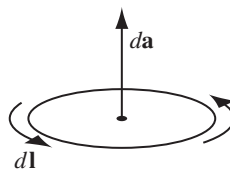


Fig. 1.32

Now, there are plenty of surfaces (infinitely many) that share any given boundary line. Twist a paper clip into a loop, and dip it in soapy water. The soap film constitutes a surface, with the wire loop as its boundary. If you blow on it, the soap film will expand, making a larger surface, with the same boundary. Ordinarily, a flux integral depends critically on what surface you integrate over, but evidently this is *not* the case with curls. For Stokes' theorem says that $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a}$ is equal to the line integral of $\mathbf{v}$ around the boundary, and the latter makes no reference to the specific surface you choose.

Corollary 1. The integral $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a}$ depends only on the boundary line, not on the particular surface used.

Corollary 2. The integral $\oint (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$ for any *closed* surface, since the boundary line, like the mouth of a balloon, shrinks down to a point, and hence the right-hand side of Eq. 1.57 vanishes.

These corollaries are analogous to the ones for the gradient theorem. We will develop the parallel further in due course.

Example 1.11. Suppose $\mathbf{v} = (2xz + 3y^2)\hat{\mathbf{y}} + (4yz^2)\hat{\mathbf{z}}$. Check Stokes' theorem for the square surface shown in Fig. 1.33.

Solution. Here,

$$\nabla \times \mathbf{v} = (4z^2 - 2x)\hat{\mathbf{x}} + 2z\hat{\mathbf{z}} \quad \text{and} \quad d\mathbf{a} = dy dz \hat{\mathbf{x}}.$$

(In saying that $d\mathbf{a}$ points in the x direction, we are committing ourselves to a counterclockwise line integral. We could as well write $d\mathbf{a} = -dy dz \hat{\mathbf{x}}$, but then we would be obliged to go clockwise.) Since $x = 0$ for this surface,

$$\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \int_0^1 \int_0^1 4z^2 dy dz = \frac{4}{3}.$$

Now, what about the line integral? We must break this up into four segments:

- (i) $x = 0, z = 0, \mathbf{v} \cdot d\mathbf{l} = 3y^2 dy, \int \mathbf{v} \cdot d\mathbf{l} = \int_0^1 3y^2 dy = 1;$
- (ii) $x = 0, y = 1, \mathbf{v} \cdot d\mathbf{l} = 4z^2 dz, \int \mathbf{v} \cdot d\mathbf{l} = \int_0^1 4z^2 dz = \frac{4}{3};$
- (iii) $x = 0, z = 1, \mathbf{v} \cdot d\mathbf{l} = 3y^2 dy, \int \mathbf{v} \cdot d\mathbf{l} = \int_1^0 3y^2 dy = -1;$
- (iv) $x = 0, y = 0, \mathbf{v} \cdot d\mathbf{l} = 0, \int \mathbf{v} \cdot d\mathbf{l} = \int_1^0 0 dz = 0.$

So

$$\oint \mathbf{v} \cdot d\mathbf{l} = 1 + \frac{4}{3} - 1 + 0 = \frac{4}{3}.$$

It checks.

A point of strategy: notice how I handled step (iii). There is a temptation to write $d\mathbf{l} = -dy \hat{\mathbf{y}}$ here, since the path goes to the left. You can get away with this, if you absolutely insist, by running the integral from $0 \rightarrow 1$. But it is much safer to say $d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}$ *always* (never any minus signs) and let the limits of the integral take care of the direction.

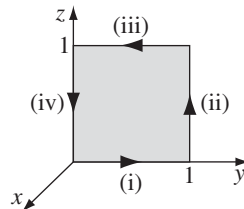


Fig. 1.33

Problem 1.34. Test Stokes' theorem for the function $\mathbf{v} = (xy)\hat{\mathbf{x}} + (2yz)\hat{\mathbf{y}} + (3zx)\hat{\mathbf{z}}$, using the triangular shaded area of Fig. 1.34.

Problem 1.35. Check Corollary 1 by using the same function and boundary line as in Ex. 1.11, but integrating over the five faces of the cube in Fig. 1.35. The back of the cube is open.

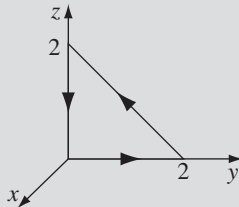


Fig. 1.34

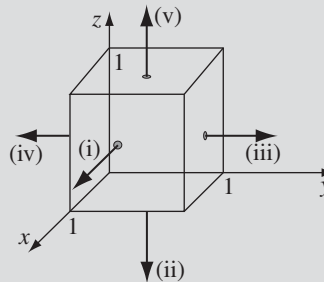


Fig. 1.35

1.3.6 Integration by Parts

The technique known (awkwardly) as **integration by parts** exploits the product rule for derivatives:

$$\frac{d}{dx}(fg) = f\left(\frac{dg}{dx}\right) + g\left(\frac{df}{dx}\right).$$

Integrating both sides, and invoking the fundamental theorem,

$$\int_a^b \frac{d}{dx}(fg) dx = fg|_a^b = \int_a^b f\left(\frac{dg}{dx}\right) dx + \int_a^b g\left(\frac{df}{dx}\right) dx,$$

or

$$\int_a^b f\left(\frac{dg}{dx}\right) dx = - \int_a^b g\left(\frac{df}{dx}\right) dx + fg|_a^b. \quad (1.58)$$

That's integration by parts. It applies to the situation in which you are called upon to integrate the product of one function (f) and the *derivative* of another (g); it says you can *transfer the derivative from g to f* , at the cost of a minus sign and a boundary term.

Example 1.12. Evaluate the integral

$$\int_0^\infty x e^{-x} dx.$$

Solution. The exponential can be expressed as a derivative:

$$e^{-x} = \frac{d}{dx}(-e^{-x}).$$

In this case, then, $f(x) = x$, $g(x) = -e^{-x}$, and $df/dx = 1$, so

$$\int_0^\infty x e^{-x} dx = \int_0^\infty e^{-x} dx - x e^{-x} \Big|_0^\infty = -e^{-x} \Big|_0^\infty = 1.$$

We can exploit the product rules of vector calculus, together with the appropriate fundamental theorems, in exactly the same way. For example, integrating

$$\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f)$$

over a volume, and invoking the divergence theorem, yields

$$\int_V \nabla \cdot (f\mathbf{A}) d\tau = \int_V f(\nabla \cdot \mathbf{A}) d\tau + \int_V \mathbf{A} \cdot (\nabla f) d\tau = \oint_S f\mathbf{A} \cdot d\mathbf{a},$$

or

$$\int_V f(\nabla \cdot \mathbf{A}) d\tau = - \int_V \mathbf{A} \cdot (\nabla f) d\tau + \oint_S f\mathbf{A} \cdot d\mathbf{a}. \quad (1.59)$$

Here again the integrand is the product of one function (f) and the derivative (in this case the *divergence*) of another ($\mathbf{A}$), and integration by parts allows us to transfer the derivative from $\mathbf{A}$ to f (where it becomes a *gradient*), at the cost of a minus sign and a boundary term (in this case a surface integral).

You might wonder how often one is likely to encounter an integral involving the product of one function and the derivative of another. The answer is ... *surprisingly* often! And integration by parts turns out to be one of the most powerful tools in vector calculus.

Problem 1.36.

(a) Show that

$$\int_S f(\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \int_S [\mathbf{A} \times (\nabla f)] \cdot d\mathbf{a} + \oint_P f \mathbf{A} \cdot d\mathbf{l}. \quad (1.60)$$

(b) Show that

$$\int_V \mathbf{B} \cdot (\nabla \times \mathbf{A}) d\tau = \int_V \mathbf{A} \cdot (\nabla \times \mathbf{B}) d\tau + \oint_S (\mathbf{A} \times \mathbf{B}) \cdot d\mathbf{a}. \quad (1.61)$$

1.4 Curvilinear Coordinates

1.4.1 Spherical Coordinates

You can label a point P by its Cartesian coordinates (x, y, z) , but sometimes it is more convenient to use **spherical** coordinates (r, θ, ϕ) ; r is the distance from the origin (the magnitude of the position vector $\mathbf{r}$), θ (the angle down from the z -axis) is called the **polar angle**, and ϕ (the angle around from the x -axis) is the **azimuthal angle**. Their relation to Cartesian coordinates can be read from Fig. 1.36:

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta. \quad (1.62)$$

Figure 1.36 also shows three unit vectors, $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, $\hat{\boldsymbol{\phi}}$, pointing in the direction of increase of the corresponding coordinates. They constitute an orthogonal (mutually perpendicular) basis set (just like $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$), and any vector $\mathbf{A}$ can be expressed in terms of them, in the usual way:

$$\mathbf{A} = A_r \hat{\mathbf{r}} + A_\theta \hat{\boldsymbol{\theta}} + A_\phi \hat{\boldsymbol{\phi}}; \quad (1.63)$$

where A_r , A_θ , and A_ϕ are the radial, polar, and azimuthal components of $\mathbf{A}$. In terms of the Cartesian unit vectors,

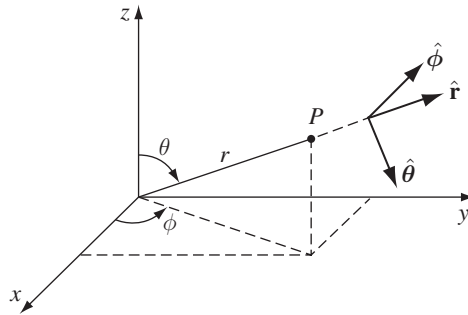


Fig. 1.36

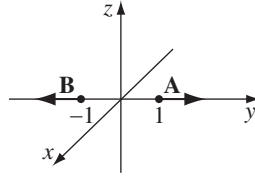


Fig. 1.37

$$\left. \begin{aligned} \hat{\mathbf{r}} &= \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}, \\ \hat{\boldsymbol{\theta}} &= \cos \theta \cos \phi \hat{\mathbf{x}} + \cos \theta \sin \phi \hat{\mathbf{y}} - \sin \theta \hat{\mathbf{z}}, \\ \hat{\boldsymbol{\phi}} &= -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}, \end{aligned} \right\} \quad (1.64)$$

as you can check for yourself (Prob. 1.38). I have put these formulas inside the back cover, for easy reference.

But there is a poisonous snake lurking here that I'd better warn you about: $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, and $\hat{\boldsymbol{\phi}}$ are associated with a *particular point* P , and they *change direction* as P moves around. For example, $\hat{\mathbf{r}}$ always points radially outward, but “radially outward” can be the x direction, the y direction, or any other direction, depending on where you are. In Fig. 1.37, $\mathbf{A} = \hat{\mathbf{y}}$ and $\mathbf{B} = -\hat{\mathbf{y}}$, and yet *both* of them would be written as $\hat{\mathbf{r}}$ in spherical coordinates. One could take account of this by explicitly indicating the point of reference: $\hat{\mathbf{r}}(\theta, \phi)$, $\hat{\boldsymbol{\theta}}(\theta, \phi)$, $\hat{\boldsymbol{\phi}}(\theta, \phi)$, but this would be cumbersome, and as long as you are alert to the problem, I don't think it will cause difficulties.⁹ In particular, do not naïvely combine the spherical components of vectors associated with different points (in Fig. 1.37, $\mathbf{A} + \mathbf{B} = \mathbf{0}$, not $2\hat{\mathbf{r}}$, and $\mathbf{A} \cdot \mathbf{B} = -1$, not $+1$). Beware of differentiating a vector that is expressed in spherical coordinates, since the unit vectors themselves are functions of position ($\partial \hat{\mathbf{r}} / \partial \theta = \hat{\boldsymbol{\theta}}$, for example). And *do not take $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, and $\hat{\boldsymbol{\phi}}$ outside an integral*, as I did with $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ in Eq. 1.53. In general, if you're uncertain about the validity of an operation, express everything in Cartesian coordinates; they are snakeproof.

An infinitesimal displacement in the $\hat{\mathbf{r}}$ direction is simply dr (Fig. 1.38a), just as an infinitesimal element of length in the x direction is dx :

$$dl_r = dr. \quad (1.65)$$

On the other hand, an infinitesimal element of length in the $\hat{\boldsymbol{\theta}}$ direction (Fig. 1.38b) is *not* just $d\theta$ (that's an *angle* – it doesn't even have the right *units* for a length); rather,

$$dl_\theta = r d\theta. \quad (1.66)$$

Similarly, an infinitesimal element of length in the $\hat{\boldsymbol{\phi}}$ direction (Fig. 1.38c) is

$$dl_\phi = r \sin \theta d\phi. \quad (1.67)$$

⁹ I claimed back at the beginning that vectors have no location, and I'll stand by that. The vectors themselves live “out there,” completely independent of our choice of coordinates. But the *notation* we use to represent them *does* depend on the point in question, in curvilinear coordinates; $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, and $\hat{\boldsymbol{\phi}}$ stand for *different vectors* at different locations.

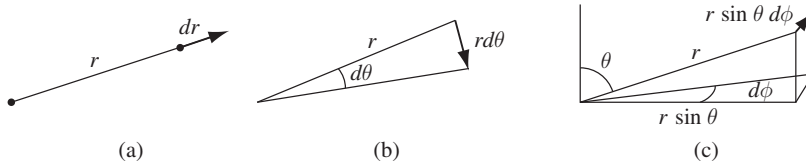


Fig. 1.38

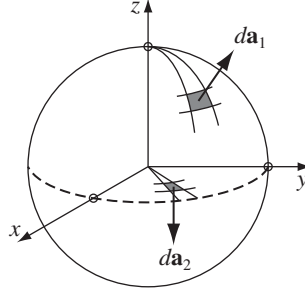


Fig. 1.39

Thus the general infinitesimal displacement $d\mathbf{l}$ is

$$d\mathbf{l} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin \theta d\phi \hat{\boldsymbol{\phi}}. \quad (1.68)$$

This plays the role (in line integrals, for example) that $d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}$ played in Cartesian coordinates.

The infinitesimal volume element $d\tau$, in spherical coordinates, is the product of the three infinitesimal displacements:

$$d\tau = dl_r dl_\theta dl_\phi = r^2 \sin \theta dr d\theta d\phi. \quad (1.69)$$

I cannot give you a general expression for *surface* elements $d\mathbf{a}$, since these depend on the orientation of the surface. You simply have to analyze the geometry for any given case (this goes for Cartesian and curvilinear coordinates alike). If you are integrating over the surface of a sphere, for instance, then r is constant, whereas θ and ϕ change (Fig. 1.39), so

$$d\mathbf{a}_1 = dl_\theta dl_\phi \hat{\mathbf{r}} = r^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}.$$

On the other hand, if the surface lies in the xy plane, say, so that θ is constant (to wit, $\pi/2$) while r and ϕ vary, then

$$d\mathbf{a}_2 = dl_r dl_\phi \hat{\boldsymbol{\theta}} = r dr d\phi \hat{\boldsymbol{\theta}}.$$

Notice, finally, that r ranges from 0 to ∞ , ϕ from 0 to 2π , and θ from 0 to π (not 2π – that would count every point twice).¹⁰

¹⁰ Alternatively, you could run ϕ from 0 to π (the “eastern hemisphere”) and cover the “western hemisphere” by extending θ from π up to 2π . But this is very bad notation, since, among other things, $\sin \theta$ will then run negative, and you’ll have to put absolute value signs around that term in volume and surface elements (area and volume being intrinsically positive quantities).

Example 1.13. Find the volume of a sphere of radius R .

Solution.

$$\begin{aligned} V &= \int d\tau = \int_{r=0}^R \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} r^2 \sin \theta \, dr \, d\theta \, d\phi \\ &= \left(\int_0^R r^2 \, dr \right) \left(\int_0^{\pi} \sin \theta \, d\theta \right) \left(\int_0^{2\pi} d\phi \right) \\ &= \left(\frac{R^3}{3} \right) (2)(2\pi) = \frac{4}{3} \pi R^3 \end{aligned}$$

(not a big surprise).

So far we have talked only about the *geometry* of spherical coordinates. Now I would like to “translate” the vector derivatives (gradient, divergence, curl, and Laplacian) into r , θ , ϕ notation. In principle, this is entirely straightforward: in the case of the gradient,

$$\nabla T = \frac{\partial T}{\partial x} \hat{\mathbf{x}} + \frac{\partial T}{\partial y} \hat{\mathbf{y}} + \frac{\partial T}{\partial z} \hat{\mathbf{z}},$$

we would first use the chain rule to expand the partials:

$$\frac{\partial T}{\partial x} = \frac{\partial T}{\partial r} \left(\frac{\partial r}{\partial x} \right) + \frac{\partial T}{\partial \theta} \left(\frac{\partial \theta}{\partial x} \right) + \frac{\partial T}{\partial \phi} \left(\frac{\partial \phi}{\partial x} \right).$$

The terms in parentheses could be worked out from Eq. 1.62 – or rather, the *inverse* of those equations (Prob. 1.37). Then we’d do the same for $\partial T/\partial y$ and $\partial T/\partial z$. Finally, we’d substitute in the formulas for $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ in terms of $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, and $\hat{\boldsymbol{\phi}}$ (Prob. 1.38). It would take an hour to figure out the gradient in spherical coordinates by this brute-force method. I suppose this is how it was first done, but there is a much more efficient indirect approach, explained in Appendix A, which has the extra advantage of treating all coordinate systems at once. I described the “straightforward” method only to convince you that there is nothing subtle or mysterious about transforming to spherical coordinates: you’re expressing the *same quantity* (gradient, divergence, or whatever) in different notation, that’s all.

Here, then, are the vector derivatives in spherical coordinates:

Gradient:

$$\nabla T = \frac{\partial T}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial T}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} \hat{\boldsymbol{\phi}}. \quad (1.70)$$

Divergence:

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta v_\theta) + \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi}. \quad (1.71)$$

Curl:

$$\begin{aligned}\nabla \times \mathbf{v} = & \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta v_\phi) - \frac{\partial v_\theta}{\partial \phi} \right] \hat{\mathbf{r}} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r} (r v_\phi) \right] \hat{\boldsymbol{\theta}} \\ & + \frac{1}{r} \left[\frac{\partial}{\partial r} (r v_\theta) - \frac{\partial v_r}{\partial \theta} \right] \hat{\boldsymbol{\phi}}.\end{aligned}\quad (1.72)$$

Laplacian:

$$\nabla^2 T = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2}. \quad (1.73)$$

For reference, these formulas are listed inside the front cover.

Problem 1.37. Find formulas for r , θ , ϕ in terms of x , y , z (the inverse, in other words, of Eq. 1.62).

- **Problem 1.38.** Express the unit vectors $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, $\hat{\boldsymbol{\phi}}$ in terms of $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$ (that is, derive Eq. 1.64). Check your answers several ways ($\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} \stackrel{?}{=} 1$, $\hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\phi}} \stackrel{?}{=} 0$, $\hat{\mathbf{r}} \times \hat{\boldsymbol{\theta}} \stackrel{?}{=} \hat{\boldsymbol{\phi}}$, ...). Also work out the inverse formulas, giving $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$ in terms of $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, $\hat{\boldsymbol{\phi}}$ (and θ , ϕ).
- **Problem 1.39.**
 - (a) Check the divergence theorem for the function $\mathbf{v}_1 = r^2 \hat{\mathbf{r}}$, using as your volume the sphere of radius R , centered at the origin.
 - (b) Do the same for $\mathbf{v}_2 = (1/r^2) \hat{\mathbf{r}}$. (If the answer surprises you, look back at Prob. 1.16.)

Problem 1.40. Compute the divergence of the function

$$\mathbf{v} = (r \cos \theta) \hat{\mathbf{r}} + (r \sin \theta) \hat{\boldsymbol{\theta}} + (r \sin \theta \cos \phi) \hat{\boldsymbol{\phi}}.$$

Check the divergence theorem for this function, using as your volume the inverted hemispherical bowl of radius R , resting on the xy plane and centered at the origin (Fig. 1.40).

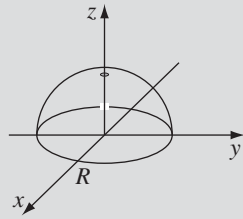


Fig. 1.40

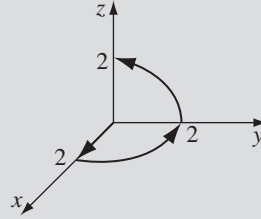


Fig. 1.41

Problem 1.41. Compute the gradient and Laplacian of the scalar function $T = r(\cos \theta + \sin \theta \cos \phi)$. Check the Laplacian by converting T to Cartesian coordinates and using Eq. 1.42. Test the gradient theorem for this function, using the path shown in Fig. 1.41, from $(0, 0, 0)$ to $(0, 0, 2)$.

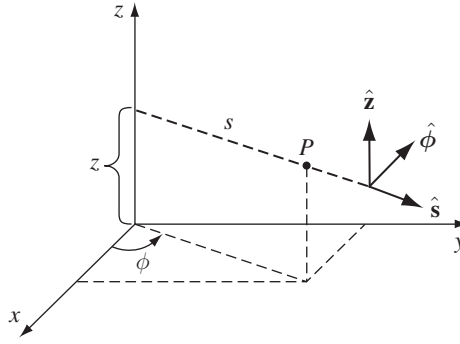


Fig. 1.42

1.4.2 Cylindrical Coordinates

The cylindrical coordinates (s, ϕ, z) of a point P are defined in Fig. 1.42. Notice that ϕ has the same meaning as in spherical coordinates, and z is the same as Cartesian; s is the distance to P from the z -axis, whereas the spherical coordinate r is the distance from the *origin*. The relation to Cartesian coordinates is

$$x = s \cos \phi, \quad y = s \sin \phi, \quad z = z. \quad (1.74)$$

The unit vectors (Prob. 1.42) are

$$\left. \begin{aligned} \hat{\mathbf{s}} &= \cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}}, \\ \hat{\boldsymbol{\phi}} &= -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}, \\ \hat{\mathbf{z}} &= \hat{\mathbf{z}}. \end{aligned} \right\} \quad (1.75)$$

The infinitesimal displacements are

$$dl_s = ds, \quad dl_\phi = s d\phi, \quad dl_z = dz, \quad (1.76)$$

so

$$d\mathbf{l} = ds \hat{\mathbf{s}} + s d\phi \hat{\boldsymbol{\phi}} + dz \hat{\mathbf{z}}, \quad (1.77)$$

and the volume element is

$$d\tau = s ds d\phi dz. \quad (1.78)$$

The range of s is $0 \rightarrow \infty$, ϕ goes from $0 \rightarrow 2\pi$, and z from $-\infty$ to ∞ .

The vector derivatives in cylindrical coordinates are

Gradient:

$$\nabla T = \frac{\partial T}{\partial s} \hat{\mathbf{s}} + \frac{1}{s} \frac{\partial T}{\partial \phi} \hat{\boldsymbol{\phi}} + \frac{\partial T}{\partial z} \hat{\mathbf{z}}. \quad (1.79)$$

Divergence:

$$\nabla \cdot \mathbf{v} = \frac{1}{s} \frac{\partial}{\partial s} (s v_s) + \frac{1}{s} \frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z}. \quad (1.80)$$

Curl:

$$\nabla \times \mathbf{v} = \left(\frac{1}{s} \frac{\partial v_z}{\partial \phi} - \frac{\partial v_\phi}{\partial z} \right) \hat{\mathbf{s}} + \left(\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s} \right) \hat{\phi} + \frac{1}{s} \left[\frac{\partial}{\partial s} (s v_\phi) - \frac{\partial v_s}{\partial \phi} \right] \hat{\mathbf{z}}. \quad (1.81)$$

Laplacian:

$$\nabla^2 T = \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial T}{\partial s} \right) + \frac{1}{s^2} \frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2}. \quad (1.82)$$

These formulas are also listed inside the front cover.

Problem 1.42. Express the cylindrical unit vectors $\hat{\mathbf{s}}, \hat{\phi}, \hat{\mathbf{z}}$ in terms of $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ (that is, derive Eq. 1.75). “Invert” your formulas to get $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ in terms of $\hat{\mathbf{s}}, \hat{\phi}, \hat{\mathbf{z}}$ (and ϕ).

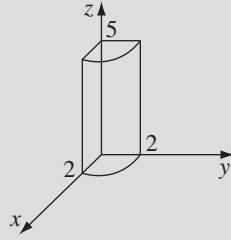


Fig. 1.43

Problem 1.43.

- (a) Find the divergence of the function

$$\mathbf{v} = s(2 + \sin^2 \phi) \hat{\mathbf{s}} + s \sin \phi \cos \phi \hat{\phi} + 3z \hat{\mathbf{z}}.$$

- (b) Test the divergence theorem for this function, using the quarter-cylinder (radius 2, height 5) shown in Fig. 1.43.
 (c) Find the curl of $\mathbf{v}$.

1.5 The Dirac Delta Function

1.5.1 The Divergence of $\hat{\mathbf{r}}/r^2$

Consider the vector function

$$\mathbf{v} = \frac{1}{r^2} \hat{\mathbf{r}}. \quad (1.83)$$

At every location, $\mathbf{v}$ is directed radially outward (Fig. 1.44); if ever there was a function that ought to have a large positive divergence, this is it. And yet, when you actually *calculate* the divergence (using Eq. 1.71), you get precisely *zero*:

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{1}{r^2} \right) = \frac{1}{r^2} \frac{\partial}{\partial r} (1) = 0. \quad (1.84)$$

Problem 1.51. For Theorem 1, show that $(d) \Rightarrow (a)$, $(a) \Rightarrow (c)$, $(c) \Rightarrow (b)$, $(b) \Rightarrow (c)$, and $(c) \Rightarrow (a)$.

Problem 1.52. For Theorem 2, show that $(d) \Rightarrow (a)$, $(a) \Rightarrow (c)$, $(c) \Rightarrow (b)$, $(b) \Rightarrow (c)$, and $(c) \Rightarrow (a)$.

Problem 1.53.

- (a) Which of the vectors in Prob. 1.15 can be expressed as the gradient of a scalar? Find a scalar function that does the job.
- (b) Which can be expressed as the curl of a vector? Find such a vector.

More Problems on Chapter 1

Problem 1.54. Check the divergence theorem for the function

$$\mathbf{v} = r^2 \cos \theta \hat{\mathbf{r}} + r^2 \cos \phi \hat{\boldsymbol{\theta}} - r^2 \cos \theta \sin \phi \hat{\boldsymbol{\phi}},$$

using as your volume one octant of the sphere of radius R (Fig. 1.48). Make sure you include the entire surface. [Answer: $\pi R^4/4$.]

Problem 1.55. Check Stokes' theorem using the function $\mathbf{v} = ay\hat{\mathbf{x}} + bx\hat{\mathbf{y}}$ (a and b are constants) and the circular path of radius R , centered at the origin in the xy plane. [Answer: $\pi R^2(b - a)$.]

Problem 1.56. Compute the line integral of

$$\mathbf{v} = 6x\hat{\mathbf{x}} + yz^2\hat{\mathbf{y}} + (3y + z)\hat{\mathbf{z}}$$

along the triangular path shown in Fig. 1.49. Check your answer using Stokes' theorem. [Answer: $8/3$.]

Problem 1.57. Compute the line integral of

$$\mathbf{v} = (r \cos^2 \theta)\hat{\mathbf{r}} - (r \cos \theta \sin \theta)\hat{\boldsymbol{\theta}} + 3r\hat{\boldsymbol{\phi}}$$

around the path shown in Fig. 1.50 (the points are labeled by their Cartesian coordinates). Do it either in cylindrical or in spherical coordinates. Check your answer, using Stokes' theorem. [Answer: $3\pi/2$.]

Problem 1.58. Check Stokes' theorem for the function $\mathbf{v} = y\hat{\mathbf{z}}$, using the triangular surface shown in Fig. 1.51. [Answer: a^2 .]

Problem 1.59. Check the divergence theorem for the function

$$\mathbf{v} = r^2 \sin \theta \hat{\mathbf{r}} + 4r^2 \cos \theta \hat{\boldsymbol{\theta}} + r^2 \tan \theta \hat{\boldsymbol{\phi}},$$

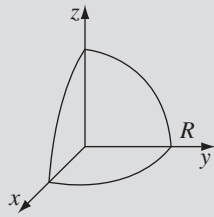


Fig. 1.48

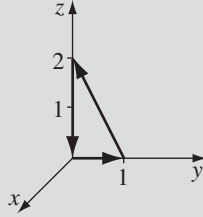


Fig. 1.49

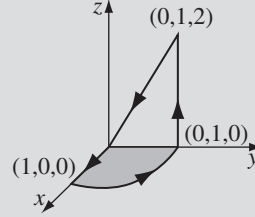


Fig. 1.50

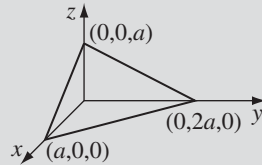


Fig. 1.51

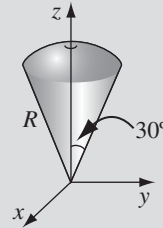


Fig. 1.52

using the volume of the “ice-cream cone” shown in Fig. 1.52 (the top surface is spherical, with radius R and centered at the origin). [Answer: $(\pi R^4/12)(2\pi + 3\sqrt{3})$.]

Problem 1.60. Here are two cute checks of the fundamental theorems:

- (a) Combine Corollary 2 to the gradient theorem with Stokes' theorem ($\mathbf{v} = \nabla T$ in this case). Show that the result is consistent with what you already knew about second derivatives.
- (b) Combine Corollary 2 to Stokes' theorem with the divergence theorem. Show that the result is consistent with what you already knew.

- **Problem 1.61.** Although the gradient, divergence, and curl theorems are the fundamental integral theorems of vector calculus, it is possible to derive a number of corollaries from them. Show the following:

- (a) $\int_V (\nabla T) \, d\tau = \oint_S T \, d\mathbf{a}$. [Hint: Let $\mathbf{v} = \mathbf{c}T$, where $\mathbf{c}$ is a constant vector, in the divergence theorem; use the product rules.]
- (b) $\int_V (\nabla \times \mathbf{v}) \, d\tau = - \oint_S \mathbf{v} \times d\mathbf{a}$. [Hint: Replace $\mathbf{v}$ by $(\mathbf{v} \times \mathbf{c})$ in the divergence theorem.]
- (c) $\int_V [T \nabla^2 U + (\nabla T) \cdot (\nabla U)] \, d\tau = \oint_S (T \nabla U) \cdot d\mathbf{a}$. [Hint: Let $\mathbf{v} = T \nabla U$ in the divergence theorem.]
- (d) $\int_V (T \nabla^2 U - U \nabla^2 T) \, d\tau = \oint_S (T \nabla U - U \nabla T) \cdot d\mathbf{a}$. [Comment: This is sometimes called **Green's second identity**; it follows from (c), which is known as **Green's identity**.]
- (e) $\int_S \nabla T \times d\mathbf{a} = - \oint_P T \, d\mathbf{l}$. [Hint: Let $\mathbf{v} = \mathbf{c}T$ in Stokes' theorem.]
- (f) $\int_S [(d\mathbf{a} \times \nabla) \times \mathbf{v}] = - \oint_P \mathbf{v} \times d\mathbf{l}$. [Hint: Replace $\mathbf{v}$ by $(\mathbf{v} \times \mathbf{c})$ in Stokes' theorem.]

- **Problem 1.62.** The integral

$$\mathbf{a} \equiv \int_S d\mathbf{a} \quad (1.106)$$

is sometimes called the **vector area** of the surface S . If S happens to be *flat*, then $|\mathbf{a}|$ is the *ordinary* (scalar) area, obviously.

- (a) Find the vector area of a hemispherical bowl of radius R .
- (b) Show that $\mathbf{a} = \mathbf{0}$ for any *closed* surface. [Hint: Use Prob. 1.61a.]
- (c) Show that $\mathbf{a}$ is the same for all surfaces sharing the same boundary.
- (d) Show that

$$\mathbf{a} = \frac{1}{2} \oint \mathbf{r} \times d\mathbf{l}, \quad (1.107)$$

where the integral is around the boundary line. [Hint: One way to do it is to draw the cone subtended by the loop at the origin. Divide the conical surface up into infinitesimal triangular wedges, each with vertex at the origin and opposite side $d\mathbf{l}$, and exploit the geometrical interpretation of the cross product (Fig. 1.8).]

- (e) Show that

$$\oint (\mathbf{c} \cdot \mathbf{r}) d\mathbf{l} = \mathbf{a} \times \mathbf{c} \quad (1.108)$$

for any constant vector $\mathbf{c}$. [Hint: Let $T = \mathbf{c} \cdot \mathbf{r}$ in Prob. 1.61e.]

- **Problem 1.63.**

- (a) Find the divergence of the function

$$\mathbf{v} = \frac{\hat{\mathbf{r}}}{r}.$$

First compute it directly, as in Eq. 1.84. Test your result using the divergence theorem, as in Eq. 1.85. Is there a delta function at the origin, as there was for $\hat{\mathbf{r}}/r^2$? What is the general formula for the divergence of $r^n \hat{\mathbf{r}}$? [Answer: $\nabla \cdot (r^n \hat{\mathbf{r}}) = (n+2)r^{n-1}$, unless $n = -2$, in which case it is $4\pi\delta^3(\mathbf{r})$; for $n < -2$, the divergence is ill defined at the origin.]

- (b) Find the *curl* of $r^n \hat{\mathbf{r}}$. Test your conclusion using Prob. 1.61b. [Answer: $\nabla \times (r^n \hat{\mathbf{r}}) = \mathbf{0}$.]

Problem 1.64. In case you're not persuaded that $\nabla^2(1/r) = -4\pi\delta^3(\mathbf{r})$ (Eq. 1.102, with $\mathbf{r}' = \mathbf{0}$ for simplicity), try replacing r by $\sqrt{r^2 + \epsilon^2}$, and watching what happens as $\epsilon \rightarrow 0$.¹⁶ Specifically, let

$$D(r, \epsilon) \equiv -\frac{1}{4\pi} \nabla^2 \frac{1}{\sqrt{r^2 + \epsilon^2}}.$$

¹⁶ This problem was suggested by Frederick Strauch.

To demonstrate that this goes to $\delta^3(\mathbf{r})$ as $\epsilon \rightarrow 0$:

- (a) show that $D(r, \epsilon) = (3\epsilon^2/4\pi)(r^2 + \epsilon^2)^{-5/2}$;
- (b) check that $D(0, \epsilon) \rightarrow \infty$ as $\epsilon \rightarrow 0$;
- (c) check that $D(r, \epsilon) \rightarrow 0$, as $\epsilon \rightarrow 0$, for all $r \neq 0$;
- (d) check that the integral of $D(r, \epsilon)$ over all space is 1.

Problem 1.65. (a) Check Stokes' theorem for the vector function

$$\mathbf{A} = \frac{-y\hat{\mathbf{x}} + x\hat{\mathbf{y}}}{x^2 + y^2},$$

using a circle of radius R in the xy plane. Diagnose the problem, and fix it by correcting the curl of $\mathbf{A}$. Use Cartesian coordinates.

- (b) Convert $\mathbf{A}$ to cylindrical coordinates, and repeat part (a), this time doing everything in terms of s , ϕ , and z .

2.1 The Electric Field

2.1.1 Introduction

The fundamental problem electrodynamics hopes to solve is this (Fig. 2.1): We have some electric charges, $q_1, q_2, q_3, \dots$ (call them **source charges**); what force do they exert on another charge, Q (call it the **test charge**)? The positions of the source charges are *given* (as functions of time); the trajectory of the test particle is *to be calculated*. In general, both the source charges and the test charge are in motion.

The solution to this problem is facilitated by the **principle of superposition**, which states that the interaction between any two charges is completely unaffected by the presence of others. This means that to determine the force on Q , we can first compute the force $\mathbf{F}_1$, due to q_1 alone (ignoring all the others); then we compute the force $\mathbf{F}_2$, due to q_2 alone; and so on. Finally, we take the vector sum of all these individual forces: $\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots$. Thus, if we can find the force on Q due to a *single* source charge q , we are, in principle, done (the rest is just a question of repeating the same operation over and over, and adding it all up).¹

Well, at first sight this looks very easy: Why don't I just write down the formula for the force on Q due to q , and be done with it? I *could*, and in Chapter 10 I shall, but you would be shocked to see it at this stage, for not only does the force on Q depend on the separation distance r between the charges (Fig. 2.2), it also depends on *both* their velocities and on the *acceleration* of q . Moreover, it is not the position, velocity, and acceleration of q *right now* that matter: electromagnetic “news” travels at the speed of light, so what concerns Q is the position, velocity, and acceleration q *had* at some earlier time, when the message left.

Therefore, in spite of the fact that the basic question (“What is the force on Q due to q ?”) is easy to state, it does not pay to confront it head on; rather, we shall go at it by stages. In the meantime, the theory we develop

¹ The principle of superposition may seem “obvious” to you, but it did not have to be so simple: if the electromagnetic force were proportional to the *square* of the total source charge, for instance, the principle of superposition would not hold, since $(q_1 + q_2)^2 \neq q_1^2 + q_2^2$ (there would be “cross terms” to consider). Superposition is not a logical necessity, but an experimental fact.

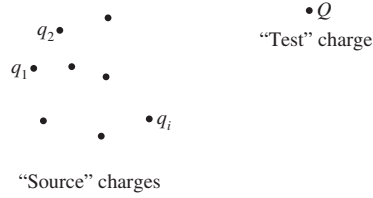


Fig. 2.1

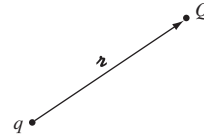


Fig. 2.2

will allow for the solution of more subtle electromagnetic problems that do not present themselves in quite this simple format. To begin with, we shall consider the special case of **electrostatics** in which all the *source charges* are *stationary* (though the test charge may be moving).

2.1.2 Coulomb's Law

What is the force on a test charge Q due to a single point charge q , that is at *rest* a distance r away? The answer (based on experiments) is given by **Coulomb's law**:

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{\mathbf{r}}. \quad (2.1)$$

The constant ϵ_0 is called (ludicrously) the **permittivity of free space**. In SI units, where force is in newtons (N), distance in meters (m), and charge in coulombs (C),

$$\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N} \cdot \text{m}^2}.$$

In words, the force is proportional to the product of the charges and inversely proportional to the square of the separation distance. As always (Section 1.1.4), $\mathbf{r}$ is the separation vector from $\mathbf{r}'$ (the location of q) to $\mathbf{r}$ (the location of Q):

$$\mathbf{r} = \mathbf{r} - \mathbf{r}'; \quad (2.2)$$

r is its magnitude, and $\hat{\mathbf{r}}$ is its direction. The force points along the line from q to Q ; it is repulsive if q and Q have the same sign, and attractive if their signs are opposite.

Coulomb's law and the principle of superposition constitute the physical input for electrostatics – the rest, except for some special properties of matter, is mathematical elaboration of these fundamental rules.

Problem 2.1.

- (a) Twelve equal charges, q , are situated at the corners of a regular 12-sided polygon (for instance, one on each numeral of a clock face). What is the net force on a test charge Q at the center?

- (b) Suppose *one* of the 12 qs is removed (the one at “6 o’clock”). What is the force on Q ? Explain your reasoning carefully.
- (c) Now 13 equal charges, q , are placed at the corners of a regular 13-sided polygon. What is the force on a test charge Q at the center?
- (d) If one of the 13 qs is removed, what is the force on Q ? Explain your reasoning.

2.1.3 The Electric Field

If we have *several* point charges $q_1, q_2, \dots, q_n$, at distances $r_1, r_2, \dots, r_n$ from Q , the total force on Q is evidently

$$\begin{aligned}\mathbf{F} &= \mathbf{F}_1 + \mathbf{F}_2 + \dots = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 Q}{r_1^2} \hat{\mathbf{r}}_1 + \frac{q_2 Q}{r_2^2} \hat{\mathbf{r}}_2 + \dots \right) \\ &= \frac{Q}{4\pi\epsilon_0} \left(\frac{q_1}{r_1^2} \hat{\mathbf{r}}_1 + \frac{q_2}{r_2^2} \hat{\mathbf{r}}_2 + \frac{q_3}{r_3^2} \hat{\mathbf{r}}_3 + \dots \right),\end{aligned}$$

or

$$\boxed{\mathbf{F} = QE}, \quad (2.3)$$

where

$$\mathbf{E}(\mathbf{r}) \equiv \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i. \quad (2.4)$$

The quantity $\mathbf{E}$ is called the **electric field** of the source charges. Notice that it is a function of position ($\mathbf{r}$), because the separation vectors $\mathbf{r}_i$ depend on the location of the **field point** P (Fig. 2.3). But it makes no reference to the test charge Q . The electric field is a vector quantity that varies from point to point and is determined by the configuration of source charges; physically, $\mathbf{E}(\mathbf{r})$ is the force per unit charge that would be exerted on a test charge, if you were to place one at P .

What exactly *is* an electric field? I have deliberately begun with what you might call the “minimal” interpretation of $\mathbf{E}$, as an intermediate step in the calculation of electric forces. But I encourage you to think of the field as a “real” physical entity, filling the space around electric

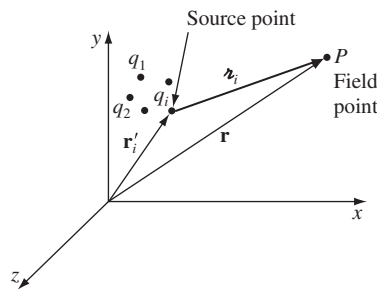


Fig. 2.3

charges. Maxwell himself came to believe that electric and magnetic fields are stresses and strains in an invisible primordial jellylike “ether.” Special relativity has forced us to abandon the notion of ether, and with it Maxwell’s mechanical interpretation of electromagnetic fields. (It is even possible, though cumbersome, to formulate classical electrodynamics as an “action-at-a-distance” theory, and dispense with the field concept altogether.) I can’t tell you, then, what a field *is* – only how to calculate it and what it can do for you once you’ve got it.

Example 2.1. Find the electric field a distance z above the midpoint between two equal charges (q), a distance d apart (Fig. 2.4a).

Solution. Let $\mathbf{E}_1$ be the field of the left charge alone, and $\mathbf{E}_2$ that of the right charge alone (Fig. 2.4b). Adding them (vectorially), the horizontal components cancel and the vertical components conspire:

$$E_z = 2 \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \cos \theta.$$

Here $r = \sqrt{z^2 + (d/2)^2}$ and $\cos \theta = z/r$, so

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{2qz}{[z^2 + (d/2)^2]^{3/2}} \hat{\mathbf{z}}.$$

Check: When $z \gg d$ you’re so far away that it just looks like a single charge $2q$, so the field should reduce to $\mathbf{E} = (2q/4\pi\epsilon_0 z^2) \hat{\mathbf{z}}$. And it *does* (just set $d \rightarrow 0$ in the formula).

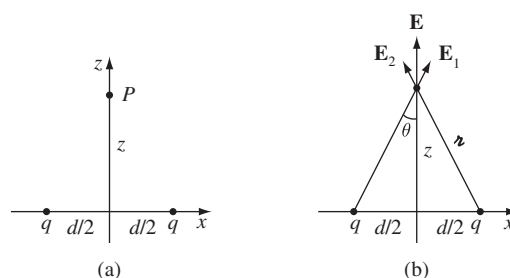


Fig. 2.4

Problem 2.2. Find the electric field (magnitude and direction) a distance z above the midpoint between equal and opposite charges ($\pm q$), a distance d apart (same as Ex. 2.1, except that the charge at $x = +d/2$ is $-q$).

2.1.4 Continuous Charge Distributions

Our definition of the electric field (Eq. 2.4) assumes that the source of the field is a collection of discrete point charges q_i . If, instead, the charge is distributed continuously over some region, the sum becomes an integral (Fig. 2.5a):

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r^2} \hat{\mathbf{r}} dq. \quad (2.5)$$

If the charge is spread out along a *line* (Fig. 2.5b), with charge-per-unit-length λ , then $dq = \lambda dl'$ (where dl' is an element of length along the line); if the charge is smeared out over a *surface* (Fig. 2.5c), with charge-per-unit-area σ , then $dq = \sigma da'$ (where da' is an element of area on the surface); and if the charge fills a *volume* (Fig. 2.5d), with charge-per-unit-volume ρ , then $dq = \rho d\tau'$ (where $d\tau'$ is an element of volume):

$$dq \rightarrow \lambda dl' \sim \sigma da' \sim \rho d\tau'.$$

Thus the electric field of a line charge is

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(\mathbf{r}')}{r^2} \hat{\mathbf{r}} dl'; \quad (2.6)$$

for a surface charge,

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma(\mathbf{r}')}{r^2} \hat{\mathbf{r}} da'; \quad (2.7)$$

and for a volume charge,

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r^2} \hat{\mathbf{r}} d\tau'. \quad (2.8)$$

Equation 2.8 itself is often referred to as “Coulomb’s law,” because it is such a short step from the original (Eq. 2.1), and because a volume charge is in some sense the most general and realistic case. Please note carefully

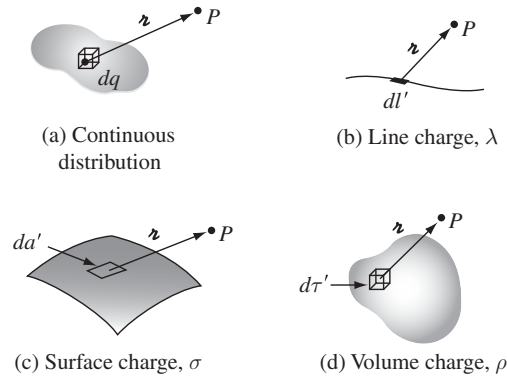


Fig. 2.5

the meaning of $\mathbf{r}$ in these formulas. Originally, in Eq. 2.4, $\mathbf{r}_i$ stood for the vector from the source charge q_i to the field point $\mathbf{r}$. Correspondingly, in Eqs. 2.5–2.8, $\mathbf{r}$ is the vector from dq (therefore from dl' , da' , or $d\tau'$) to the field point $\mathbf{r}$.²

Example 2.2. Find the electric field a distance z above the midpoint of a straight line segment of length $2L$ that carries a uniform line charge λ (Fig. 2.6).

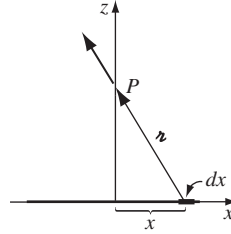


Fig. 2.6

Solution. The simplest method is to chop the line into symmetrically placed pairs (at $\pm x$), quote the result of Ex. 2.1 (with $d/2 \rightarrow x$, $q \rightarrow \lambda dx$), and integrate ($x: 0 \rightarrow L$). But here's a more general approach:³

$$\begin{aligned} \mathbf{r} &= z\hat{\mathbf{z}}, \quad \mathbf{r}' = x\hat{\mathbf{x}}, \quad dl' = dx; \\ \mathbf{r} &= \mathbf{r} - \mathbf{r}' = z\hat{\mathbf{z}} - x\hat{\mathbf{x}}, \quad r = \sqrt{z^2 + x^2}, \quad \hat{\mathbf{r}} = \frac{\mathbf{r}}{r} = \frac{z\hat{\mathbf{z}} - x\hat{\mathbf{x}}}{\sqrt{z^2 + x^2}}, \\ \mathbf{E} &= \frac{1}{4\pi\epsilon_0} \int_{-L}^L \frac{\lambda}{z^2 + x^2} \frac{z\hat{\mathbf{z}} - x\hat{\mathbf{x}}}{\sqrt{z^2 + x^2}} dx \\ &= \frac{\lambda}{4\pi\epsilon_0} \left[z\hat{\mathbf{z}} \int_{-L}^L \frac{1}{(z^2 + x^2)^{3/2}} dx - \hat{\mathbf{x}} \int_{-L}^L \frac{x}{(z^2 + x^2)^{3/2}} dx \right] \\ &= \frac{\lambda}{4\pi\epsilon_0} \left[z\hat{\mathbf{z}} \left(\frac{x}{z^2\sqrt{z^2 + x^2}} \right) \Big|_{-L}^L - \hat{\mathbf{x}} \left(-\frac{1}{\sqrt{z^2 + x^2}} \right) \Big|_{-L}^L \right] \\ &= \frac{1}{4\pi\epsilon_0} \frac{2\lambda L}{z\sqrt{z^2 + L^2}} \hat{\mathbf{z}}. \end{aligned}$$

For points far from the line ($z \gg L$),

$$E \approx \frac{1}{4\pi\epsilon_0} \frac{2\lambda L}{z^2}.$$

² *Warning:* The unit vector $\hat{\mathbf{r}}$ is not constant; its direction depends on the source point $\mathbf{r}'$, and hence it cannot be taken outside the integrals (Eqs. 2.5–2.8). In practice, you must work with Cartesian components ($\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$ are constant, and do come out), even if you use curvilinear coordinates to perform the integration.

³ Ordinarily I'll put a prime on the source coordinates, but where no confusion is likely to arise I'll remove the prime to simplify the notation.

This makes sense: From far away the line looks like a point charge $q = 2\lambda L$. In the limit $L \rightarrow \infty$, on the other hand, we obtain the field of an infinite straight wire:

$$E = \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{z}. \quad (2.9)$$

Problem 2.3. Find the electric field a distance z above one end of a straight line segment of length L (Fig. 2.7) that carries a uniform line charge λ . Check that your formula is consistent with what you would expect for the case $z \gg L$.

Problem 2.4. Find the electric field a distance z above the center of a square loop (side length a) carrying a uniform line charge λ (Fig. 2.8). [Hint: Use the result of Ex. 2.2.]

Problem 2.5. Find the electric field a distance z above the center of a circular loop of radius r (Fig. 2.9) that carries a uniform line charge λ .

Problem 2.6. Find the electric field a distance z above the center of a flat circular disk of radius R (Fig. 2.10) that carries a uniform surface charge σ . What does your formula give in the limit $R \rightarrow \infty$? Also check the case $z \gg R$.

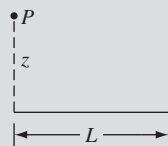


Fig. 2.7

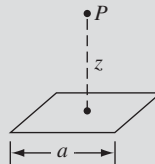


Fig. 2.8

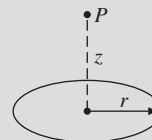


Fig. 2.9

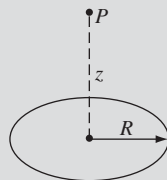


Fig. 2.10

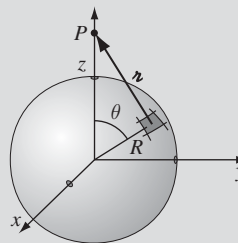


Fig. 2.11

! **Problem 2.7.** Find the electric field a distance z from the center of a spherical surface of radius R (Fig. 2.11) that carries a uniform charge density σ . Treat the case $z < R$ (inside) as well as $z > R$ (outside). Express your answers in terms of the total charge q on the sphere. [Hint: Use the law of cosines to write r in terms of R and θ . Be sure to take the *positive* square root: $\sqrt{R^2 + z^2 - 2Rz} = (R - z)$ if $R > z$, but it's $(z - R)$ if $R < z$.]

Problem 2.8. Use your result in Prob. 2.7 to find the field inside and outside a solid sphere of radius R that carries a uniform volume charge density ρ . Express your answers in terms of the total charge of the sphere, q . Draw a graph of $|\mathbf{E}|$ as a function of the distance from the center.

2.2 Divergence and Curl of Electrostatic Fields

2.2.1 Field Lines, Flux, and Gauss's Law

In principle, we are *done* with the subject of electrostatics. Equation 2.8 tells us how to compute the field of a charge distribution, and Eq. 2.3 tells us what the force on a charge Q placed in this field will be. Unfortunately, as you may have discovered in working Prob. 2.7, the integrals involved in computing $\mathbf{E}$ can be formidable, even for reasonably simple charge distributions. Much of the rest of electrostatics is devoted to assembling a bag of tools and tricks for avoiding these integrals. It all begins with the divergence and curl of $\mathbf{E}$. I shall calculate the divergence of $\mathbf{E}$ directly from Eq. 2.8, in Section 2.2.2, but first I want to show you a more qualitative, and perhaps more illuminating, intuitive approach.

Let's begin with the simplest possible case: a single point charge q , situated at the origin:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}. \quad (2.10)$$

To get a “feel” for this field, I might sketch a few representative vectors, as in Fig. 2.12a. Because the field falls off like $1/r^2$, the vectors get shorter as you go farther away from the origin; they always point radially outward. But there is a nicer way to represent this field, and that's to connect up the arrows, to form **field lines** (Fig. 2.12b). You might think that I have thereby thrown away information about the *strength* of the field, which was contained in the lengths of the arrows. But actually I have not. The magnitude of the field is indicated by the *density* of the field lines: it's strong

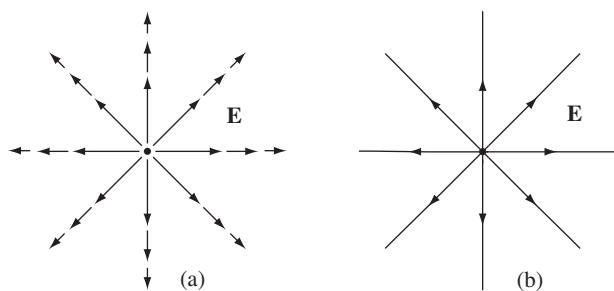


Fig. 2.12

near the center where the field lines are close together, and weak farther out, where they are relatively far apart.

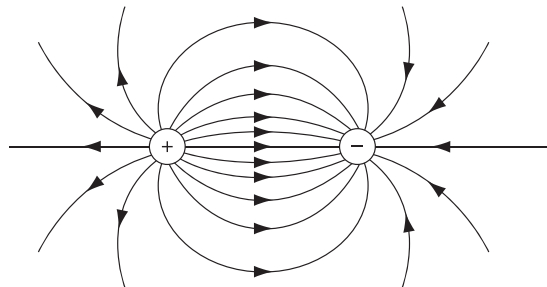
In truth, the field-line diagram is deceptive, when I draw it on a two-dimensional surface, for the density of lines passing through a circle of radius r is the total number divided by the circumference ($n/2\pi r$), which goes like $(1/r)$, not $(1/r^2)$.⁴ But if you imagine the model in three dimensions (a pincushion with needles sticking out in all directions), then the density of lines is the total number divided by the *area* of the sphere ($n/4\pi r^2$), which *does* go like $(1/r^2)$.

Such diagrams are also convenient for representing more complicated fields. Of course, the number of lines you draw depends on how lazy you are (and how sharp your pencil is), though you ought to include enough to get an accurate sense of the field, and you must be consistent: if q gets 8 lines, then $2q$ deserves 16. And you must space them fairly – they emanate from a point charge symmetrically in all directions. Field lines begin on positive charges and end on negative ones; they cannot simply terminate in midair,⁵ though they may extend out to infinity. Moreover, field lines can never cross – at the intersection, the field would have two different directions at once! With all this in mind, it is easy to sketch the field of any simple configuration of point charges: begin by drawing the lines in the neighborhood of each charge, and then connect them up or extend them to infinity (Figs. 2.13 and 2.14).

In this model, the *flux* of $\mathbf{E}$ through a surface $\mathcal{S}$,

$$\Phi_E \equiv \int_{\mathcal{S}} \mathbf{E} \cdot d\mathbf{a}, \quad (2.11)$$

is a measure of the “number of field lines” passing through $\mathcal{S}$. I put this in quotes because of course we can only draw a representative *sample* of the field lines – the *total* number would be infinite. But *for a given sampling*

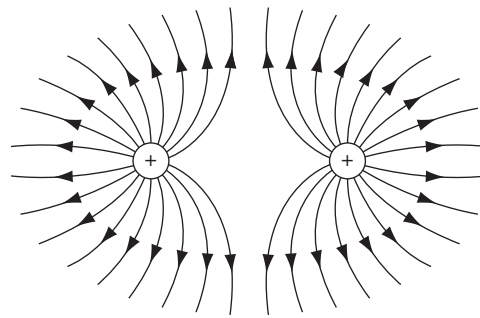


Opposite charges

Fig. 2.13

⁴ For a critique of electric field-line diagrams, see A. Wolf et al., *Am. J. Phys.* **64**, 714 (1996).

⁵ If they *did*, the divergence of $\mathbf{E}$ would not be zero, and (as we shall soon see) that cannot happen in empty space. Of course, in a region with charge density $\rho \neq 0$ some field lines *will* terminate.



Equal charges

Fig. 2.14

rate the flux is *proportional* to the number of lines drawn, because the field strength, remember, is proportional to the density of field lines (the number per unit area), and hence $\mathbf{E} \cdot d\mathbf{a}$ is proportional to the number of lines passing through the infinitesimal area $d\mathbf{a}$. (The dot product picks out the component of $d\mathbf{a}$ along the direction of $\mathbf{E}$, as indicated in Fig. 2.15. It is the area *in the plane perpendicular to $\mathbf{E}$* that we have in mind when we say that the density of field lines is the number per unit area.)

This suggests that the flux through any *closed* surface is a measure of the total charge inside, for the field lines that originate on a positive charge must either pass out through the surface or else terminate on a negative charge inside (Fig. 2.16a). On the other hand, a charge *outside* the surface will contribute nothing to the total flux, since its field lines pass in one side and out the other (Fig. 2.16b). This is the *essence* of **Gauss's law**. Now let's make it quantitative.

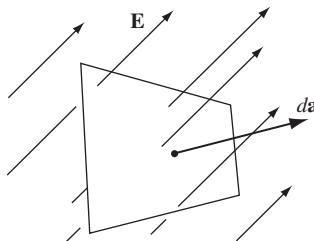


Fig. 2.15

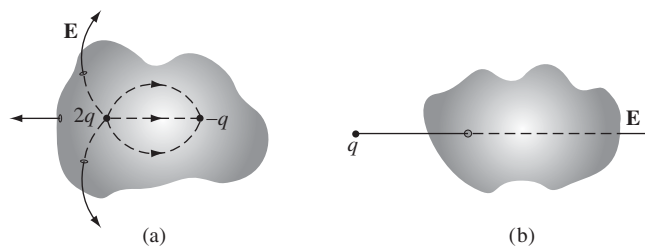


Fig. 2.16

In the case of a point charge q at the origin, the flux of $\mathbf{E}$ through a spherical surface of radius r is

$$\oint \mathbf{E} \cdot d\mathbf{a} = \int \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r^2} \hat{\mathbf{r}} \right) \cdot (r^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}) = \frac{1}{\epsilon_0} q. \quad (2.12)$$

Notice that the radius of the sphere cancels out, for while the surface area goes *up* as r^2 , the field goes *down* as $1/r^2$, so the product is constant. In terms of the field-line picture, this makes good sense, since the same number of field lines pass through any sphere centered at the origin, regardless of its size. In fact, it didn't have to be a sphere – *any* closed surface, whatever its shape, would be pierced by the same number of field lines. Evidently, *the flux through any surface enclosing the charge is q/ϵ_0 .*

Now suppose that instead of a single charge at the origin, we have a bunch of charges scattered about. According to the principle of superposition, the total field is the (vector) sum of all the individual fields:

$$\mathbf{E} = \sum_{i=1}^n \mathbf{E}_i.$$

The flux through a surface that encloses them all is

$$\oint \mathbf{E} \cdot d\mathbf{a} = \sum_{i=1}^n \left(\oint \mathbf{E}_i \cdot d\mathbf{a} \right) = \sum_{i=1}^n \left(\frac{1}{\epsilon_0} q_i \right).$$

For any closed surface, then,

$$\boxed{\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}}, \quad (2.13)$$

where Q_{enc} is the total charge enclosed within the surface.⁶ This is the quantitative statement of Gauss's law. Although it contains no information that was not already present in Coulomb's law plus the principle of superposition, it is of almost magical power, as you will see in Section 2.2.3. Notice that it all hinges on the $1/r^2$ character of Coulomb's law; without that the crucial cancelation of the rs in Eq. 2.12 would not occur, and the total flux of $\mathbf{E}$ would depend on the surface chosen, not merely on the total charge enclosed. Other $1/r^2$ forces (I am thinking particularly of Newton's law of universal gravitation) will obey "Gauss's laws" of their own, and the applications we develop here carry over directly.

As it stands, Gauss's law is an *integral* equation, but we can easily turn it into a *differential* one, by applying the divergence theorem:

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \int_V (\nabla \cdot \mathbf{E}) d\tau.$$

Rewriting Q_{enc} in terms of the charge density ρ , we have

$$Q_{\text{enc}} = \int_V \rho d\tau.$$

⁶ If there's some more charge, *outside* the surface, it will contribute to the field, of course, but not to the flux.

So Gauss's law becomes

$$\int_{\mathcal{V}} (\nabla \cdot \mathbf{E}) d\tau = \int_{\mathcal{V}} \left(\frac{\rho}{\epsilon_0} \right) d\tau.$$

And since this holds for *any* volume, the integrands must *themselves* be equal:⁷

$$\boxed{\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho.} \quad (2.14)$$

Equation 2.14 carries the same message as Eq. 2.13; it is **Gauss's law in differential form**. The differential version is tidier, but the integral form has the advantage in that it accommodates point, line, and surface charges more naturally.

Problem 2.9. Suppose the electric field in some region is found to be $\mathbf{E} = kr^3 \hat{\mathbf{r}}$, in spherical coordinates (k is some constant).

- Find the charge density ρ .
- Find the total charge contained in a sphere of radius R , centered at the origin. (Do it two different ways.)

Problem 2.10. A charge q sits at the back corner of a cube, as shown in Fig. 2.17. What is the flux of $\mathbf{E}$ through the shaded side?

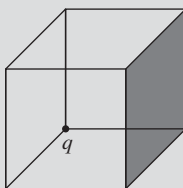


Fig. 2.17

Problem 2.11. Draw a field-line diagram for a uniformly charged solid sphere.

2.2.2 The Divergence of $\mathbf{E}$

Let's go back, now, and calculate the divergence of $\mathbf{E}$ directly from Eq. 2.8:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_{\text{all space}} \frac{\hat{\mathbf{z}}}{r^2} \rho(\mathbf{r}') d\tau'. \quad (2.15)$$

⁷ Think about it: Suppose the integrands were *different*, in the neighborhood of some point P . Then I could choose $\mathcal{V}$ to be a tiny volume centered at P , and the *integrals* would not be equal.

(Originally the integration was over the volume occupied by the charge, but I may as well extend it to all space, since $\rho = 0$ in the exterior region anyway.) Noting that the $\mathbf{r}$ -dependence is contained in $\mathbf{r} = \mathbf{r} - \mathbf{r}'$, we have

$$\nabla \cdot \mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) \rho(\mathbf{r}') d\tau'.$$

This is precisely the divergence we calculated in Eq. 1.100:

$$\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi\delta^3(\mathbf{r}).$$

Thus

$$\nabla \cdot \mathbf{E} = \frac{1}{4\pi\epsilon_0} \int 4\pi\delta^3(\mathbf{r} - \mathbf{r}') \rho(\mathbf{r}') d\tau' = \frac{1}{\epsilon_0} \rho(\mathbf{r}), \quad (2.16)$$

which is Gauss's law in differential form (Eq. 2.14). To recover the integral form (Eq. 2.13), we run the previous argument in reverse – integrate over a volume and apply the divergence theorem:

$$\int_V \nabla \cdot \mathbf{E} d\tau = \oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} \int_V \rho d\tau = \frac{1}{\epsilon_0} Q_{\text{enc}}.$$

2.2.3 Applications of Gauss's Law

I must interrupt the theoretical development at this point to show you the extraordinary power of Gauss's law, in integral form. When symmetry permits, it affords *by far* the quickest and easiest way of computing electric fields. I'll illustrate the method with a series of examples.

Example 2.3. Find the field outside a uniformly charged solid sphere of radius R and total charge q .

Solution. Imagine a spherical surface at radius $r > R$ (Fig. 2.18); this is called a **Gaussian surface** in the trade. Gauss's law says that

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}},$$

and in this case $Q_{\text{enc}} = q$. At first glance this doesn't seem to get us very far, because the quantity we want ($\mathbf{E}$) is buried inside the surface integral. Luckily, symmetry allows us to extract $\mathbf{E}$ from under the integral sign: $\mathbf{E}$ certainly points radially outward,⁸ as does $d\mathbf{a}$, so we can drop the dot product,

⁸ If you doubt that $\mathbf{E}$ is radial, consider the alternative. Suppose, say, that it points due east, at the "equator." But the orientation of the equator is perfectly arbitrary – nothing is spinning here, so there is no natural "north–south" axis – any argument purporting to show that $\mathbf{E}$ points east could just as well be used to show it points west, or north, or any other direction. The only *unique* direction on the surface of a sphere is radial.

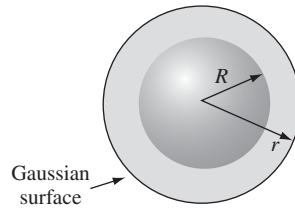


Fig. 2.18

$$\int_S \mathbf{E} \cdot d\mathbf{a} = \int_S |\mathbf{E}| da,$$

and the *magnitude* of $\mathbf{E}$ is constant over the Gaussian surface, so it comes outside the integral:

$$\int_S |\mathbf{E}| da = |\mathbf{E}| \int_S da = |\mathbf{E}| 4\pi r^2.$$

Thus

$$|\mathbf{E}| 4\pi r^2 = \frac{1}{\epsilon_0} q,$$

or

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}.$$

Notice a remarkable feature of this result: the field outside the sphere is exactly *the same as it would have been if all the charge had been concentrated at the center!*

Gauss's law is always *true*, but it is not always *useful*. If ρ had not been uniform (or, at any rate, not spherically symmetrical), or if I had chosen some other shape for my Gaussian surface, it would still have been true that the flux of $\mathbf{E}$ is q/ϵ_0 , but $\mathbf{E}$ would not have pointed in the same direction as $d\mathbf{a}$, and its magnitude would not have been constant over the surface, and without that I cannot pull $|\mathbf{E}|$ outside the integral. *Symmetry is crucial* to this application of Gauss's law. As far as I know, there are only three kinds of symmetry that work:

- (1) *Spherical symmetry*. Make your Gaussian surface a concentric sphere.
- (2) *Cylindrical symmetry*. Make your Gaussian surface a coaxial cylinder (Fig. 2.19).
- (3) *Plane symmetry*. Use a Gaussian "pillbox" that straddles the surface (Fig. 2.20).

Although (2) and (3) technically require infinitely long cylinders, and infinite planes, we shall often use them to get approximate answers for "long" cylinders or "large" planes, as long as we are not near the edges.

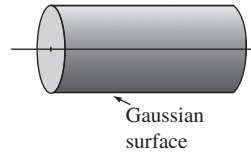


Fig. 2.19

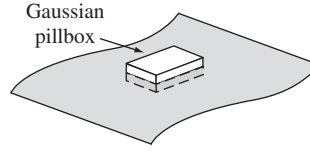


Fig. 2.20

Example 2.4. A long cylinder (Fig. 2.21) carries a charge density that is proportional to the distance from the axis: $\rho = ks$, for some constant k . Find the electric field inside this cylinder.

Solution. Draw a Gaussian cylinder of length l and radius s . For this surface, Gauss's law states that

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}.$$

The enclosed charge is

$$Q_{\text{enc}} = \int \rho \, d\tau = \int (ks')(s' \, ds' \, d\phi \, dz) = 2\pi kl \int_0^s s'^2 \, ds' = \frac{2}{3}\pi kls^3.$$

(I used the volume element appropriate to cylindrical coordinates, Eq. 1.78, and integrated ϕ from 0 to 2π , dz from 0 to l . I put a prime on the integration variable s' , to distinguish it from the radius s of the Gaussian surface.)

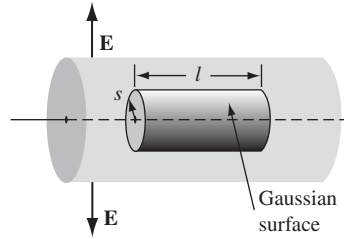


Fig. 2.21

Now, symmetry dictates that $\mathbf{E}$ must point radially outward, so for the curved portion of the Gaussian cylinder we have

$$\int \mathbf{E} \cdot d\mathbf{a} = \int |\mathbf{E}| \, da = |\mathbf{E}| \int da = |\mathbf{E}| 2\pi sl,$$

while the two ends contribute nothing (here $\mathbf{E}$ is perpendicular to $d\mathbf{a}$). Thus,

$$|\mathbf{E}| 2\pi sl = \frac{1}{\epsilon_0} \frac{2}{3}\pi kls^3,$$

or, finally,

$$\mathbf{E} = \frac{1}{3\epsilon_0} ks^2 \hat{\mathbf{s}}.$$

Example 2.5. An infinite plane carries a uniform surface charge σ . Find its electric field.

Solution. Draw a “Gaussian pillbox,” extending equal distances above and below the plane (Fig. 2.22). Apply Gauss’s law to this surface:

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}.$$

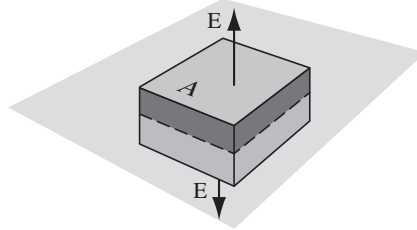


Fig. 2.22

In this case, $Q_{\text{enc}} = \sigma A$, where A is the area of the lid of the pillbox. By symmetry, $\mathbf{E}$ points away from the plane (upward for points above, downward for points below). So the top and bottom surfaces yield

$$\int \mathbf{E} \cdot d\mathbf{a} = 2A|\mathbf{E}|,$$

whereas the sides contribute nothing. Thus

$$2A|\mathbf{E}| = \frac{1}{\epsilon_0} \sigma A,$$

or

$$\mathbf{E} = \frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}}, \quad (2.17)$$

where $\hat{\mathbf{n}}$ is a unit vector pointing away from the surface. In Prob. 2.6, you obtained this same result by a much more laborious method.

It seems surprising, at first, that the field of an infinite plane is *independent of how far away you are*. What about the $1/r^2$ in Coulomb’s law? The point is that as you move farther and farther away from the plane, more and more charge comes into your “field of view” (a cone shape extending out from your eye), and this compensates for the diminishing influence of any particular piece. The electric field of a sphere falls off like $1/r^2$; the electric field of an infinite line falls off like $1/s$; and the electric field of an infinite plane does not fall off at all (you cannot escape from an infinite plane).

Although the direct use of Gauss’s law to compute electric fields is limited to cases of spherical, cylindrical, and planar symmetry, we can put together *combinations* of objects possessing such symmetry, even if the arrangement as a whole is not symmetrical. For example, invoking the principle of superposition, we could find the field in the vicinity of two uniformly charged parallel cylinders, or a sphere near an infinite charged plane.

Example 2.6. Two infinite parallel planes carry equal but opposite uniform charge densities $\pm\sigma$ (Fig. 2.23). Find the field in each of the following three regions: (i) to the left of both, (ii) between them, (iii) to the right of both.

Solution. The left-hand plate produces a field $(1/2\epsilon_0)\sigma$, which points away from it – to the left in region (i) and to the right in regions (ii) and (iii) (Fig. 2.24). The right-hand plate, being negatively charged, produces a field $(1/2\epsilon_0)\sigma$, which points *toward* it – to the right in regions (i) and (ii) and to the left in region (iii). The two fields cancel in regions (i) and (iii); they conspire in region (ii). *Conclusion:* The field between the plates is σ/ϵ_0 , and points to the right; elsewhere it is zero.

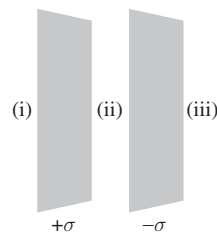


Fig. 2.23

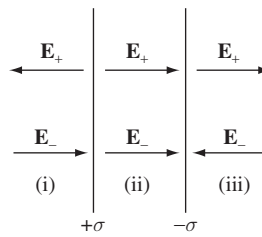


Fig. 2.24

Problem 2.12. Use Gauss's law to find the electric field inside and outside a spherical shell of radius R that carries a uniform surface charge density σ . Compare your answer to Prob. 2.7; notice how much quicker and easier Gauss's method is.

Problem 2.13. Use Gauss's law to find the electric field inside a uniformly charged solid sphere (charge density ρ). Compare your answer to Prob. 2.8.

Problem 2.14. Find the electric field a distance s from an infinitely long straight wire that carries a uniform line charge λ . Compare Eq. 2.9.

Problem 2.15. Find the electric field inside a sphere that carries a charge density proportional to the distance from the center, $\rho = kr$, for some constant k . [Caution: This charge density is *not* uniform, and you must *integrate* to get the enclosed charge.]

Problem 2.16. A thick spherical shell carries charge density

$$\rho = \frac{k}{r^2} \quad (a \leq r \leq b)$$

(Fig. 2.25). Find the electric field in the following three regions: (i) $r < a$, (ii) $a < r < b$, (iii) $r > b$. Plot $|\mathbf{E}|$ as a function of r , for the case $b = 2a$.

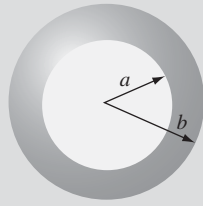


Fig. 2.25

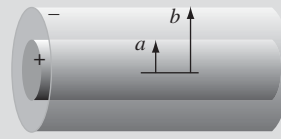


Fig. 2.26

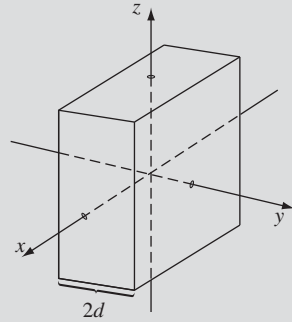


Fig. 2.27

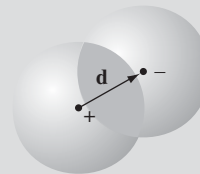


Fig. 2.28

Problem 2.17. A long coaxial cable (Fig. 2.26) carries a uniform *volume* charge density ρ on the inner cylinder (radius a), and a uniform *surface* charge density σ on the outer cylindrical shell (radius b). This surface charge is negative and is of just the right magnitude that the cable as a whole is electrically neutral. Find the electric field in each of the following three regions: (i) inside the inner cylinder ($s < a$), (ii) between the cylinders ($a < s < b$), (iii) outside the cable ($s > b$). Plot $|\mathbf{E}|$ as a function of s .

Problem 2.18. An infinite plane slab, of thickness $2d$, carries a uniform volume charge density ρ (Fig. 2.27). Find the electric field, as a function of y , where $y = 0$ at the center. Plot E versus y , calling E positive when $\mathbf{E}$ points in the $+y$ direction and negative when it points in the $-y$ direction.

- **Problem 2.19.** Two spheres, each of radius R and carrying uniform volume charge densities $+\rho$ and $-\rho$, respectively, are placed so that they partially overlap (Fig. 2.28). Call the vector from the positive center to the negative center $\mathbf{d}$. Show that the field in the region of overlap is constant, and find its value. [Hint: Use the answer to Prob. 2.13.]

2.2.4 The Curl of $\mathbf{E}$

I'll calculate the curl of $\mathbf{E}$, as I did the divergence in Section 2.2.1, by studying first the simplest possible configuration: a point charge at the origin. In this case,

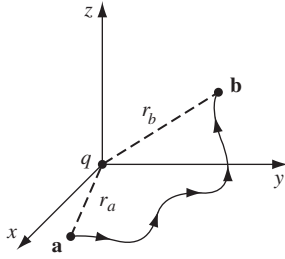


Fig. 2.29

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}.$$

Now, a glance at Fig. 2.12 should convince you that the curl of this field has to be zero, but I suppose we ought to come up with something a little more rigorous than that. What if we calculate the line integral of this field from some point **a** to some other point **b** (Fig. 2.29):

$$\int_a^b \mathbf{E} \cdot d\mathbf{l}.$$

In spherical coordinates, $d\mathbf{l} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin \theta d\phi \hat{\boldsymbol{\phi}}$, so

$$\mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr.$$

Therefore,

$$\int_a^b \mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \int_a^b \frac{q}{r^2} dr = \frac{-1}{4\pi\epsilon_0} \frac{q}{r} \Big|_{r_a}^{r_b} = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_a} - \frac{q}{r_b} \right), \quad (2.18)$$

where r_a is the distance from the origin to the point **a** and r_b is the distance to **b**. The integral around a *closed* path is evidently zero (for then $r_a = r_b$):

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0, \quad (2.19)$$

and hence, applying Stokes' theorem (and noting that Eq. 2.19 applies to *any* closed path),

$$\nabla \times \mathbf{E} = \mathbf{0}. \quad (2.20)$$

Now, I proved Eqs. 2.19 and 2.20 only for the field of a single point charge at the origin, but these results make no reference to what is, after all, a perfectly arbitrary choice of coordinates; they hold no matter *where* the charge is located. Moreover, if we have many charges, the principle of superposition says that the total field is the vector sum of their individual fields:

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \cdots,$$

so

$$\nabla \times \mathbf{E} = \nabla \times (\mathbf{E}_1 + \mathbf{E}_2 + \cdots) = (\nabla \times \mathbf{E}_1) + (\nabla \times \mathbf{E}_2) + \cdots = \mathbf{0}.$$

Thus, Eqs. 2.19 and 2.20 hold for *any* (static) charge distribution whatever.

Problem 2.20. Calculate $\nabla \times \mathbf{E}$ directly from Eq. 2.8, by the method of Section 2.2.2. Refer to Prob. 1.63b if you get stuck.

2.3 Electric Potential

2.3.1 Introduction to Potential

The electric field $\mathbf{E}$ is not just *any* old vector function. It is a very special *kind* of vector function: one whose curl is zero. The vector $\mathbf{E} = y\hat{\mathbf{x}}$, for example, could not possibly be an electrostatic field; *no* set of charges, regardless of their sizes and positions, could ever produce such a field. We're going to exploit this special property of electric fields to reduce a *vector* problem (finding $\mathbf{E}$) to a much simpler *scalar* problem. The first theorem in Section 1.6.2 asserts that any vector whose curl is zero is equal to the gradient of some scalar. What I'm going to do now amounts to a proof of that claim, in the context of electrostatics.

Because $\nabla \times \mathbf{E} = \mathbf{0}$, the line integral of $\mathbf{E}$ around any closed loop is zero (that follows from Stokes' theorem). Because $\oint \mathbf{E} \cdot d\mathbf{l} = 0$, the line integral of $\mathbf{E}$ from point $\mathbf{a}$ to point $\mathbf{b}$ is the same for all paths (otherwise you could go *out* along path (i) and return along path (ii) – Fig. 2.30 – and obtain $\oint \mathbf{E} \cdot d\mathbf{l} \neq 0$). Because the line integral is independent of path, we can define a function⁹

$$V(\mathbf{r}) \equiv - \int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}. \quad (2.21)$$

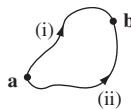


Fig. 2.30

⁹ To avoid any possible ambiguity, I should perhaps put a prime on the integration variable:

$$V(\mathbf{r}) = - \int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E}(\mathbf{r}') \cdot d\mathbf{l}'.$$

But this makes for cumbersome notation, and I prefer whenever possible to reserve the primes for source points. However, when (as in Ex. 2.7) we calculate such integrals explicitly, I will restore the primes.

Here $\mathcal{O}$ is some standard reference point on which we have agreed beforehand; V then depends only on the point $\mathbf{r}$. It is called the **electric potential**.

The potential *difference* between two points $\mathbf{a}$ and $\mathbf{b}$ is

$$\begin{aligned} V(\mathbf{b}) - V(\mathbf{a}) &= - \int_{\mathcal{O}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} + \int_{\mathcal{O}}^{\mathbf{a}} \mathbf{E} \cdot d\mathbf{l} \\ &= - \int_{\mathcal{O}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} - \int_{\mathbf{a}}^{\mathcal{O}} \mathbf{E} \cdot d\mathbf{l} = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l}. \end{aligned} \quad (2.22)$$

Now, the fundamental theorem for gradients states that

$$V(\mathbf{b}) - V(\mathbf{a}) = \int_{\mathbf{a}}^{\mathbf{b}} (\nabla V) \cdot d\mathbf{l},$$

so

$$\int_{\mathbf{a}}^{\mathbf{b}} (\nabla V) \cdot d\mathbf{l} = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l}.$$

Since, finally, this is true for *any* points $\mathbf{a}$ and $\mathbf{b}$, the integrands must be equal:

$$\boxed{\mathbf{E} = -\nabla V.} \quad (2.23)$$

Equation 2.23 is the differential version of Eq. 2.21; it says that the electric field is the gradient of a scalar potential, which is what we set out to prove.

Notice the subtle but crucial role played by path independence (or, equivalently, the fact that $\nabla \times \mathbf{E} = \mathbf{0}$) in this argument. If the line integral of $\mathbf{E}$ depended on the path taken, then the “definition” of V , Eq. 2.21, would be nonsense. It would not define a function, since changing the path would alter the value of $V(\mathbf{r})$. By the way, don’t let the minus sign in Eq. 2.23 distract you; it carries over from Eq. 2.21 and is largely a matter of convention.

Problem 2.21. One of these is an impossible electrostatic field. Which one?

- (a) $\mathbf{E} = k[xy\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + 3xz\hat{\mathbf{z}}].$
- (b) $\mathbf{E} = k[y^2\hat{\mathbf{x}} + (2xy + z^2)\hat{\mathbf{y}} + 2yz\hat{\mathbf{z}}].$

Here k is a (nonzero) constant with the appropriate units. For the *possible* one, find the potential, using the origin as your reference point. Check your answer by computing ∇V . [Hint: You must select a specific path to integrate along. It doesn’t matter *what* path you choose, since the answer is path independent, but you simply cannot integrate unless you have a definite path in mind.]

2.3.2 Comments on Potential

(i) **The name.** The word “potential” is a hideous misnomer because it inevitably reminds you of potential *energy*. This is particularly insidious, because there *is* a connection between “potential” and “potential energy,” as you will see in Section 2.4. I’m sorry that it is impossible to escape this word. The best I can do is to insist once and for all that “potential” and “potential energy” are completely different terms and should, by all rights, have different names. Incidentally, a region (typically a surface) over which the potential is constant is called an **equipotential**.

(ii) **Advantage of the potential formulation.** If you know V , you can easily get $\mathbf{E}$ – just take the gradient: $\mathbf{E} = -\nabla V$. This is quite extraordinary when you stop to think about it, for $\mathbf{E}$ is a *vector* quantity (three components), but V is a *scalar* (one component). How can *one* function possibly contain all the information that *three* independent functions carry? The answer is that the three components of $\mathbf{E}$ are not really as independent as they look; in fact, they are explicitly interrelated by the very condition we started with, $\nabla \times \mathbf{E} = \mathbf{0}$. In terms of the components,

$$\frac{\partial E_x}{\partial y} = \frac{\partial E_y}{\partial x}, \quad \frac{\partial E_z}{\partial y} = \frac{\partial E_y}{\partial z}, \quad \frac{\partial E_x}{\partial z} = \frac{\partial E_z}{\partial x}.$$

This brings us back to my observation at the beginning of Section 2.3.1: $\mathbf{E}$ is a *very special kind of vector*. What the potential formulation does is to exploit this feature to maximum advantage, reducing a vector problem to a scalar one, in which there is no need to fuss with components.

(iii) **The reference point $\mathcal{O}$.** There is an essential ambiguity in the definition of potential, since the choice of reference point $\mathcal{O}$ was arbitrary. Changing reference points amounts to adding a constant K to the potential:

$$V'(\mathbf{r}) = - \int_{\mathcal{O}'}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l} = - \int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l} - \int_{\mathcal{O}}^{\mathcal{O}'} \mathbf{E} \cdot d\mathbf{l} = K + V(\mathbf{r}),$$

where K is the line integral of $\mathbf{E}$ from the old reference point $\mathcal{O}$ to the new one $\mathcal{O}'$. Of course, adding a constant to V will not affect the potential *difference* between two points:

$$V'(\mathbf{b}) - V'(\mathbf{a}) = V(\mathbf{b}) - V(\mathbf{a}),$$

since the K s cancel out. (Actually, it was already clear from Eq. 2.22 that the potential difference is independent of $\mathcal{O}$, because it can be written as the line integral of $\mathbf{E}$ from $\mathbf{a}$ to $\mathbf{b}$, with no mention of $\mathcal{O}$.) Nor does the ambiguity affect the gradient of V :

$$\nabla V' = \nabla V,$$

since the derivative of a constant is zero. That’s why all such V s, differing only in their choice of reference point, correspond to the same field $\mathbf{E}$.

Potential as such carries no real physical significance, for at any given point we can adjust its value at will by a suitable relocation of $\mathcal{O}$. In this

sense, it is rather like altitude: If I ask you how high Denver is, you will probably tell me its height above sea level, because that is a convenient and traditional reference point. But we could as well agree to measure altitude above Washington, DC, or Greenwich, or wherever. That would add (or, rather, subtract) a fixed amount from all our sea-level readings, but it wouldn't change anything about the real world. The only quantity with intrinsic meaning is the *difference* in altitude between two points, and *that* is the same *whatever* your reference level.

Having said this, however, there *is* a “natural” spot to use for $\mathcal{O}$ in electrostatics – analogous to sea level for altitude – and that is a point infinitely far from the charge. Ordinarily, then, we “set the zero of potential at infinity.” (Since $V(\mathcal{O}) = 0$, choosing a reference point is equivalent to selecting a place where V is to be zero.) But I must warn you that there is one special circumstance in which this convention fails: when the charge distribution itself extends to infinity. The symptom of trouble, in such cases, is that the potential blows up. For instance, the field of a uniformly charged plane is $(\sigma/2\epsilon_0)\hat{n}$, as we found in Ex. 2.5; if we naively put $\mathcal{O} = \infty$, then the potential at height z above the plane becomes

$$V(z) = - \int_{\infty}^z \frac{1}{2\epsilon_0} \sigma dz = -\frac{1}{2\epsilon_0} \sigma (z - \infty).$$

The remedy is simply to choose some other reference point (in this example you might use a point on the plane). Notice that this difficulty occurs only in textbook problems; in “real life” there is no such thing as a charge distribution that goes on forever, and we can *always* use infinity as our reference point.

(iv) Potential obeys the superposition principle. The original superposition principle pertains to the force on a test charge Q . It says that the total force on Q is the vector sum of the forces attributable to the source charges individually:

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \cdots.$$

Dividing through by Q , we find that the electric field, too, obeys the superposition principle:

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \cdots.$$

Integrating from the common reference point to $\mathbf{r}$, it follows that the potential also satisfies such a principle:

$$V = V_1 + V_2 + \cdots.$$

That is, the potential at any given point is the sum of the potentials due to all the source charges separately. Only this time it is an *ordinary* sum, not a *vector* sum, which makes it a lot easier to work with.

(v) Units of Potential. In our units, force is measured in newtons and charge in coulombs, so electric fields are in newtons per coulomb. Accordingly, potential is newton-meters per coulomb, or joules per coulomb. A joule per coulomb is a **volt**.

Example 2.7. Find the potential inside and outside a spherical shell of radius R (Fig. 2.31) that carries a uniform surface charge. Set the reference point at infinity.

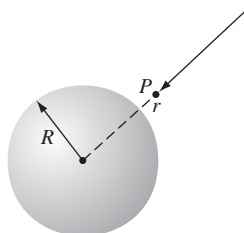


Fig. 2.31

Solution. From Gauss's law, the field outside is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}},$$

where q is the total charge on the sphere. The field inside is zero. For points outside the sphere ($r > R$),

$$V(r) = - \int_{\infty}^r \mathbf{E} \cdot d\mathbf{l} = - \frac{1}{4\pi\epsilon_0} \int_{\infty}^r \frac{q}{r'^2} dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{r'} \Big|_{\infty}^r = \frac{1}{4\pi\epsilon_0} \frac{q}{r}.$$

To find the potential *inside* the sphere ($r < R$), we must break the integral into two pieces, using in each region the field that prevails there:

$$V(r) = \frac{-1}{4\pi\epsilon_0} \int_{\infty}^R \frac{q}{r'^2} dr' - \int_R^r (0) dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{r'} \Big|_{\infty}^R + 0 = \frac{1}{4\pi\epsilon_0} \frac{q}{R}.$$

Notice that the potential is *not* zero inside the shell, even though the field vanishes! To be sure, V is a *constant* in this region, so that $\nabla V = \mathbf{0}$ – that's what matters. In problems of this type, you must always *work your way in from the reference point*; that's where the potential is “nailed down.” It is tempting to suppose that you could figure out the potential inside the sphere on the basis of the field there alone, but this is false: the potential inside the sphere is sensitive to what's going on outside the sphere as well. If I placed a second uniformly charged shell out at radius $R' > R$, the potential inside R would change, even though the field would still be zero. Gauss's law guarantees that charge exterior to a given point (that is, at larger r) produces no net *field* at that point, provided it is spherically or cylindrically symmetric, but there is no such rule for *potential*, when infinity is used as the reference point.

Problem 2.22. Find the potential inside and outside a uniformly charged solid sphere whose radius is R and whose total charge is q . Use infinity as your reference point. Compute the gradient of V in each region, and check that it yields the correct field. Sketch $V(r)$.

- **Problem 2.23.** Find the potential a distance s from an infinitely long straight wire that carries a uniform line charge λ . Compute the gradient of your potential, and check that it yields the correct field.

Problem 2.24. For the charge configuration of Prob. 2.16, find the potential at the center, using infinity as your reference point.

Problem 2.25. For the configuration of Prob. 2.17, find the potential difference between a point on the axis and a point on the outer cylinder. Note that it is not necessary to commit yourself to a particular reference point, if you use Eq. 2.22.

2.3.3 Poisson's Equation and Laplace's Equation

We found in Section 2.3.1 that the electric field can be written as the gradient of a scalar potential,

$$\mathbf{E} = -\nabla V.$$

The following question arises: What do the divergence and curl of $\mathbf{E}$,

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \text{and} \quad \nabla \times \mathbf{E} = \mathbf{0},$$

tell us about V ? Well, $\nabla \cdot \mathbf{E} = \nabla \cdot (-\nabla V) = -\nabla^2 V$, so, apart from that persistent minus sign, the divergence of $\mathbf{E}$ is the Laplacian of V . Gauss's law, then, says

$$\boxed{\nabla^2 V = -\frac{\rho}{\epsilon_0}} \quad (2.24)$$

This is known as **Poisson's equation**. In regions where there is no charge, so $\rho = 0$, Poisson's equation reduces to **Laplace's equation**,

$$\nabla^2 V = 0. \quad (2.25)$$

We'll explore this equation more fully in Chapter 3.

So much for Gauss's law. What about the curl law? This says that

$$\nabla \times \mathbf{E} = \nabla \times (-\nabla V) = \mathbf{0}.$$

But that's no condition on V ; curl of gradient is *always* zero. Of course, we *used* the curl law to show that $\mathbf{E}$ could be expressed as the gradient of a scalar, so it's not really surprising that this works out: $\nabla \times \mathbf{E} = \mathbf{0}$ *permits* $\mathbf{E} = -\nabla V$; in return, $\mathbf{E} = -\nabla V$ *guarantees* $\nabla \times \mathbf{E} = \mathbf{0}$. It takes only *one* differential equation (Poisson's) to determine V , because V is a scalar; for $\mathbf{E}$ we needed *two*, the divergence and the curl.

2.3.4 The Potential of a Localized Charge Distribution

I defined V in terms of $\mathbf{E}$ (Eq. 2.21). Ordinarily, though, it's $\mathbf{E}$ that we're *looking* for (if we already knew $\mathbf{E}$, there wouldn't be much point in calculating V). The idea is that it might be easier to get V first, and then calculate $\mathbf{E}$ by taking the gradient. Typically, then, we know where the charge is (that is, we know ρ), and we want to find V . Now, Poisson's equation relates V and ρ , but unfortunately it's “the wrong way around”: it would give us ρ , if we knew V , whereas we want V , knowing ρ . What we must do, then, is “invert” Poisson's equation. That's the program for this section, although I shall do it by roundabout means, beginning, as always, with a point charge at the origin.

The electric field is $\mathbf{E} = (1/4\pi\epsilon_0)(q/r^2)\hat{\mathbf{r}}$, and $d\mathbf{l} = dr\hat{\mathbf{r}} + r d\theta\hat{\boldsymbol{\theta}} + r \sin\theta d\phi\hat{\boldsymbol{\phi}}$ (Eq. 1.68), so

$$\mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr.$$

Setting the reference point at infinity, the potential of a point charge q at the origin is

$$V(r) = - \int_{\infty}^r \mathbf{E} \cdot d\mathbf{l} = \frac{-1}{4\pi\epsilon_0} \int_{\infty}^r \frac{q}{r'^2} dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \Big|_{\infty}^r = \frac{1}{4\pi\epsilon_0} \frac{q}{r}.$$

(You see here the advantage of using infinity for the reference point: it kills the lower limit on the integral.) Notice the sign of V ; presumably the conventional minus sign in the definition (Eq. 2.21) was chosen in order to make the potential of a positive charge come out positive. It is useful to remember that regions of positive charge are potential “hills,” regions of negative charge are potential “valleys,” and the electric field points “downhill,” from plus toward minus.

In general, the potential of a point charge q is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r}, \quad (2.26)$$

where r , as always, is the distance from q to $\mathbf{r}$ (Fig. 2.32). Invoking the superposition principle, then, the potential of a collection of charges is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}, \quad (2.27)$$

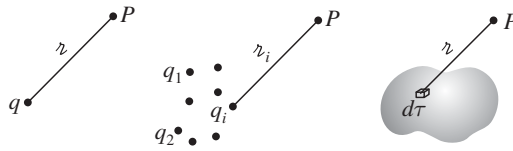


Fig. 2.32

or, for a continuous distribution,

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{z} dq. \quad (2.28)$$

In particular, for a volume charge, it is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{z} d\tau'. \quad (2.29)$$

This is the equation we were looking for, telling us how to compute V when we know ρ ; it is, if you like, the “solution” to Poisson’s equation, for a localized charge distribution.¹⁰ Compare Eq. 2.29 with the corresponding formula for the electric *field* in terms of ρ (Eq. 2.8):

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{z^2} \hat{\mathbf{z}} d\tau'.$$

The main point to notice is that the pesky unit vector $\hat{\mathbf{z}}$ is gone, so you don’t have to worry about components. The potentials of line and surface charges are

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(\mathbf{r}')}{z} dl' \quad \text{and} \quad V = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma(\mathbf{r}')}{z} da'. \quad (2.30)$$

I should warn you that everything in this section is predicated on the assumption that the reference point is at infinity. This is certainly not obvious in Eq. 2.29, but remember that we *got* that equation from the potential of a point charge at the origin, $(1/4\pi\epsilon_0)(q/r)$, which is valid only when $\mathcal{O} = \infty$. If you try to apply these formulas to one of those artificial problems in which the charge itself extends to infinity, the integral will diverge – that’s the alarm bell, telling you to go back and fix the reference point.

Example 2.8. Find the potential of a uniformly charged spherical shell of radius R (Fig. 2.33).

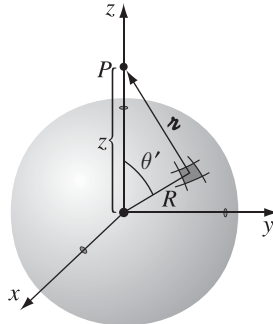


Fig. 2.33

¹⁰ Equation 2.29 is an example of the Helmholtz theorem (Appendix B).

Solution. This is the same problem we solved in Ex. 2.7, but this time let's do it using Eq. 2.30:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma}{r} da'.$$

We might as well set the point P on the z -axis and use the law of cosines to express r :

$$r^2 = R^2 + z^2 - 2Rz \cos \theta'.$$

An element of surface area on the sphere is $R^2 \sin \theta' d\theta' d\phi'$, so

$$\begin{aligned} 4\pi\epsilon_0 V(z) &= \sigma \int \frac{R^2 \sin \theta' d\theta' d\phi'}{\sqrt{R^2 + z^2 - 2Rz \cos \theta'}} \\ &= 2\pi R^2 \sigma \int_0^\pi \frac{\sin \theta'}{\sqrt{R^2 + z^2 - 2Rz \cos \theta'}} d\theta' \\ &= 2\pi R^2 \sigma \left(\frac{1}{Rz} \sqrt{R^2 + z^2 - 2Rz \cos \theta'} \right) \Big|_0^\pi \\ &= \frac{2\pi R \sigma}{z} (\sqrt{R^2 + z^2 + 2Rz} - \sqrt{R^2 + z^2 - 2Rz}) \\ &= \frac{2\pi R \sigma}{z} \left[\sqrt{(R+z)^2} - \sqrt{(R-z)^2} \right]. \end{aligned}$$

At this stage, we must be very careful to take the *positive* root. For points *outside* the sphere, z is greater than R , and hence $\sqrt{(R-z)^2} = z - R$; for points *inside* the sphere, $\sqrt{(R-z)^2} = R - z$. Thus,

$$V(z) = \begin{cases} \frac{R\sigma}{2\epsilon_0 z} [(R+z) - (z-R)] = \frac{R^2\sigma}{\epsilon_0 z} & \text{outside,} \\ \frac{R\sigma}{2\epsilon_0 z} [(R+z) - (R-z)] = \frac{R\sigma}{\epsilon_0} & \text{inside.} \end{cases}$$

In terms of r and the total charge on the shell, $q = 4\pi R^2 \sigma$,

$$V(r) = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{q}{r} & (r \geq R), \\ \frac{1}{4\pi\epsilon_0} \frac{q}{R} & (r \leq R). \end{cases}$$

Of course, in this particular case, it was easier to get V by using Eq. 2.21 than Eq. 2.30, because Gauss's law gave us $\mathbf{E}$ with so little effort. But if you compare Ex. 2.8 with Prob. 2.7, you will appreciate the power of the potential formulation.

Problem 2.26. Using Eqs. 2.27 and 2.30, find the potential at a distance z above the center of the charge distributions in Fig. 2.34. In each case, compute $\mathbf{E} = -\nabla V$, and compare your answers with Ex. 2.1, Ex. 2.2, and Prob. 2.6, respectively. Suppose that we changed the

right-hand charge in Fig. 2.34a to $-q$: What then is the potential at P ? What field does that suggest? Compare your answer to Prob. 2.2, and explain carefully any discrepancy.

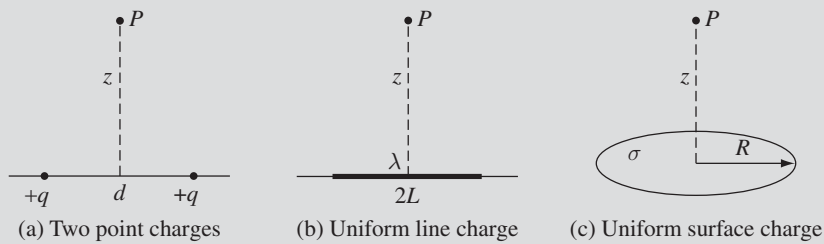


Fig. 2.34

Problem 2.27. A conical surface (an empty ice-cream cone) carries a uniform surface charge σ . The height of the cone is h , as is the radius of the top. Find the potential difference between points **a** (the vertex) and **b** (the center of the top).

Problem 2.28. Find the potential on the axis of a uniformly charged solid cylinder, a distance z from the center. The length of the cylinder is L , its radius is R , and the charge density is ρ . Use your result to calculate the electric field at this point. (Assume that $z > L/2$.)

Problem 2.29. Use Eq. 2.29 to calculate the potential inside a uniformly charged solid sphere of radius R and total charge q . Compare your answer to Prob. 2.22.

Problem 2.30. Check that Eq. 2.29 satisfies Poisson's equation, by applying the Laplacian and using Eq. 1.102.

2.3.5 Boundary Conditions

In the typical electrostatic problem you are given a source charge distribution ρ , and you want to find the electric field $\mathbf{E}$ it produces. Unless the symmetry of the problem allows a solution by Gauss's law, it is generally to your advantage to calculate the potential first, as an intermediate step. These are the three fundamental quantities of electrostatics: ρ , $\mathbf{E}$, and V . We have, in the course of our discussion, derived all six formulas interrelating them. These equations are neatly summarized in Fig. 2.35. We began with just two experimental observations: (1) the principle of superposition – a broad general rule applying to *all* electromagnetic forces, and (2) Coulomb's law – the fundamental law of electrostatics. From these, all else followed.

You may have noticed, in studying Exs. 2.5 and 2.6, or working problems such as 2.7, 2.12, and 2.17, that the electric field always undergoes a discontinuity when you cross a surface charge σ . In fact, it is a simple matter to find the *amount* by which $\mathbf{E}$ changes at such a boundary. Suppose we draw

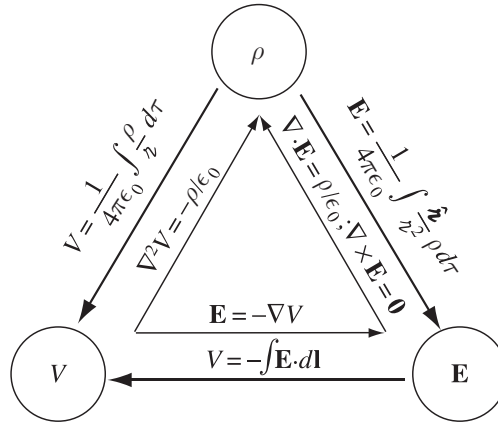


Fig. 2.35

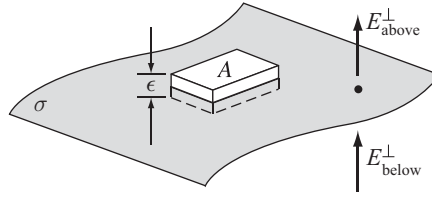


Fig. 2.36

a wafer-thin Gaussian pillbox, extending just barely over the edge in each direction (Fig. 2.36). Gauss's law says that

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{1}{\epsilon_0} \sigma A,$$

where A is the area of the pillbox lid. (If σ varies from point to point or the surface is curved, we must pick A to be extremely small.) Now, the *sides* of the pillbox contribute nothing to the flux, in the limit as the thickness ϵ goes to zero, so we are left with

$$E_{\text{above}}^{\perp} - E_{\text{below}}^{\perp} = \frac{1}{\epsilon_0} \sigma, \quad (2.31)$$

where $E_{\text{above}}^{\perp}$ denotes the component of $\mathbf{E}$ that is perpendicular to the surface immediately above, and $E_{\text{below}}^{\perp}$ is the same, only just below the surface. For consistency, we let “upward” be the positive direction for both. *Conclusion: The normal component of $\mathbf{E}$ is discontinuous by an amount σ/ϵ_0 at any boundary.* In particular, where there is *no* surface charge, $E^{\perp}$ is continuous, as for instance at the surface of a uniformly charged solid sphere.

The *tangential* component of $\mathbf{E}$, by contrast, is *always* continuous. For if we apply Eq. 2.19,

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0,$$

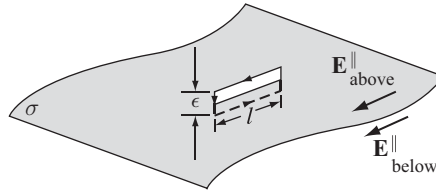


Fig. 2.37

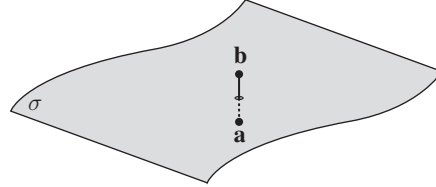


Fig. 2.38

to the thin rectangular loop of Fig. 2.37, the ends give nothing (as $\epsilon \rightarrow 0$), and the sides give $(E_{\text{above}}^{\parallel} l - E_{\text{below}}^{\parallel} l)$, so

$$\mathbf{E}_{\text{above}}^{\parallel} = \mathbf{E}_{\text{below}}^{\parallel}, \quad (2.32)$$

where $\mathbf{E}^{\parallel}$ stands for the components of $\mathbf{E}$ *parallel* to the surface. The boundary conditions on $\mathbf{E}$ (Eqs. 2.31 and 2.32) can be combined into a single formula:

$$\mathbf{E}_{\text{above}} - \mathbf{E}_{\text{below}} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}, \quad (2.33)$$

where $\hat{\mathbf{n}}$ is a unit vector perpendicular to the surface, pointing from “below” to “above.”¹¹

The potential, meanwhile, is continuous across *any* boundary (Fig. 2.38), since

$$V_{\text{above}} - V_{\text{below}} = - \int_a^b \mathbf{E} \cdot d\mathbf{l};$$

as the path length shrinks to zero, so too does the integral:

$$V_{\text{above}} = V_{\text{below}}. \quad (2.34)$$

However, the *gradient* of V inherits the discontinuity in $\mathbf{E}$; since $\mathbf{E} = -\nabla V$, Eq. 2.33 implies that

$$\nabla V_{\text{above}} - \nabla V_{\text{below}} = -\frac{1}{\epsilon_0} \sigma \hat{\mathbf{n}}, \quad (2.35)$$

¹¹ Notice that it doesn't matter which side you call “above” and which “below,” since reversal would switch the direction of $\hat{\mathbf{n}}$. Incidentally, if you're only interested in the field due to the (essentially flat) local patch of surface charge itself, the answer is $(\sigma/2\epsilon_0)\hat{\mathbf{n}}$ immediately above the surface, and $-(\sigma/2\epsilon_0)\hat{\mathbf{n}}$ immediately below. This follows from Ex. 2.5, for if you are close enough to the patch it “looks” like an infinite plane. Evidently, the entire discontinuity in $\mathbf{E}$ is attributable to this local patch of surface charge.

or, more conveniently,

$$\frac{\partial V_{\text{above}}}{\partial n} - \frac{\partial V_{\text{below}}}{\partial n} = -\frac{1}{\epsilon_0} \sigma, \quad (2.36)$$

where

$$\frac{\partial V}{\partial n} = \nabla V \cdot \hat{\mathbf{n}} \quad (2.37)$$

denotes the **normal derivative** of V (that is, the rate of change in the direction perpendicular to the surface).

Please note that these boundary conditions relate the fields and potentials *just* above and *just* below the surface. For example, the derivatives in Eq. 2.36 are the *limiting* values as we approach the surface from either side.

Problem 2.31.

- (a) Check that the results of Exs. 2.5 and 2.6, and Prob. 2.12, are consistent with Eq. 2.33.
- (b) Use Gauss's law to find the field inside and outside a long hollow cylindrical tube, which carries a uniform surface charge σ . Check that your result is consistent with Eq. 2.33.
- (c) Check that the result of Ex. 2.8 is consistent with boundary conditions 2.34 and 2.36.

2.4 Work and Energy in Electrostatics

2.4.1 The Work It Takes to Move a Charge

Suppose you have a stationary configuration of source charges, and you want to move a test charge Q from point **a** to point **b** (Fig. 2.39). *Question:* How much work will you have to do? At any point along the path, the electric force on Q is $\mathbf{F} = QE$; the force *you* must exert, in opposition to this electrical force, is $-QE$. (If the sign bothers you, think about lifting a brick: gravity exerts a force mg *downward*, but *you* exert a force mg *upward*. Of course, you *could* apply an even greater force – then the brick would accelerate, and part of your effort would be “wasted” generating kinetic energy. What we’re interested in here is the *minimum* force you must exert to do the job.) The work you do is therefore

$$W = \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{l} = -Q \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} = Q[V(\mathbf{b}) - V(\mathbf{a})].$$

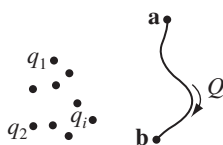


Fig. 2.39

Notice that the answer is independent of the path you take from **a** to **b**; in mechanics, then, we would call the electrostatic force “conservative.” Dividing through by Q , we have

$$V(\mathbf{b}) - V(\mathbf{a}) = \frac{W}{Q}. \quad (2.38)$$

In words, *the potential difference between points **a** and **b** is equal to the work it takes, per unit charge, to carry a particle from **a** to **b***. In particular, if you want to bring Q in from far away and stick it at point **r**, the work you must do is

$$W = Q[V(\mathbf{r}) - V(\infty)],$$

so, if you have set the reference point at infinity,

$$W = QV(\mathbf{r}). \quad (2.39)$$

In this sense, *potential* is potential *energy* (the work required to create the system) *per unit charge* (just as the *field* is the *force* per unit charge).

2.4.2 The Energy of a Point Charge Distribution

How much work would it take to assemble an entire *collection* of point charges? Imagine bringing in the charges, one by one, from far away (Fig. 2.40). The first charge, q_1 , takes *no* work, since there is no field yet to fight against. Now bring in q_2 . According to Eq. 2.39, this will cost you $q_2 V_1(\mathbf{r}_2)$, where V_1 is the potential due to q_1 , and $\mathbf{r}_2$ is the place we’re putting q_2 :

$$W_2 = \frac{1}{4\pi\epsilon_0} q_2 \left(\frac{q_1}{r_{12}} \right)$$

(r_{12} is the distance between q_1 and q_2 once they are in position). As you bring in each charge, nail it down in its final location, so it doesn’t move when you bring in the next charge. Now bring in q_3 ; this requires work $q_3 V_{1,2}(\mathbf{r}_3)$, where $V_{1,2}$ is the potential due to charges q_1 and q_2 , namely, $(1/4\pi\epsilon_0)(q_1/r_{13} + q_2/r_{23})$. Thus

$$W_3 = \frac{1}{4\pi\epsilon_0} q_3 \left(\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}} \right).$$

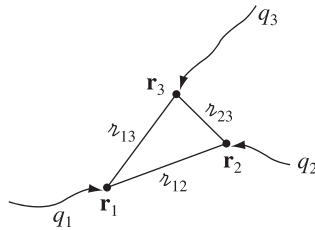


Fig. 2.40

Similarly, the extra work to bring in q_4 will be

$$W_4 = \frac{1}{4\pi\epsilon_0} q_4 \left(\frac{q_1}{r_{14}} + \frac{q_2}{r_{24}} + \frac{q_3}{r_{34}} \right).$$

The *total* work necessary to assemble the first four charges, then, is

$$W = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_1 q_4}{r_{14}} + \frac{q_2 q_3}{r_{23}} + \frac{q_2 q_4}{r_{24}} + \frac{q_3 q_4}{r_{34}} \right).$$

You see the general rule: take the product of each pair of charges, divide by their separation distance, and add it all up:

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \sum_{j>i}^n \frac{q_i q_j}{r_{ij}}. \quad (2.40)$$

The stipulation $j > i$ is to remind you not to count the same pair twice. A nicer way to accomplish this is *intentionally* to count each pair twice, and then divide by 2:

$$W = \frac{1}{8\pi\epsilon_0} \sum_{i=1}^n \sum_{j \neq i}^n \frac{q_i q_j}{r_{ij}} \quad (2.41)$$

(we must still exclude $i = j$, of course). Notice that in this form the answer plainly does not depend on the *order* in which you assemble the charges, since every pair occurs in the sum.

Finally, let's pull out the factor q_i :

$$W = \frac{1}{2} \sum_{i=1}^n q_i \left(\sum_{j \neq i}^n \frac{1}{4\pi\epsilon_0} \frac{q_j}{r_{ij}} \right).$$

The term in parentheses is the potential at point $\mathbf{r}_i$ (the position of q_i) due to all the *other* charges – *all* of them, now, not just the ones that were present at some stage during the assembly. Thus,

$$W = \frac{1}{2} \sum_{i=1}^n q_i V(\mathbf{r}_i). \quad (2.42)$$

That's how much work it takes to assemble a configuration of point charges; it's also the amount of work you'd get back if you dismantled the system. In the meantime, it represents energy stored in the configuration ("potential" energy, if you insist, though for obvious reasons I prefer to avoid that word in this context).

Problem 2.32.

- (a) Three charges are situated at the corners of a square (side length a), as shown in Fig. 2.41. How much work does it take to bring in another charge, $+q$, from far away and place it in the fourth corner?
- (b) How much work does it take to assemble the entire configuration of four charges?

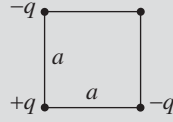


Fig. 2.41

Problem 2.33. Two positive point charges, q_A and q_B (masses m_A and m_B) are at rest, held together by a massless string of length a . Now the string is cut, and the particles fly off in opposite directions. How fast is each one going, when they are far apart?

Problem 2.34. Consider an infinite chain of point charges, $\pm q$ (with alternating signs), strung out along the x -axis, each a distance a from its nearest neighbors. Find the work per particle required to assemble this system. [Partial answer: $-\alpha q^2/(4\pi\epsilon_0 a)$, for some dimensionless number α ; your problem is to determine α . It is known as the **Madelung constant**. Calculating the Madelung constant for two- and three-dimensional arrays is much more subtle and difficult.]

2.4.3 The Energy of a Continuous Charge Distribution

For a volume charge density ρ , Eq. 2.42 becomes

$$W = \frac{1}{2} \int \rho V d\tau. \quad (2.43)$$

(The corresponding integrals for line and surface charges would be $\int \lambda V dl$ and $\int \sigma V da$.) There is a lovely way to rewrite this result, in which ρ and V are eliminated in favor of $\mathbf{E}$. First use Gauss's law to express ρ in terms of $\mathbf{E}$:

$$\rho = \epsilon_0 \nabla \cdot \mathbf{E}, \quad \text{so } W = \frac{\epsilon_0}{2} \int (\nabla \cdot \mathbf{E}) V d\tau.$$

Now use integration by parts (Eq. 1.59) to transfer the derivative from $\mathbf{E}$ to V :

$$W = \frac{\epsilon_0}{2} \left[- \int_V \mathbf{E} \cdot (\nabla V) d\tau + \oint_S V \mathbf{E} \cdot d\mathbf{a} \right].$$

But $\nabla V = -\mathbf{E}$, so

$$W = \frac{\epsilon_0}{2} \left(\int_V E^2 d\tau + \oint_S V \mathbf{E} \cdot d\mathbf{a} \right). \quad (2.44)$$

But what volume $\mathcal{V}$ is this we're integrating over? Let's go back to the formula we started with, Eq. 2.43. From its derivation, it is clear that we should integrate over the region where the charge is located. But actually, any *larger* volume would do just as well: the "extra" territory we throw in will contribute nothing to the integral, since $\rho = 0$ out there. With this in mind, we return to Eq. 2.44. What happens *here*, as we enlarge the volume beyond the minimum necessary to trap all the charge? Well, the integral of

E^2 can only increase (the integrand being positive); evidently the surface integral must decrease correspondingly, to leave the sum intact. (In fact, at large distances from the charge, E goes like $1/r^2$ and V like $1/r$, while the surface area grows like r^2 ; roughly speaking, then, the surface integral goes down like $1/r$.) Please understand: Eq. 2.44 gives you the correct energy W , *whatever* volume you use (as long as it encloses all the charge), but the contribution from the volume integral goes up, and that of the surface integral goes down, as you take larger and larger volumes. In particular, why not integrate over *all* space? Then the surface integral goes to zero, and we are left with

$$W = \frac{\epsilon_0}{2} \int E^2 d\tau \quad (\text{all space}). \quad (2.45)$$

Example 2.9. Find the energy of a uniformly charged spherical shell of total charge q and radius R .

Solution 1. Use Eq. 2.43, in the version appropriate to surface charges:

$$W = \frac{1}{2} \int \sigma V da.$$

Now, the potential at the surface of this sphere is $(1/4\pi\epsilon_0)q/R$, a constant (Ex. 2.7), so

$$W = \frac{1}{8\pi\epsilon_0} \frac{q}{R} \int \sigma da = \frac{1}{8\pi\epsilon_0} \frac{q^2}{R}.$$

Solution 2. Use Eq. 2.45. Inside the sphere, $\mathbf{E} = \mathbf{0}$; outside,

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}, \quad \text{so } E^2 = \frac{q^2}{(4\pi\epsilon_0)^2 r^4}.$$

Therefore,

$$\begin{aligned} W &= \frac{\epsilon_0}{2(4\pi\epsilon_0)^2} \int_{\text{outside}} \left(\frac{q^2}{r^4}\right) (r^2 \sin \theta dr d\theta d\phi) \\ &= \frac{1}{32\pi^2\epsilon_0} q^2 4\pi \int_R^\infty \frac{1}{r^2} dr = \frac{1}{8\pi\epsilon_0} \frac{q^2}{R}. \end{aligned}$$

Problem 2.35. Find the energy stored in a uniformly charged *solid* sphere of radius R and charge q . Do it three different ways:

- (a) Use Eq. 2.43. You found the potential in Prob. 2.22.
- (b) Use Eq. 2.45. Don't forget to integrate over *all* space.
- (c) Use Eq. 2.44. Take a spherical volume of radius a . What happens as $a \rightarrow \infty$?

Problem 2.36. Here is a fourth way of computing the energy of a uniformly charged solid sphere (you might call it the **method of assembly**): Build it up like a snowball, layer by layer, each time bringing in an infinitesimal charge dq from far away and smearing it uniformly over the surface, thereby increasing the radius. How much work dW does it take to advance the radius from r to $r + dr$? Integrate your result to find the work necessary to create the entire sphere of radius R and total charge q .

2.4.4 Comments on Electrostatic Energy

(i) **A perplexing “inconsistency.”** Equation 2.45 clearly implies that the energy of a stationary charge distribution is always *positive*. On the other hand, Eq. 2.42 (from which 2.45 was derived), can be positive or negative. For instance, according to Eq. 2.42, the energy of two equal but opposite charges a distance z apart is $-(1/4\pi\epsilon_0)(q^2/z)$. What’s gone wrong? Which equation is correct?

The answer is that *both* are correct, but they address slightly different questions. Equation 2.42 does not take into account the work necessary to *make* the point charges in the first place; we *started* with point charges and simply found the work required to bring them together. This is wise strategy, since Eq. 2.45 indicates that the energy of a point charge is in fact *infinite*:

$$W = \frac{\epsilon_0}{2(4\pi\epsilon_0)^2} \int \left(\frac{q^2}{r^4}\right) (r^2 \sin \theta \, dr \, d\theta \, d\phi) = \frac{q^2}{8\pi\epsilon_0} \int_0^\infty \frac{1}{r^2} \, dr = \infty.$$

Equation 2.45 is more *complete*, in the sense that it tells you the *total* energy stored in a charge configuration, but Eq. 2.42 is more appropriate when you’re dealing with point charges, because we prefer (for good reason!) to leave out that portion of the total energy that is attributable to the fabrication of the point charges themselves. In practice, after all, the point charges (electrons, say) are *given* to us ready-made; all *we* do is move them around. Since we did not put them together, and we cannot take them apart, it is immaterial how much work the process would involve. (Still, the infinite energy of a point charge is a recurring source of embarrassment for electromagnetic theory, afflicting the quantum version as well as the classical. We shall return to the problem in Chapter 11.)

Now, you may wonder where the inconsistency crept into an ostensibly watertight derivation. The “flaw” lies between Eqs. 2.42 and 2.43: in the former, $V(\mathbf{r}_i)$ represents the potential due to all the *other* charges *but not* q_i , whereas in the latter, $V(\mathbf{r})$ is the *full* potential. For a *continuous* distribution, there is no distinction, since the amount of charge *right at the point* $\mathbf{r}$ is vanishingly small, and its contribution to the potential is zero. But in the presence of point charges you’d better stick with Eq. 2.42.

(ii) **Where is the energy stored?** Equations 2.43 and 2.45 offer two different ways of calculating the same thing. The first is an integral over

the charge distribution; the second is an integral over the field. These can involve completely different regions. For instance, in the case of the spherical shell (Ex. 2.9), the charge is confined to the surface, whereas the electric field is everywhere *outside* this surface. Where *is* the energy, then? Is it stored in the field, as Eq. 2.45 suggests, or is it stored in the charge, as Eq. 2.43 implies? At the present stage this is simply an unanswerable question: I can tell you what the total energy is, and I can provide you with several different ways to calculate it, but it is pointless to worry about *where* the energy is located. In the context of radiation theory (Chapter 11) it is useful (and in general relativity it is *essential*) to regard the energy as stored in the field, with a density

$$\frac{\epsilon_0}{2} E^2 = \text{energy per unit volume.} \quad (2.46)$$

But in electrostatics one could just as well say it is stored in the charge, with a density $\frac{1}{2}\rho V$. The difference is purely a question of bookkeeping.¹²

(iii) The superposition principle. Because electrostatic energy is *quadratic* in the fields, it does *not* obey a superposition principle. The energy of a compound system is *not* the sum of the energies of its parts considered separately – there are also “cross terms”:

$$\begin{aligned} W_{\text{tot}} &= \frac{\epsilon_0}{2} \int E^2 d\tau = \frac{\epsilon_0}{2} \int (\mathbf{E}_1 + \mathbf{E}_2)^2 d\tau \\ &= \frac{\epsilon_0}{2} \int (E_1^2 + E_2^2 + 2\mathbf{E}_1 \cdot \mathbf{E}_2) d\tau \\ &= W_1 + W_2 + \epsilon_0 \int \mathbf{E}_1 \cdot \mathbf{E}_2 d\tau. \end{aligned} \quad (2.47)$$

For example, if you double the charge everywhere, you *quadruple* the total energy.

Problem 2.37. Consider two concentric spherical shells, of radii a and b . Suppose the inner one carries a charge q , and the outer one a charge $-q$ (both of them uniformly distributed over the surface). Calculate the energy of this configuration, (a) using Eq. 2.45, and (b) using Eq. 2.47 and the results of Ex. 2.9.

Problem 2.38. Find the interaction energy ($\epsilon_0 \int \mathbf{E}_1 \cdot \mathbf{E}_2 d\tau$ in Eq. 2.47) for two point charges, q_1 and q_2 , a distance a apart. [Hint: Put q_1 at the origin and q_2 on the z -axis; use spherical coordinates, and do the r integral first.]

¹² Ever since Maxwell, physicists have agonized over the “location” of electrostatic energy. Feynman, in his inimitable way, lays the matter to rest: “Who cares? What’s the meaning of such a question? ... If we restrict ourselves to electrostatics there is really no way to tell where the energy is located.” R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, Vol. 2, Addison-Wesley, Reading, MA (1967), Chap. 8, p. 8.

2.5 Conductors

2.5.1 Basic Properties

In an **insulator**, such as glass or rubber, each electron is on a short leash, attached to a particular atom. In a metallic **conductor**, by contrast, one or more electrons per atom are free to roam. (In liquid conductors such as salt water, it is ions that do the moving.) A *perfect* conductor would contain an *unlimited* supply of free charges. In real life there are no perfect conductors, but metals come pretty close, for most purposes.

From this definition, the basic electrostatic properties of ideal conductors immediately follow:

(i) **$E = 0$ inside a conductor.** Why? Because if there *were* any field, those free charges would move, and it wouldn't be *electrostatics* any more. Hmm ... that's hardly a satisfactory explanation; maybe all it proves is that you can't have electrostatics when conductors are present. We had better examine what happens when you put a conductor into an external electric field E_0 (Fig. 2.42). Initially, the field will drive any free positive charges to the right, and negative ones to the left. (In practice, it's the negative charges – electrons – that do the moving, but when they depart, the right-hand side is left with a net positive charge – the stationary nuclei – so it doesn't really matter which charges move; the effect is the same.) When they come to the edge of the material, the charges pile up: plus on the right-hand side, minus on the left. Now, these **induced charges** produce a field of their own, E_1 , which, as you can see from the figure, is in the *opposite direction* to E_0 . That's the crucial point, for it means that the field of the induced charges *tends to cancel the original field*. Charge will continue to flow until this cancellation is complete, and the resultant field inside the conductor is precisely zero.¹³ The whole process is practically instantaneous.

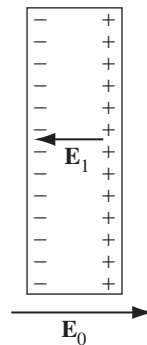


Fig. 2.42

¹³ Outside the conductor the field is *not* zero, for here E_0 and E_1 do *not* tend to cancel.

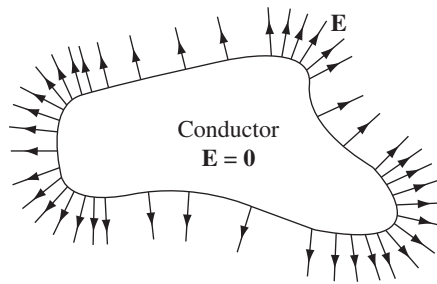


Fig. 2.43

(ii) $\rho = 0$ **inside a conductor.** This follows from Gauss's law: $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$. If $\mathbf{E}$ is zero, so also is ρ . There is still charge around, but exactly as much plus as minus, so the *net* charge density in the interior is zero.

(iii) **Any net charge resides on the surface.** That's the only place left.

(iv) **A conductor is an equipotential.** For if $\mathbf{a}$ and $\mathbf{b}$ are any two points within (or at the surface of) a given conductor, $V(\mathbf{b}) - V(\mathbf{a}) = -\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} = 0$, and hence $V(\mathbf{a}) = V(\mathbf{b})$.

(v) **$\mathbf{E}$ is perpendicular to the surface, just outside a conductor.** Otherwise, as in (i), charge will immediately flow around the surface until it kills off the tangential component (Fig. 2.43). (*Perpendicular* to the surface, charge cannot flow, of course, since it is confined to the conducting object.)

I think it is astonishing that the charge on a conductor flows to the surface. Because of their mutual repulsion, the charges naturally spread out as much as possible, but for *all* of them to go to the surface seems like a waste of the interior space. Surely we could do better, from the point of view of making each charge as far as possible from its neighbors, to sprinkle *some* of them throughout the volume Well, it simply *is not so*. You do best to put *all* the charge on the surface, and this is true regardless of the size or shape of the conductor.¹⁴

The problem can also be phrased in terms of energy. Like any other free dynamical system, the charge on a conductor will seek the configuration that minimizes its potential energy. Property (iii) asserts that the electrostatic energy of a solid object (with specified shape and total charge) is a minimum when that charge is spread over the surface. For instance, the energy of a sphere is $(1/8\pi\epsilon_0)(q^2/R)$ if the charge is uniformly distributed over the surface, as we found in Ex. 2.9, but it is greater, $(3/20\pi\epsilon_0)(q^2/R)$, if the charge is uniformly distributed throughout the volume (Prob. 2.35).

¹⁴ By the way, the one- and two-dimensional analogs are quite different: The charge on a conducting *disk* does *not* all go to the perimeter (R. Friedberg, *Am. J. Phys.* **61**, 1084 (1993)), nor does the charge on a conducting *needle* go to the two ends (D. J. Griffiths and Y. Li, *Am. J. Phys.* **64**, 706 (1996)) – see Prob. 2.58. Moreover, if the exponent of r in Coulomb's law were not precisely 2, the charge on a solid conductor would not all go to the surface – see D. J. Griffiths and D. Z. Uvanovic, *Am. J. Phys.* **69**, 435 (2001), and Prob. 2.55g.

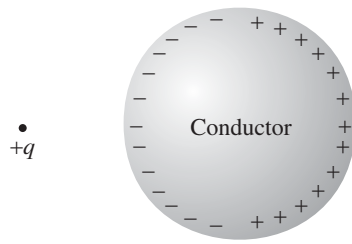


Fig. 2.44

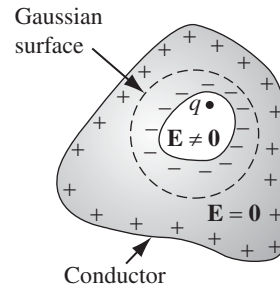


Fig. 2.45

2.5.2 Induced Charges

If you hold a charge $+q$ near an uncharged spherical conductor (Fig. 2.44), the two will attract one another. The reason for this is that q will pull minus charges over to the near side and repel plus charges to the far side. (Another way to think of it is that the charge moves around in such a way as to kill off the field of q for points inside the conductor, where the total field must be zero.) Since the negative induced charge is closer to q , there is a net force of attraction. (In Chapter 3 we shall calculate this force explicitly, for the case of a spherical conductor.)

When I speak of the field, charge, or potential “inside” a conductor, I mean in the “meat” of the conductor; if there is some hollow *cavity* in the conductor, and within that cavity you put some charge, then the field *in the cavity* will *not* be zero. But in a remarkable way the cavity and its contents are electrically isolated from the outside world by the surrounding conductor (Fig. 2.45). No external fields penetrate the conductor; they are canceled at the outer surface by the induced charge there. Similarly, the field due to charges within the cavity is canceled, for all exterior points, by the induced charge on the inner surface. However, the compensating charge left over on the *outer* surface of the conductor effectively “communicates” the presence of q to the outside world. The total charge induced on the cavity wall is equal and opposite to the charge inside, for if we surround the cavity with a Gaussian surface, all points of which are in the conductor (Fig. 2.45), $\oint \mathbf{E} \cdot d\mathbf{a} = 0$, and hence (by Gauss’s law) the net enclosed charge must be zero. But $Q_{\text{enc}} = q + q_{\text{induced}}$, so $q_{\text{induced}} = -q$. And if the conductor as a whole is electrically neutral, there must be a charge $+q$ on its outer surface.

Example 2.10. An uncharged spherical conductor, centered at the origin, has a cavity of some weird shape carved out of it (Fig. 2.46). Somewhere within the cavity is a charge q . *Question:* What is the field outside the sphere?

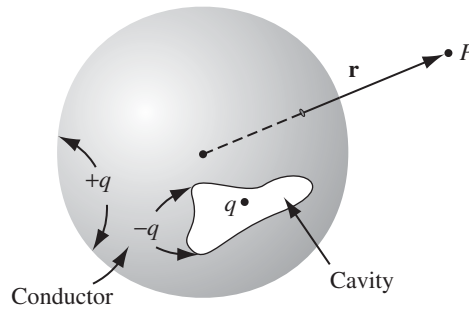


Fig. 2.46

Solution. At first glance, it would appear that the answer depends on the shape of the cavity and the location of the charge. But that's wrong: the answer is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

regardless. The conductor conceals from us all information concerning the nature of the cavity, revealing only the total charge it contains. How can this be? Well, the charge $+q$ induces an opposite charge $-q$ on the wall of the cavity, which distributes itself in such a way that its field cancels that of q , for all points exterior to the cavity. Since the conductor carries no net charge, this leaves $+q$ to distribute itself uniformly over the surface of the sphere. (It's *uniform* because the asymmetrical influence of the point charge $+q$ is negated by that of the induced charge $-q$ on the inner surface.) For points outside the sphere, then, the only thing that survives is the field of the leftover $+q$, uniformly distributed over the outer surface.

It may occur to you that in one respect this argument is open to challenge: There are actually *three* fields at work here: $\mathbf{E}_q$, $\mathbf{E}_{\text{induced}}$, and $\mathbf{E}_{\text{leftover}}$. All we know for certain is that the sum of the three is zero inside the conductor, yet I claimed that the first two *alone* cancel, while the third is separately zero there. Moreover, even if the first two cancel *within* the conductor, who is to say they still cancel for points *outside*? They do not, after all, cancel for points *inside* the cavity. I cannot give you a completely satisfactory answer at the moment, but this much at least is true: There *exists* a way of distributing $-q$ over the inner surface so as to cancel the field of q at *all* points exterior to the cavity. For that same cavity could have been carved out of a *huge* spherical conductor with a radius of 27 miles, or light years, or whatever. In that case, the leftover $+q$ on the outer surface is simply too far away to produce a significant field, and the other two fields would *have* to accomplish the cancellation by themselves. So we know they *can* do it ... but are we sure they *choose* to? Perhaps for small spheres nature

prefers some complicated three-way cancellation. Nope: As we'll see in the uniqueness theorems of Chapter 3, electrostatics is very stingy with its options; there is always precisely one way – no more – of distributing the charge on a conductor so as to make the field inside zero. Having found a *possible* way, we are guaranteed that no alternative exists, even in principle.

If a cavity surrounded by conducting material is itself empty of charge, then the field within the cavity is zero. For any field line would have to begin and end on the cavity wall, going from a plus charge to a minus charge (Fig. 2.47). Letting that field line be part of a closed loop, the rest of which is entirely inside the conductor (where $\mathbf{E} = \mathbf{0}$), the integral $\oint \mathbf{E} \cdot d\mathbf{l}$ is distinctly *positive*, in violation of Eq. 2.19. It follows that $\mathbf{E} = \mathbf{0}$ within an *empty* cavity, and there is in fact *no* charge on the surface of the cavity. (This is why you are relatively safe inside a metal car during a thunderstorm – you may get *cooked*, if lightning strikes, but you will not be *electrocuted*. The same principle applies to the placement of sensitive apparatus inside a grounded **Faraday cage**, to shield out stray electric fields. In practice, the enclosure doesn't even have to be a solid conductor – chicken wire will often suffice.)

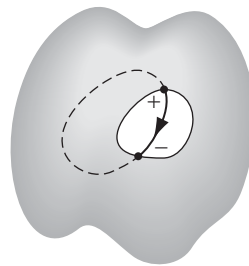


Fig. 2.47

Problem 2.39. A metal sphere of radius R , carrying charge q , is surrounded by a thick concentric metal shell (inner radius a , outer radius b , as in Fig. 2.48). The shell carries no net charge.

- (a) Find the surface charge density σ at R , at a , and at b .
- (b) Find the potential at the center, using infinity as the reference point.
- (c) Now the outer surface is touched to a grounding wire, which drains off charge and lowers its potential to zero (same as at infinity). How do your answers to (a) and (b) change?

Problem 2.40. Two spherical cavities, of radii a and b , are hollowed out from the interior of a (neutral) conducting sphere of radius R (Fig. 2.49). At the center of each cavity a point charge is placed – call these charges q_a and q_b .

- (a) Find the surface charge densities σ_a , σ_b , and σ_R .
- (b) What is the field outside the conductor?

- (c) What is the field within each cavity?
 (d) What is the force on q_a and q_b ?
 (e) Which of these answers would change if a third charge, q_c , were brought near the conductor (but outside it)?

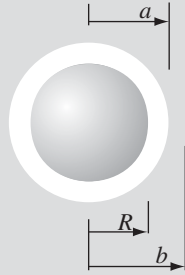


Fig. 2.48

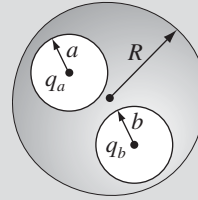


Fig. 2.49

Problem 2.41.

- (a) A point charge q is inside a cavity in an uncharged conductor (Fig. 2.45). Is the force on q necessarily zero?¹⁵
 (b) Is the force between a point charge and a nearby uncharged conductor always attractive?¹⁶

2.5.3 Surface Charge and the Force on a Conductor

Because the field inside a conductor is zero, Eq. 2.33 requires that the field immediately *outside* is

$$\mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}, \quad (2.48)$$

consistent with our earlier conclusion that the field is normal to the surface. In terms of potential, Eq. 2.36 yields

$$\sigma = -\epsilon_0 \frac{\partial V}{\partial n}. \quad (2.49)$$

These equations enable you to calculate the surface charge on a conductor, if you can determine $\mathbf{E}$ or V ; we shall use them frequently in the next chapter.

In the presence of an electric field, any surface charge will experience a force; the force per unit area, $\mathbf{f}$, is $\sigma \mathbf{E}$. But there's a problem here, for the electric field is *discontinuous* at a surface charge, so what are we supposed to use: $\mathbf{E}_{\text{above}}$, $\mathbf{E}_{\text{below}}$, or something in between? The answer is that we should use the *average* of the two:

$$\mathbf{f} = \sigma \mathbf{E}_{\text{average}} = \frac{1}{2} \sigma (\mathbf{E}_{\text{above}} + \mathbf{E}_{\text{below}}). \quad (2.50)$$

¹⁵ This problem was suggested by Nelson Christensen.

¹⁶ See M. Levin and S. G. Johnson, *Am. J. Phys.* **79**, 843 (2011).

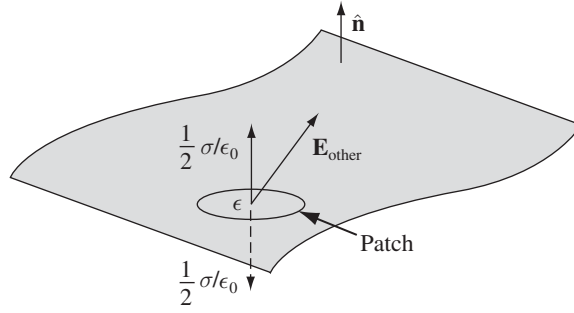


Fig. 2.50

Why the average? The reason is very simple, though the telling makes it sound complicated: Let's focus our attention on a tiny patch of surface surrounding the point in question (Fig. 2.50). (Make it small enough so it is essentially flat and the surface charge on it essentially constant.) The *total* field consists of two parts – that attributable to the patch itself, and that due to everything else (other regions of the surface, as well as any external sources that may be present):

$$\mathbf{E} = \mathbf{E}_{\text{patch}} + \mathbf{E}_{\text{other}}.$$

Now, the patch cannot exert a force on itself, any more than you can lift yourself by standing in a basket and pulling up on the handles. The force on the patch, then, is due exclusively to $\mathbf{E}_{\text{other}}$, and *this* suffers *no* discontinuity (if we removed the patch, the field in the “hole” would be perfectly smooth). The discontinuity is due entirely to the charge on the patch, which puts out a field $(\sigma/2\epsilon_0)$ on either side, pointing away from the surface. Thus,

$$\begin{aligned}\mathbf{E}_{\text{above}} &= \mathbf{E}_{\text{other}} + \frac{\sigma}{2\epsilon_0} \hat{n}, \\ \mathbf{E}_{\text{below}} &= \mathbf{E}_{\text{other}} - \frac{\sigma}{2\epsilon_0} \hat{n},\end{aligned}$$

and hence

$$\mathbf{E}_{\text{other}} = \frac{1}{2}(\mathbf{E}_{\text{above}} + \mathbf{E}_{\text{below}}) = \mathbf{E}_{\text{average}}.$$

Averaging is really just a device for removing the contribution of the patch itself.

That argument applies to *any* surface charge; in the particular case of a conductor, the field is zero inside and $(\sigma/\epsilon_0)\hat{n}$ outside (Eq. 2.48), so the average is $(\sigma/2\epsilon_0)\hat{n}$, and the force per unit area is

$$\mathbf{f} = \frac{1}{2\epsilon_0} \sigma^2 \hat{n}. \quad (2.51)$$

This amounts to an outward **electrostatic pressure** on the surface, tending to draw the conductor into the field, regardless of the sign of σ . Expressing the pressure in terms of the field just outside the surface,

$$P = \frac{\epsilon_0}{2} E^2. \quad (2.52)$$

Problem 2.42. Two large metal plates (each of area A) are held a small distance d apart. Suppose we put a charge Q on each plate; what is the electrostatic pressure on the plates?

Problem 2.43. A metal sphere of radius R carries a total charge Q . What is the force of repulsion between the “northern” hemisphere and the “southern” hemisphere?

2.5.4 Capacitors

Suppose we have *two* conductors, and we put charge $+Q$ on one and $-Q$ on the other (Fig. 2.51). Since V is constant over a conductor, we can speak unambiguously of the potential difference between them:

$$V = V_+ - V_- = - \int_{(-)}^{(+)} \mathbf{E} \cdot d\mathbf{l}.$$

We don’t know how the charge distributes itself over the two conductors, and calculating the field would be a nightmare if their shapes are complicated, but this much we *do* know: $\mathbf{E}$ is *proportional* to Q , for $\mathbf{E}$ is given by Coulomb’s law:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\rho}{r^2} \hat{\mathbf{r}} d\tau,$$

so if you double ρ , you double $\mathbf{E}$. [Wait a minute! How do we know that doubling Q (and also $-Q$) simply doubles ρ ? Maybe the charge *moves around* into a completely different configuration, quadrupling ρ in some places and halving it in others, just so the *total* charge on each conductor is doubled. The *fact* is that this concern is unwarranted – doubling Q *does* double ρ everywhere; it *doesn’t* shift the charge around. The proof of this will come in Chapter 3; for now you’ll just have to trust me.]

Since $\mathbf{E}$ is proportional to Q , so also is V . The constant of proportionality is called the **capacitance** of the arrangement:

$$C \equiv \frac{Q}{V}. \quad (2.53)$$

Capacitance is a purely geometrical quantity, determined by the sizes, shapes, and separation of the two conductors. In SI units, C is measured in **farads** (F); a farad is a coulomb-per-volt. Actually, this turns out to be inconveniently large; more practical units are the microfarad (10^{-6} F) and the picofarad (10^{-12} F).



Fig. 2.51

Notice that V is, by definition, the potential of the *positive* conductor minus that of the negative one; likewise, Q is the charge on the *positive* conductor. Accordingly, capacitance is an intrinsically positive quantity. (By the way, you will occasionally hear someone speak of the capacitance of a *single* conductor. In this case the “second conductor,” with the negative charge, is an imaginary spherical shell of infinite radius surrounding the one conductor. It contributes nothing to the field, so the capacitance is given by Eq. 2.53, where V is the potential with infinity as the reference point.)

Example 2.11. Find the capacitance of a **parallel-plate capacitor** consisting of two metal surfaces of area A held a distance d apart (Fig. 2.52).

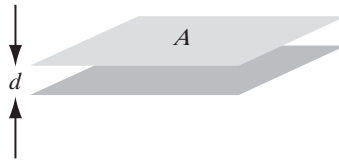


Fig. 2.52

Solution. If we put $+Q$ on the top and $-Q$ on the bottom, they will spread out uniformly over the two surfaces, provided the area is reasonably large and the separation small.¹⁷ The surface charge density, then, is $\sigma = Q/A$ on the top plate, and so the field, according to Ex. 2.6, is $(1/\epsilon_0)Q/A$. The potential difference between the plates is therefore

$$V = \frac{Q}{A\epsilon_0}d,$$

and hence

$$C = \frac{A\epsilon_0}{d}. \quad (2.54)$$

If, for instance, the plates are square, with sides 1 cm long, and they are held 1 mm apart, then the capacitance is 9×10^{-13} F.

Example 2.12. Find the capacitance of two concentric spherical metal shells, with radii a and b .

Solution. Place charge $+Q$ on the inner sphere, and $-Q$ on the outer one. The field between the spheres is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}},$$

¹⁷ The *exact* solution is not easy – even for the simpler case of circular plates. See G. T. Carlson and B. L. Illman, *Am. J. Phys.* **62**, 1099 (1994).

so the potential difference between them is

$$V = - \int_b^a \mathbf{E} \cdot d\mathbf{l} = - \frac{Q}{4\pi\epsilon_0} \int_b^a \frac{1}{r^2} dr = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right).$$

As promised, V is proportional to Q ; the capacitance is

$$C = \frac{Q}{V} = 4\pi\epsilon_0 \frac{ab}{(b-a)}.$$

To “charge up” a capacitor, you have to remove electrons from the positive plate and carry them to the negative plate. In doing so, you fight against the electric field, which is pulling them back toward the positive conductor and pushing them away from the negative one. How much work does it take, then, to charge the capacitor up to a final amount Q ? Suppose that at some intermediate stage in the process the charge on the positive plate is q , so that the potential difference is q/C . According to Eq. 2.38, the work you must do to transport the next piece of charge, dq , is

$$dW = \left(\frac{q}{C} \right) dq.$$

The total work necessary, then, to go from $q = 0$ to $q = Q$, is

$$W = \int_0^Q \left(\frac{q}{C} \right) dq = \frac{1}{2} \frac{Q^2}{C},$$

or, since $Q = CV$,

$$W = \frac{1}{2} CV^2, \quad (2.55)$$

where V is the final potential of the capacitor.

Problem 2.44. Find the capacitance per unit length of two coaxial metal cylindrical tubes, of radii a and b (Fig. 2.53).



Fig. 2.53

Problem 2.45. Suppose the plates of a parallel-plate capacitor move closer together by an infinitesimal distance ϵ , as a result of their mutual attraction.

- (a) Use Eq. 2.52 to express the work done by electrostatic forces, in terms of the field E , and the area of the plates, A .

(b) Use Eq. 2.46 to express the energy lost by the field in this process.

(This problem is supposed to be easy, but it contains the embryo of an alternative derivation of Eq. 2.52, using conservation of energy.)

More Problems on Chapter 2

Problem 2.46. Find the electric field at a height z above the center of a square sheet (side length a) carrying a uniform surface charge σ . Check your result for the limiting cases $a \rightarrow \infty$ and $z \gg a$. [Answer: $(\sigma/2\epsilon_0)\{(4/\pi) \tan^{-1} \sqrt{1 + (a^2/2z^2)} - 1\}$.]

Problem 2.47. If the electric field in some region is given (in spherical coordinates) by the expression

$$\mathbf{E}(\mathbf{r}) = \frac{k}{r} [3\hat{\mathbf{r}} + 2 \sin \theta \cos \theta \sin \phi \hat{\boldsymbol{\theta}} + \sin \theta \cos \phi \hat{\boldsymbol{\phi}}],$$

for some constant k , what is the charge density? [Answer: $3k\epsilon_0(1 + \cos 2\theta \sin \phi)/r^2$.]

Problem 2.48. Find the net force that the southern hemisphere of a uniformly charged solid sphere exerts on the northern hemisphere. Express your answer in terms of the radius R and the total charge Q . [Answer: $(1/4\pi\epsilon_0)(3Q^2/16R^2)$.]

Problem 2.49. An inverted hemispherical bowl of radius R carries a uniform surface charge density σ . Find the potential difference between the “north pole” and the center. [Answer: $(R\sigma/2\epsilon_0)(\sqrt{2} - 1)$.]

Problem 2.50. A sphere of radius R carries a charge density $\rho(r) = kr$ (where k is a constant). Find the energy of the configuration. Check your answer by calculating it in at least two different ways. [Answer: $\pi k^2 R^7/7\epsilon_0$.]

Problem 2.51. The electric potential of some configuration is given by the expression

$$V(\mathbf{r}) = A \frac{e^{-\lambda r}}{r},$$

where A and λ are constants. Find the electric field $\mathbf{E}(\mathbf{r})$, the charge density $\rho(r)$, and the total charge Q . [Answer: $\rho = \epsilon_0 A(4\pi\delta^3(\mathbf{r}) - \lambda^2 e^{-\lambda r}/r)$.]

Problem 2.52. Find the potential on the rim of a uniformly charged disk (radius R , charge density σ). [Hint: First show that $V = k(\sigma R/\pi\epsilon_0)$, for some dimensionless number k , which you can express as an integral. Then evaluate k analytically, if you can, or by computer.]

! **Problem 2.53.** Two infinitely long wires running parallel to the x -axis carry uniform charge densities $+\lambda$ and $-\lambda$ (Fig. 2.54).

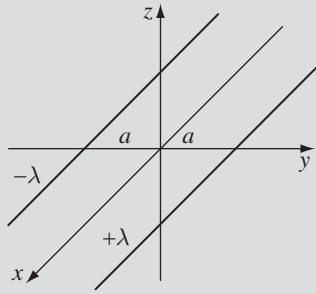


Fig. 2.54

- (a) Find the potential at any point (x, y, z) , using the origin as your reference.
- (b) Show that the equipotential surfaces are circular cylinders, and locate the axis and radius of the cylinder corresponding to a given potential V_0 .

! **Problem 2.54.** In a vacuum diode, electrons “boil” off a hot **cathode**, at potential zero, and accelerate across a gap to the **anode**, which is held at positive potential V_0 . The cloud of moving electrons within the gap (called **space charge**) quickly builds up to the point where it reduces the field at the surface of the cathode to zero. From then on, a steady current I flows between the plates.

Suppose the plates are large relative to the separation ($A \gg d^2$ in Fig. 2.55), so that edge effects can be neglected. Then V , ρ , and v (the speed of the electrons) are all functions of x alone.

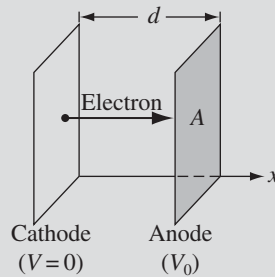


Fig. 2.55

- (a) Write Poisson's equation for the region between the plates.
- (b) Assuming the electrons start from rest at the cathode, what is their speed at point x , where the potential is $V(x)$?
- (c) In the steady state, I is independent of x . What, then, is the relation between ρ and v ?
- (d) Use these three results to obtain a differential equation for V , by eliminating ρ and v .
- (e) Solve this equation for V as a function of x , V_0 , and d . Plot $V(x)$, and compare it to the potential *without* space charge. Also, find ρ and v as functions of x .
- (f) Show that

$$I = KV_0^{3/2}, \quad (2.56)$$

and find the constant K . (Equation 2.56 is called the **Child–Langmuir law**. It holds for other geometries as well, whenever space charge limits the current. Notice that the space-charge-limited diode is *nonlinear* – it does not obey Ohm’s law.)

- ! **Problem 2.55.** Imagine that new and extraordinarily precise measurements have revealed an error in Coulomb’s law. The *actual* force of interaction between two point charges is found to be

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \left(1 + \frac{r}{\lambda}\right) e^{-(r/\lambda)} \hat{\mathbf{r}},$$

where λ is a new constant of nature (it has dimensions of length, obviously, and is a huge number – say half the radius of the known Universe – so the correction is small, which is why no one ever noticed the discrepancy before). You are charged with the task of reformulating electrostatics to accommodate the new discovery. Assume the principle of superposition still holds.

- What is the electric field of a charge distribution ρ (replacing Eq. 2.8)?
- Does this electric field admit a scalar potential? Explain briefly how you reached your conclusion. (No formal proof necessary – just a persuasive argument.)
- Find the potential of a point charge q – the analog to Eq. 2.26. (If your answer to (b) was “no,” better go back and change it!) Use ∞ as your reference point.
- For a point charge q at the origin, show that

$$\oint_S \mathbf{E} \cdot d\mathbf{a} + \frac{1}{\lambda^2} \int_V V d\tau = \frac{1}{\epsilon_0} q,$$

where S is the surface and V the volume of any sphere centered at q .

- Show that this result generalizes:

$$\oint_S \mathbf{E} \cdot d\mathbf{a} + \frac{1}{\lambda^2} \int_V V d\tau = \frac{1}{\epsilon_0} Q_{\text{enc}},$$

for *any* charge distribution. (This is the next best thing to Gauss’s law, in the new “electrostatics.”)

- Draw the triangle diagram (like Fig. 2.35) for this world, putting in all the appropriate formulas. (Think of Poisson’s equation as the formula for ρ in terms of V , and Gauss’s law (differential form) as an equation for ρ in terms of $\mathbf{E}$.)
- Show that *some* of the charge on a conductor distributes itself (uniformly!) over the volume, with the remainder on the surface. [Hint: $\mathbf{E}$ is still zero, inside a conductor.]

Problem 2.56. Suppose an electric field $\mathbf{E}(x, y, z)$ has the form

$$E_x = ax, \quad E_y = 0, \quad E_z = 0,$$

where a is a constant. What is the charge density? How do you account for the fact that the field points in a particular direction, when the charge density is uniform? [This is a more subtle problem than it looks, and worthy of careful thought.]

Problem 2.57. All of electrostatics follows from the $1/r^2$ character of Coulomb's law, together with the principle of superposition. An analogous theory can therefore be constructed for Newton's law of universal gravitation. What is the gravitational energy of a sphere, of mass M and radius R , assuming the density is uniform? Use your result to estimate the gravitational energy of the Sun (look up the relevant numbers). Note that the energy is *negative* – masses *attract*, whereas (like) electric charges *repel*. As matter “falls in,” to create the Sun, its kinetic energy is converted into other forms (typically thermal), and is eventually released as radiation. The Sun radiates at a rate of 3.86×10^{26} W; if all this came from gravitational energy, how long would the Sun last? [The Sun is in fact much older than that, so evidently this is *not* the source of its energy.¹⁸]

! **Problem 2.58.** We know that the charge on a conductor goes to the surface, but just how it distributes itself there is not easy to determine. One famous example in which the surface charge density can be calculated explicitly is the ellipsoid:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

In this case,¹⁹

$$\sigma = \frac{Q}{4\pi abc} \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right)^{-1/2}, \quad (2.57)$$

where Q is the total charge. By choosing appropriate values for a , b , and c , obtain (from Eq. 2.57) (a) the net (both sides) surface charge density $\sigma(r)$ on a circular disk of radius R ; (b) the net surface charge density $\sigma(x)$ on an infinite conducting “ribbon” in the xy plane, which straddles the y -axis from $x = -a$ to $x = a$ (let Λ be the total charge per unit length of ribbon); (c) the net charge per unit length $\lambda(x)$ on a conducting “needle,” running from $x = -a$ to $x = a$. In each case, sketch the graph of your result.



Problem 2.59.

- (a) Consider an equilateral triangle, inscribed in a circle of radius a , with a point charge q at each vertex. The electric field is zero (obviously) at the center, but (surprisingly) there are three *other* points inside the triangle where the field is zero. Where are they? [Answer: $r = 0.285a$ – you'll probably need a computer to get it.]
- (b) For a regular n -sided polygon there are n points (in addition to the center) where the field is zero.²⁰ Find their distance from the center for $n = 4$ and $n = 5$. What do you suppose happens as $n \rightarrow \infty$?

¹⁸ Lord Kelvin used this argument to counter Darwin's theory of evolution, which called for a much older Earth. Of course, we now know that the source of the Sun's energy is nuclear fusion, not gravity.

¹⁹ For the derivation (which is a real tour de force), see W. R. Smythe, *Static and Dynamic Electricity*, 3rd ed., Hemisphere, New York (1989), Sect. 5.02.

²⁰ S. D. Baker, *Am. J. Phys.* **52**, 265 (1984); D. Kiang and D. A. Tindall, *Am. J. Phys.* **53**, 593 (1985).

Problem 2.60. Prove or disprove (with a counterexample) the following:

Theorem. Suppose a conductor carrying a net charge Q , when placed in an external electric field $\mathbf{E}_e$, experiences a force $\mathbf{F}$; if the external field is now reversed ($\mathbf{E}_e \rightarrow -\mathbf{E}_e$), the force also reverses ($\mathbf{F} \rightarrow -\mathbf{F}$).

What if we stipulate that the external field is *uniform*?

Problem 2.61. A point charge q is at the center of an uncharged spherical conducting shell, of inner radius a and outer radius b . *Question:* How much work would it take to move the charge out to infinity (through a tiny hole drilled in the shell)? [*Answer:* $(q^2/8\pi\epsilon_0)(1/a - 1/b)$.]



Problem 2.62. What is the minimum-energy configuration for a system of N equal point charges placed on or inside a circle of radius R ?²¹ Because the charge on a conductor goes to the surface, you might think the N charges would arrange themselves (uniformly) around the circumference. Show (to the contrary) that for $N = 12$ it is better to place 11 on the circumference and 1 at the center. How about for $N = 11$ (is the energy lower if you put all 11 around the circumference, or if you put 10 on the circumference and 1 at the center)? [*Hint:* Do it numerically – you'll need at least 4 significant digits. Express all energies as multiples of $q^2/4\pi\epsilon_0 R$.]

! **Problem 2.63.**

- Find the potential on the rim of a uniformly charged circular disk of radius r and charge density σ . [*Hint:* Chop the disk into circular arcs of radius r and width dr , running from $-\theta$ to $+\theta$, and integrate over θ .]
- Use your result in (a) to calculate the energy of a uniformly charged disk of radius R and total charge Q .
- A circular *conducting* disk of radius R carries a total charge Q . In Prob. 2.58a you showed that the charge density is

$$\sigma(r) = \frac{Q}{2\pi R} \frac{1}{\sqrt{R^2 - r^2}}.$$

Find the potential of the disk.

- Use your result in (c) to calculate the energy of a conducting disk of radius R and total charge Q . Compare your answer to (b); which is smaller, and why?²²



Problem 2.64. Many programming languages include packages for drawing field diagrams (like Fig. 2.12a). You input $\mathbf{E}(x, y, z)$, and the program generates a figure showing arrows representing the field at an array of representative points. In Mathematica it's called `VectorPlot`. Suppose, for example, you want to sketch the field of a point charge q at the origin; in the xy plane the field is

²¹ M. G. Calkin, D. Kiang, and D. A. Tindall, *Am. J. Phys.* **55**, 157 (1987).

²² For applications of these results see T. M. Minter, *Eur. J. Phys.* **35**, 035022 (2014).

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{\hat{\mathbf{r}}}{r^2} = \frac{q}{4\pi\epsilon_0} \frac{\mathbf{r}}{r^3} = \frac{q}{4\pi\epsilon_0} \frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}}}{(x^2 + y^2)^{3/2}},$$

so, setting $q/4\pi\epsilon_0 \rightarrow 1$ and using the Mathematica syntax for defining a function, you type in

```
EF[{x_,y_}] := {x,y}/((x^2 + y^2)^(3/2))
```

and invoke

```
VectorPlot[EF[{x,y}], {x,-1,1}, {y,-1,1}]
```

(the terms in curly brackets specify the region of interest: $-1 \leq x \leq 1$ and $-1 \leq y \leq 1$).

(a) Try it. If you want to change the number of arrows drawn (say, to 10 per row/column), type

```
VectorPlot[EF[{x,y}], {x,-1,1}, {y,-1,1}, VectorPoints->10]
```

Notice that in Mathematica the *strength* of the field is represented by the *color* of the arrows; if you prefer to indicate it by their *length*, insert (after a comma, as always) `VectorScaling->Automatic`; to turn off the color coding altogether, also insert `VectorColorFunction->None`. If you want to draw in the x - and y -axes, insert `Axes->True`. Construct field diagrams for *two* charges, equal (as in Fig. 2.14), and opposite (as in Fig. 2.13).

(b) Mathematica can also draw the full three-dimensional field pattern, using

```
VectorPlot3D[EF[{x,y,z}], {x,-1,1}, {y,-1,1}, {z,-1,1},  
VectorScaling->Automatic]
```

Try it, for the field of a point charge at the origin. The result is rather congested; I prefer to insert `VectorMarkers->Arrow`, which makes it a little easier to read.



Problem 2.65. In Prob. 2.64 you drew field *vectors* at representative points; suppose instead you want to construct a field *line* passing through the point (x_0, y_0) . In Mathematica you can do it using `StreamPlot`:

```
StreamPlot[EF[{x,y}], {x,-1,1}, {y,-1,1}, StreamPoints->{{x0,y0}},  
StreamScale->Full, Axes->True]
```

(a) For a point charge at the origin, draw the field line passing through $(0.5, 0.5)$.

(b) Draw a typical field line for the configurations in Figs. 2.13 and 2.14.

(c) If you change the argument of `StreamPoints` to a *number* (for instance, writing `StreamPoints->20`), `StreamPlot` will draw 20 representative field lines. Try it for the configurations in part (b).



Problem 2.66. In Prob. 2.65 we used `StreamPlot` to draw electric field lines in two dimensions; it's quick and easy, but it doesn't tell you what the computer is actually *doing*. The basic idea is very simple: to get from one point on a field line ($\mathbf{r}(t)$) to the next ($\mathbf{r}(t + dt)$), you take a short step in the direction of the field:

$$\mathbf{r}(t + dt) = \mathbf{r}(t) + \mathbf{E}(\mathbf{r}(t)) dt.$$

Here t is any convenient parameter – perhaps arc length – that increases monotonically along the curve. In the limit of infinitesimal step size,

$$\frac{d\mathbf{r}}{dt} = \mathbf{E}.$$

This is an ordinary first-order differential equation for $\mathbf{r}(t)$, and Mathematica can solve it (numerically) using `NDSolve`. First specify your electric field ($\mathbf{EF}$), as in Prob. 2.64. Next, instruct the computer to calculate the coordinates $(x(t), y(t), z(t))$ of the field line:

```
FieldLine = NDSolve[{
  x'[t]==EF[{x[t],y[t],z[t]}][[1]],
  y'[t]==EF[{x[t],y[t],z[t]}][[2]],
  z'[t]==EF[{x[t],y[t],z[t]}][[3]],
  x[0]==x0,y[0]==y0,z[0]==z0},{x,y,z},{t,0,tmax}];
```

In the last line you will insert appropriate numbers for the starting point (x_0, y_0, z_0) , and choose how far out ($t_{\max}$) you want to go. Finally, to draw the field line, type

```
ParametricPlot3D[{x[t],y[t],z[t]}/.FieldLine,{t,0,tmax}]
```

Beware: If the field line wanders too close to a point charge, where the field is huge and rapidly changing, the numerical solution may be unstable, and you'll either get an error message or the plot will develop weird kinks.

- To make sure the code is working, reproduce the figures from Prob. 2.65a and b (but now in three dimensions). You might start out with $t_{\max} = 10$, but be prepared to adjust it as necessary.
- The field lines in part (a) all lie in a plane. Construct an electric field line that does *not* lie in a plane, using a configuration of three point charges.



Problem 2.67. A point charge q of mass m moves in an electric field $\mathbf{E}(x, y, z)$. It starts out at $\mathbf{x}_0 = (x_0, y_0, z_0)$, with velocity $\mathbf{v}_0 = (v_{x0}, v_{y0}, v_{z0})$, and you want to determine its trajectory. The following Mathematica code will do the job:

```
Etraj[x0_, v0_, m_, q_, dt_, n_, EF_] :=
Module[{X, V, x, v, xprev, j},
  X = Table[0., {i, 1, n}];
  X[[1]] = x0;
  V = Table[0., {i, 1, n}];
  V[[1]] = v0;
  x = x0;
  v = v0;
  For[j = 2, j <= n, j = j+1,
    xprev = x;
    x = x + v dt + 1/(2m) q EF[x] dt^2;
    v = 1/(2m) (q EF[xprev] + q EF[x]) dt;
    X[[j]] = x;
    V[[j]] = v;
  ];
  Return[{X, V}];
]
ShowTrajectory[pts_] :=
  Show[Graphics3D[Table[Line[{pts[[jj]], pts[[jj+1]]},
    {jj, 1, Length[pts]-1}]]]
```

- (a) `Etraj` takes as input the vectors $\mathbf{x}_0$ and $\mathbf{v}_0$, the values q, m, dt, n , and the electric field `EF`; it returns two lists: X (the position of the charge) and V (its velocity), at a succession of n times, separated by small intervals dt . The crucial steps are

$$\begin{aligned} \mathbf{x} &= \mathbf{x} + \mathbf{v} \, dt + \frac{1}{2m} q \, \mathbf{EF}[\mathbf{x}] \, dt^2; \\ \mathbf{v} &= \mathbf{v} + \frac{1}{m} (q \, \mathbf{EF}[\mathbf{x}_{\text{prev}}] + q \, \mathbf{EF}[\mathbf{x}]) \, dt; \end{aligned}$$

Explain what they do. To generate the plot, you type

```
{X1,V1} = Etraj[...];
ShowTrajectory[X1]
```

- (b) As an example, suppose the field is produced by a (negative) point charge Q at the origin. This is mathematically identical to the classical Kepler problem; the trajectories are *ellipses* (if the total energy – kinetic plus potential – is negative), *hyperbolic* (if the energy is positive), and *parabolic* (if it is zero). Generate an example of each case. (You might as well set $q = 1, m = 1$, and $Q/4\pi\epsilon_0 = -1$.)
- (c) For something a little more exotic, try *two* (equal, negative) source charges, at $(\pm 0.5, 0, 0)$; start q at the origin, with initial velocity $\mathbf{v} = (1.02, 1.10, 0)$, and set $dt = 1/20,000, n = 32,000$.

5.1 The Lorentz Force Law

5.1.1 Magnetic Fields

Remember the basic problem of classical electrodynamics: we have a collection of charges $q_1, q_2, q_3, \dots$ (the “source” charges), and we want to calculate the force they exert on some other charge Q (the “test” charge – Fig. 2.1). According to the principle of superposition, it is sufficient to find the force of a *single* source charge – the total is then the vector sum of all the individual forces. Up to now, we have confined our attention to the simplest case, *electrostatics*, in which the source charge is *at rest* (though the test charge need not be). The time has come to consider the forces between charges *in motion*.

To give you some sense of what is in store, imagine that I set up the following demonstration: Two wires hang from the ceiling, a few centimeters apart; when I turn on a current, so that it passes up one wire and back down the other, the wires jump apart – they repel one another (Fig. 5.1a). How do we explain this? You might suppose that the battery (or whatever drives the current) is actually charging up the wire, and that the force is simply due to the electrical repulsion of like charges. But this is incorrect. I could hold up a test charge near these wires, and there would be *no* force on it,¹ for the wires are in fact electrically neutral. (It’s true that electrons are flowing down the line – that’s what a current *is* – but there are just as many stationary plus charges as moving minus charges on any given segment.) Moreover, if I hook up my demonstration so as to make the current flow up *both* wires (Fig. 5.1b), they are found to *attract*!

Whatever force accounts for the attraction of parallel currents and the repulsion of antiparallel ones is *not* electrostatic in nature. It is our first encounter with a *magnetic* force. Whereas a *stationary* charge produces only an electric field $\mathbf{E}$ in the space around it, a *moving* charge generates, in addition, a magnetic field $\mathbf{B}$. In fact, magnetic fields are a lot easier to detect, in practice – all you need is a Boy Scout compass. How these devices work is irrelevant at the moment; it is enough to know that the needle points in the direction of the local magnetic field. Ordinarily, this means *north*, in response to the Earth’s magnetic field, but in the laboratory, where typical

¹ This is not precisely true, as we shall see in Prob. 7.45

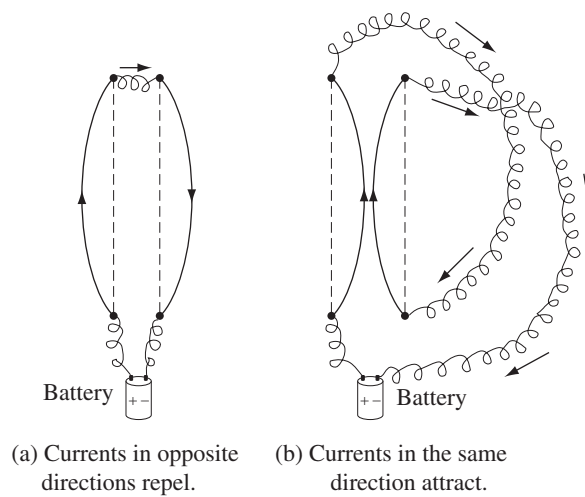


Fig. 5.1

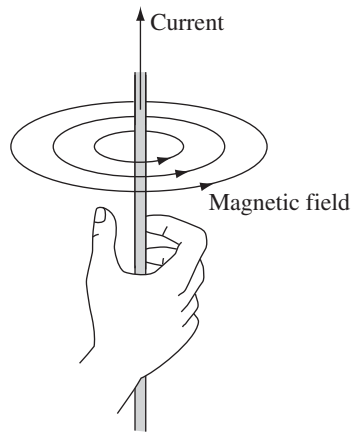


Fig. 5.2

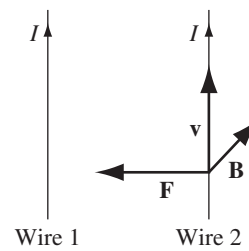


Fig. 5.3

fields may be hundreds of times stronger than that, the compass indicates the direction of whatever magnetic field is present.

Now, if you hold up a tiny compass in the vicinity of a current-carrying wire, you quickly discover a very peculiar thing: the field does not point *toward* the wire, nor *away* from it, but rather it *circles around the wire*. In fact, if you grab the wire with your right hand – thumb in the direction of the current – your fingers curl around in the direction of the magnetic field (Fig. 5.2). How can such a field lead to a force of attraction on a nearby parallel current? At the second wire, the magnetic field points *into the page* (Fig. 5.3), the current is *upward*, and yet the resulting force is *to the left*! It's going to take a strange law to account for these directions.

5.1.2 Magnetic Forces

In fact, this combination of directions is just right for a cross product: the magnetic force on a charge Q , moving with velocity $\mathbf{v}$ in a magnetic field $\mathbf{B}$, is²

$$\mathbf{F}_{\text{mag}} = Q(\mathbf{v} \times \mathbf{B}). \quad (5.1)$$

This is known as the **Lorentz force law**.³ In the presence of both electric *and* magnetic fields, the net force on Q would be

$$\mathbf{F} = Q[\mathbf{E} + (\mathbf{v} \times \mathbf{B})]. \quad (5.2)$$

I do not pretend to have *derived* Eq. 5.1, of course; it is a fundamental axiom of the theory, whose justification is to be found in experiments such as the one I described in Section 5.1.1.

Our main job from now on is to calculate the magnetic field $\mathbf{B}$ (and for that matter the electric field $\mathbf{E}$ as well; the rules are more complicated when the source charges are in motion). But before we proceed, it is worthwhile to take a closer look at the Lorentz force law itself; it is a peculiar law, and it leads to some truly bizarre particle trajectories.

Example 5.1. (Cyclotron motion) The archetypical motion of a charged particle in a magnetic field is circular, with the magnetic force providing the centripetal acceleration. In Fig. 5.4, a uniform magnetic field points *into* the page; if the charge Q moves counterclockwise, with speed v , around a circle of radius R , the magnetic force points *inward*, and has a fixed magnitude QvB – just right to sustain uniform circular motion:

$$QvB = m \frac{v^2}{R}, \quad \text{or } p = QBR, \quad (5.3)$$

where m is the particle's mass and $p = mv$ is its momentum. Equation 5.3 is known as the **cyclotron formula** because it describes the motion of a particle in a cyclotron – the first of the modern particle accelerators. It also suggests a simple experimental technique for finding the momentum of a charged particle: send it through a region of known magnetic field, and measure the radius of its trajectory. This is in fact the standard means for determining the momenta of elementary particles.

I assumed that the charge moves in a plane perpendicular to $\mathbf{B}$. If it starts out with some additional speed $v_{\parallel}$ *parallel* to $\mathbf{B}$, this component of the motion is unaffected by the magnetic field, and the particle moves in a *helix* (Fig. 5.5). The radius is still given by Eq. 5.3, but the velocity in question is now the component perpendicular to $\mathbf{B}$: call it $v_{\perp}$.

² Since $\mathbf{F}$ and $\mathbf{v}$ are vectors, $\mathbf{B}$ is actually a *pseudovector*.

³ Actually, it is due to Oliver Heaviside.

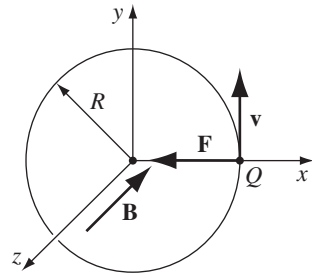


Fig. 5.4

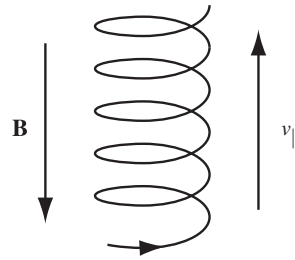


Fig. 5.5

Example 5.2. (Cycloid motion) A more exotic trajectory occurs if we include a uniform electric field, at right angles to the magnetic one. Suppose, for instance, that $\mathbf{B}$ points in the x direction, and $\mathbf{E}$ in the z direction, as shown in Fig. 5.6. A positive charge is released from the origin; what path will it follow?

Solution. Let's think it through qualitatively, first. Initially, the particle is at rest, so the magnetic force is zero, and the electric field accelerates the charge in the z direction. As it picks up speed, a magnetic force develops which, according to Eq. 5.1, pulls the charge around to the right. The faster it goes, the stronger F_{mag} becomes; eventually, it curves the particle back around toward the y -axis. At this point the charge is moving *against* the electrical force, so it begins to slow down – the magnetic force then decreases, and the electrical force takes over, bringing the particle to rest at point a , in Fig. 5.6. There the entire process commences anew, carrying the particle over to point b , and so on.

Now let's do it quantitatively. There being no force in the x direction, the position of the particle at any time t can be described by the vector $\mathbf{r}(t) = (0, y(t), z(t))$; the velocity is therefore

$$\mathbf{v} = (0, \dot{y}, \dot{z}),$$

where the dots denote time derivatives. Thus

$$\mathbf{v} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & \dot{y} & \dot{z} \\ B & 0 & 0 \end{vmatrix} = B\dot{z}\hat{\mathbf{y}} - B\dot{y}\hat{\mathbf{z}},$$

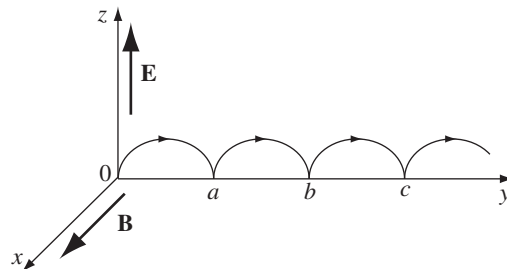


Fig. 5.6

and hence, applying Newton's second law,

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = Q(E\hat{\mathbf{z}} + B\dot{y}\hat{\mathbf{x}} - B\dot{z}\hat{\mathbf{y}}) = m\mathbf{a} = m(\ddot{y}\hat{\mathbf{x}} + \ddot{z}\hat{\mathbf{z}}).$$

Or, treating the $\hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$ components separately,

$$QB\dot{z} = m\ddot{y}, \quad QE - QB\dot{y} = m\ddot{z}.$$

For convenience, let

$$\omega \equiv \frac{QB}{m}. \quad (5.4)$$

(This is the **cyclotron frequency**, at which the particle would revolve in the absence of any electric field.) Then the equations of motion take the form

$$\ddot{y} = \omega\dot{z}, \quad \ddot{z} = \omega\left(\frac{E}{B} - \dot{y}\right). \quad (5.5)$$

Their general solution⁴ is

$$\left. \begin{aligned} y(t) &= C_1 \cos \omega t + C_2 \sin \omega t + (E/B)t + C_3, \\ z(t) &= C_2 \cos \omega t - C_1 \sin \omega t + C_4. \end{aligned} \right\} \quad (5.6)$$

But the particle started from rest ($\dot{y}(0) = \dot{z}(0) = 0$), at the origin ($y(0) = z(0) = 0$); these four conditions determine the constants C_1 , C_2 , C_3 , and C_4 :

$$y(t) = \frac{E}{\omega B}(\omega t - \sin \omega t), \quad z(t) = \frac{E}{\omega B}(1 - \cos \omega t). \quad (5.7)$$

In this form, the answer is not terribly enlightening,⁵ but if we let

$$R \equiv \frac{E}{\omega B}, \quad (5.8)$$

and eliminate the sines and cosines by exploiting the trigonometric identity $\sin^2 \omega t + \cos^2 \omega t = 1$, we find that

$$(y - R\omega t)^2 + (z - R)^2 = R^2. \quad (5.9)$$

This is the formula for a *circle*, of radius R , whose center $(0, R\omega t, R)$ travels in the y direction at a constant speed

$$u = \omega R = \frac{E}{B}. \quad (5.10)$$

The particle moves as though it were a spot on the rim of a wheel rolling along the y -axis. The curve generated in this way is called a **cycloid**. Notice that the overall motion is *not* in the direction of $\mathbf{E}$, as you might expect, but *perpendicular* to it.

⁴ As coupled differential equations, they are easily solved by differentiating the first and using the second to eliminate $\ddot{z}$.

⁵ Notice that $v^2 = \dot{y}^2 + \dot{z}^2 = [(2E/B) \sin(\omega t/2)]^2$, so this solution (which was based on the *nonrelativistic* form of Newton's second law) is valid only when $2E \ll cB$, where c is the speed of light.

One implication of the Lorentz force law (Eq. 5.1) deserves special attention:

magnetic forces do no work.

For if Q moves an amount $d\mathbf{l} = \mathbf{v} dt$, the work done is

$$dW_{\text{mag}} = \mathbf{F}_{\text{mag}} \cdot d\mathbf{l} = Q(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} dt = 0. \quad (5.11)$$

This follows because $(\mathbf{v} \times \mathbf{B})$ is perpendicular to $\mathbf{v}$, so $(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} = 0$. Magnetic forces may alter the *direction* in which a particle moves, but they cannot speed it up or slow it down. The fact that magnetic forces do no work is an elementary and direct consequence of the Lorentz force law, but there are many situations in which it *appears* so manifestly false that one's confidence is bound to waver. When a magnetic crane lifts the carcass of a junked car, for instance, *something* is obviously doing work, and it seems perverse to deny that the magnetic force is responsible. Well, perverse or not, deny it we must, and it can be a very subtle matter to figure out who *does* deserve the credit in such circumstances. We'll see a cute example in the next section, but the full story will have to await Chapter 8.

Problem 5.1. A particle of charge q enters a region of uniform magnetic field $\mathbf{B}$ (pointing *into* the page). The field deflects the particle a distance d above the original line of flight, as shown in Fig. 5.7. Is the charge positive or negative? In terms of a , d , B , and q , find the momentum of the particle.

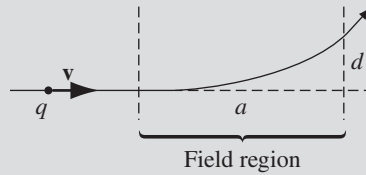


Fig. 5.7

Problem 5.2. Find and sketch the trajectory of the particle in Ex. 5.2, if it starts at the origin with velocity

- (a) $\mathbf{v}(0) = (E/B)\hat{\mathbf{y}}$;
- (b) $\mathbf{v}(0) = (E/2B)\hat{\mathbf{y}}$;
- (c) $\mathbf{v}(0) = (E/B)(\hat{\mathbf{y}} + \hat{\mathbf{z}})$.

Problem 5.3. In 1897, J. J. Thomson “discovered” the electron by measuring the charge-to-mass ratio of “cathode rays” (actually, streams of electrons, with charge q and mass m) as follows:

- (a) First he passed the beam through uniform crossed electric and magnetic fields $\mathbf{E}$ and $\mathbf{B}$ (mutually perpendicular, and both of them perpendicular to the beam), and adjusted the

electric field until he got zero deflection. What, then, was the speed of the particles (in terms of E and B)?

- (b) Then he turned off the electric field, and measured the radius of curvature, R , of the beam, as deflected by the magnetic field alone. In terms of E , B , and R , what is the charge-to-mass ratio (q/m) of the particles?

5.1.3 Currents

The **current** in a wire is the *charge per unit time* passing a given point. By definition, negative charges moving to the left count the same as positive ones to the right. This conveniently reflects the *physical* fact that almost all phenomena involving moving charges depend on the *product* of charge and velocity – if you reverse the signs of q and $\mathbf{v}$, you get the same answer, so it doesn't really matter which you have. (The Lorentz force law is a case in point; the Hall effect (Prob. 5.44) is a notorious exception.) In practice, it is ordinarily the negatively charged electrons that do the moving – in the direction *opposite* to the electric current. To avoid the petty complications this entails, I shall often pretend it's the positive charges that move, as in fact everyone assumed they did for a century or so after Benjamin Franklin established his unfortunate convention.⁶ Current is measured in coulombs per second, or **amperes** (A):

$$1 \text{ A} = 1 \text{ C/s.} \quad (5.12)$$

A line charge λ traveling down a wire at speed v (Fig. 5.8) constitutes a current

$$I = \lambda v, \quad (5.13)$$

because a segment of length $v\Delta t$, carrying charge $\lambda v\Delta t$, passes point P in a time interval Δt . Current is actually a *vector*:

$$\mathbf{I} = \lambda \mathbf{v}. \quad (5.14)$$

Because the path of the flow is dictated by the shape of the wire, one doesn't ordinarily bother to display the direction of $\mathbf{I}$ explicitly, but when it comes to surface and volume currents we cannot afford to be so casual, and for the sake of notational consistency it is a good idea to acknowledge the

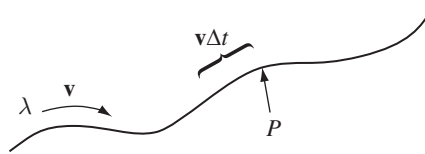


Fig. 5.8

⁶ If we called the electron plus and the proton minus, the problem would never arise. In the context of Franklin's experiments with cat's fur and glass rods, the choice was completely arbitrary.

vectorial character of currents right from the start.⁷ A neutral wire, of course, contains as many stationary positive charges as mobile negative ones. The former do not contribute to the current – the charge density λ in Eq. 5.13 refers only to the *moving* charges. In the unusual situation where *both* types move, $\mathbf{I} = \lambda_+ \mathbf{v}_+ + \lambda_- \mathbf{v}_-$.⁸

The magnetic force on a segment of current-carrying wire is

$$\mathbf{F}_{\text{mag}} = \int (\mathbf{v} \times \mathbf{B}) dq = \int (\mathbf{v} \times \mathbf{B}) \lambda dl = \int (\mathbf{I} \times \mathbf{B}) dl. \quad (5.15)$$

Inasmuch as $\mathbf{I}$ and $d\mathbf{l}$ both point in the same direction, we can just as well write this as

$$\mathbf{F}_{\text{mag}} = \int I(d\mathbf{l} \times \mathbf{B}). \quad (5.16)$$

Typically, the current is constant (in magnitude) along the wire, and in that case I comes outside the integral:

$$\mathbf{F}_{\text{mag}} = I \int (d\mathbf{l} \times \mathbf{B}). \quad (5.17)$$

Example 5.3. A rectangular loop of wire, supporting a mass m , hangs vertically with one end in a uniform magnetic field $\mathbf{B}$, which points into the page in the shaded region of Fig. 5.9. For what current I , in the loop, would the magnetic force upward exactly balance the gravitational force downward?

Solution. First of all, the current must circulate clockwise, in order for $(\mathbf{I} \times \mathbf{B})$ in the horizontal segment to point upward. The force is

$$F_{\text{mag}} = IBa,$$

where a is the width of the loop. (The magnetic forces on the two vertical segments cancel.) For F_{mag} to balance the weight (mg), we must therefore have

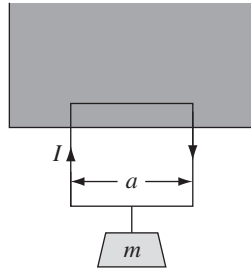


Fig. 5.9

⁷ If it bothers you to treat current as a vector, please go to my website and read the note “Is Current a Vector?”: www.reed.edu/physics/faculty/griffiths/VectorCurrent.pdf.

⁸ In principle there could even be multiple species of charge (some positive and some negative), each with its own density and velocity; in that case $\mathbf{I} = \sum_i \lambda_i \mathbf{v}_i$. But ordinarily the simple form (5.14) will suffice.

$$I = \frac{mg}{Ba}. \quad (5.18)$$

The weight just *hangs* there, suspended in midair!

What happens if we now *increase* the current? Then the upward magnetic force *exceeds* the downward force of gravity, and the loop rises, lifting the weight. *Somebody's* doing work, and it sure looks as though the magnetic force is responsible. Indeed, one is tempted to write

$$W_{\text{mag}} = F_{\text{mag}}h = IBah, \quad (5.19)$$

where h is the distance the loop rises. But we know that magnetic forces *never* do work. What's going on here?

Well, when the loop starts to rise, the charges in the wire are no longer moving horizontally – their velocity now acquires an upward component u , the speed of the loop (Fig. 5.10), in addition to the horizontal component w associated with the current ($I = \lambda w$). The magnetic force, which is always perpendicular to the velocity, no longer points straight up, but tilts back. It is perpendicular to the *net* displacement of the charge (which is in the direction of $\mathbf{v}$), and therefore *it does no work on q* . It does have a vertical component (qwB); the net vertical force on all the charge (λa) in the upper segment of the loop is

$$F_{\text{vert}} = \lambda awB = IBa \quad (5.20)$$

(as before); but now it also has a *horizontal* component (quB), which opposes the flow of current. Whoever is in charge of maintaining that current, therefore, must now *push* those charges along, against the backward component of the magnetic force.

The total horizontal force on the top segment is

$$F_{\text{horiz}} = \lambda auB. \quad (5.21)$$

In a time dt , the charges move a (horizontal) distance $w dt$, so the work done by this agency (presumably a battery or a generator) is

$$W_{\text{battery}} = \lambda aB \int uw dt = IBah,$$

which is precisely what we naïvely attributed to the *magnetic* force in Eq. 5.19. Was work done in this process? Absolutely! Who did it?

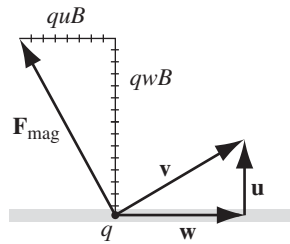


Fig. 5.10

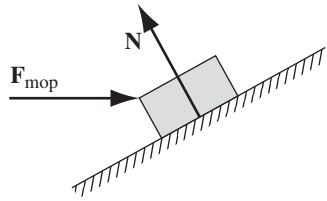


Fig. 5.11

The battery! What, then, was the role of the magnetic force? Well, it *redirected* the horizontal force of the battery into the *vertical* motion of the loop and the weight.⁹

It may help to consider a mechanical analogy. Imagine you're sliding a trunk up a frictionless ramp, by pushing on it horizontally with a mop (Fig. 5.11). The normal force ($\mathbf{N}$) does no work, because it is perpendicular to the displacement. But it *does* have a vertical component (which in fact is what lifts the trunk), and a (backward) horizontal component (which you have to overcome by pushing on the mop). Who is doing the work here? *You* are, obviously – and yet your *force* (which is purely horizontal) is not (at least, not directly) what lifts the box. The normal force plays the same passive (but crucial) role as the magnetic force in Ex. 5.3: while doing no work itself, it *redirects* the efforts of the active agent (you, or the battery, as the case may be), from horizontal to vertical.

When charge flows over a *surface*, we describe it by the **surface current density**, $\mathbf{K}$, defined as follows: Consider a “ribbon” of infinitesimal width $dl_{\perp}$, running parallel to the flow (Fig. 5.12). If the current in this ribbon is dI , the surface current density is

$$\mathbf{K} \equiv \frac{d\mathbf{I}}{dl_{\perp}}. \quad (5.22)$$

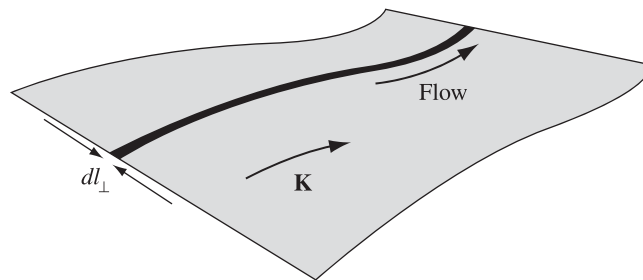


Fig. 5.12

⁹ If you like, the *vertical* component of $\mathbf{F}_{\text{mag}}$ does work lifting the weight, but the *horizontal* component does equal *negative* work opposing the current. However you look at it, the *net* work done by the magnetic force is *zero*.

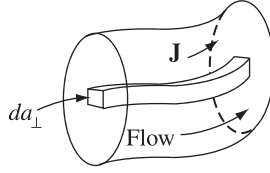


Fig. 5.13

In words, K is the *current per unit width*. In particular, if the (mobile) surface charge density is σ and its velocity is $\mathbf{v}$, then

$$\mathbf{K} = \sigma \mathbf{v}. \quad (5.23)$$

In general, $\mathbf{K}$ will vary from point to point over the surface, reflecting variations in σ and/or $\mathbf{v}$. The magnetic force on the surface current is

$$\mathbf{F}_{\text{mag}} = \int (\mathbf{v} \times \mathbf{B}) \sigma da = \int (\mathbf{K} \times \mathbf{B}) da. \quad (5.24)$$

Caveat: Just as $\mathbf{E}$ suffers a discontinuity at a surface *charge*, so $\mathbf{B}$ is discontinuous at a surface *current*. In Eq. 5.24, you must be careful to use the *average* field, just as we did in Section 2.5.3.

When the flow of charge is distributed throughout a three-dimensional region, we describe it by the **volume current density**, $\mathbf{J}$, defined as follows: Consider a “tube” of infinitesimal cross section $da_{\perp}$, running parallel to the flow (Fig. 5.13). If the current in this tube is dI , the volume current density is

$$\mathbf{J} \equiv \frac{d\mathbf{I}}{da_{\perp}}. \quad (5.25)$$

In words, J is the *current per unit area*. If the (mobile) volume charge density is ρ and the velocity is $\mathbf{v}$, then

$$\mathbf{J} = \rho \mathbf{v}. \quad (5.26)$$

The magnetic force on a volume current is therefore

$$\mathbf{F}_{\text{mag}} = \int (\mathbf{v} \times \mathbf{B}) \rho d\tau = \int (\mathbf{J} \times \mathbf{B}) d\tau. \quad (5.27)$$

Example 5.4.

- (a) A current I is uniformly distributed over a wire of circular cross section, with radius a (Fig. 5.14). Find the volume current density J .

Solution. The area (perpendicular to the flow) is πa^2 , so

$$J = \frac{I}{\pi a^2}.$$

This was trivial because the current density was uniform.

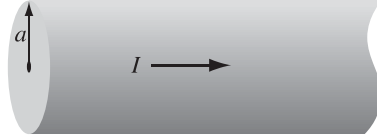


Fig. 5.14

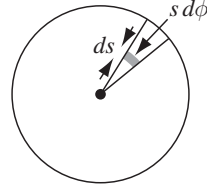


Fig. 5.15

- (b) Suppose the current density in the wire is proportional to the distance from the axis,

$$J = ks$$

(for some constant k). Find the total current in the wire.

Solution. Because J varies with s , we must *integrate* Eq. 5.25. The current through the shaded patch (Fig. 5.15) is $J da_{\perp}$, and $da_{\perp} = s ds d\phi$. So

$$I = \int (ks)(s ds d\phi) = 2\pi k \int_0^a s^2 ds = \frac{2\pi ka^3}{3}.$$

According to Eq. 5.25, the total current crossing a surface S can be written as

$$I = \int_S J da_{\perp} = \int_S \mathbf{J} \cdot d\mathbf{a}. \quad (5.28)$$

(The dot product serves neatly to pick out the perpendicular component of $d\mathbf{a}$.) In particular, the charge per unit time leaving a volume $\mathcal{V}$ is

$$\oint_S \mathbf{J} \cdot d\mathbf{a} = \int_{\mathcal{V}} (\nabla \cdot \mathbf{J}) d\tau.$$

Because charge is conserved, whatever flows out through the surface must come at the expense of what remains inside:

$$\int_{\mathcal{V}} (\nabla \cdot \mathbf{J}) d\tau = -\frac{d}{dt} \int_{\mathcal{V}} \rho d\tau = -\int_{\mathcal{V}} \left(\frac{\partial \rho}{\partial t}\right) d\tau.$$

(The minus sign reflects the fact that an *outward* flow *decreases* the charge left in $\mathcal{V}$.) Since this applies to *any* volume, we conclude that

$$\boxed{\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}}. \quad (5.29)$$

This is the precise mathematical statement of local charge conservation; it is called the **continuity equation**.

For future reference, let me summarize the “dictionary” we have developed for translating equations into the forms appropriate to point, line, surface, and volume currents:

$$\sum_{i=1}^n () q_i \mathbf{v}_i \sim \int_{\text{line}} () \mathbf{I} dl \sim \int_{\text{surface}} () \mathbf{K} da \sim \int_{\text{volume}} () \mathbf{J} d\tau. \quad (5.30)$$

This correspondence, which is analogous to $q \sim \lambda dl \sim \sigma da \sim \rho d\tau$ for the various charge distributions, generates Eqs. 5.15, 5.24, and 5.27 from the original Lorentz force law (5.1).

Problem 5.4. Suppose that the magnetic field in some region has the form

$$\mathbf{B} = kz\hat{\mathbf{x}}$$

(where k is a constant). Find the force on a square loop (side length a), lying in the yz plane and centered at the origin, if it carries a current I , flowing counterclockwise, when you look down the x -axis.

Problem 5.5. A current I flows down a wire of radius a .

- If it is uniformly distributed over the surface, what is the surface current density K ?
- If it is distributed in such a way that the volume current density is inversely proportional to the distance from the axis, what is $J(s)$?

Problem 5.6.

- A phonograph record carries a uniform density of “static electricity” σ . If it rotates at angular velocity ω , what is the surface current density K at a distance r from the center?
- A uniformly charged solid sphere, of radius R and total charge Q , is centered at the origin and spinning at a constant angular velocity ω about the z -axis. Find the current density $\mathbf{J}$ at any point (r, θ, ϕ) within the sphere.

Problem 5.7. For a configuration of charges and currents confined within a volume $\mathcal{V}$, show that

$$\int_{\mathcal{V}} \mathbf{J} d\tau = d\mathbf{p}/dt, \quad (5.31)$$

where $\mathbf{p}$ is the total dipole moment. [Hint: Evaluate $\int_{\mathcal{V}} \nabla \cdot (x\mathbf{J}) d\tau$.]

5.2 The Biot–Savart Law

5.2.1 Steady Currents

Stationary charges produce electric fields that are constant in time; hence the term **electrostatics**.¹⁰ *Steady currents* produce magnetic fields that are constant in time; the theory of steady currents is called **magnetostatics**.

¹⁰ Actually, it is not necessary that the charges be stationary, but only that the charge *density* at each point be constant. For example, the sphere in Prob. 5.6b produces an electrostatic field $1/4\pi\epsilon_0(Q/r^2)\hat{\mathbf{r}}$, even though it is rotating, because ρ does not depend on t .

Stationary charges	$\Rightarrow$	constant electric fields: electrostatics.
Steady currents	$\Rightarrow$	constant magnetic fields: magnetostatics.

By **steady current** I mean a continuous flow that has been going on forever, without change and without charge piling up anywhere. (Some people call them “stationary currents”; to my ear, that’s an oxymoron.) Formally, electro/magnetostatics is the régime

$$\frac{\partial \rho}{\partial t} = 0, \quad \frac{\partial \mathbf{J}}{\partial t} = \mathbf{0}, \quad (5.32)$$

at all places and all times. Of course, there’s no such thing in practice as a *truly* steady current, any more than there is a *truly* stationary charge. In this sense, both electrostatics and magnetostatics describe artificial worlds that exist only in textbooks. However, they represent suitable *approximations* as long as the actual fluctuations are remote, or gradual – in fact, for most purposes magnetostatics applies very well to household currents, which alternate 120 times a second!

Notice, however, that a moving *point* charge *cannot possibly constitute a steady current*. If it’s here one instant, it’s gone the next. This may seem like a minor thing to *you*, but it’s a major headache for *me*. I developed each topic in electrostatics by starting out with the simple case of a point charge at rest; then I generalized to an arbitrary charge distribution by invoking the superposition principle. This approach is not open to us in magnetostatics because a moving point charge does not produce a static field in the first place. We are *forced* to deal with extended current distributions right from the start, and, as a result, the arguments are bound to be more cumbersome.

When a steady current flows in a wire, its magnitude I must be the same all along the line; otherwise, charge would be piling up somewhere, and it wouldn’t be a steady current. More generally, since $\partial \rho / \partial t = 0$ in magnetostatics, the continuity equation (5.29) becomes

$$\nabla \cdot \mathbf{J} = 0. \quad (5.33)$$

5.2.2 The Magnetic Field of a Steady Current

The magnetic field of a steady line current is given by the **Biot–Savart law**:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I} \times \hat{\mathbf{r}}}{r^2} dl' = \frac{\mu_0}{4\pi} I \int \frac{d\mathbf{l}' \times \hat{\mathbf{r}}}{r^2}. \quad (5.34)$$

The integration is along the current path, in the direction of the flow; $d\mathbf{l}'$ is an element of length along the wire, and $\hat{\mathbf{r}}$, as always, is the vector from the

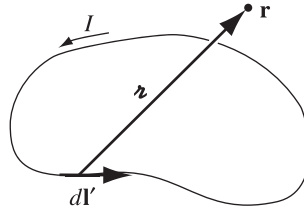


Fig. 5.16

source to the point $\mathbf{r}$ (Fig. 5.16). The constant μ_0 is called the **permeability of free space**.¹¹

$$\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2. \quad (5.35)$$

These units are such that $\mathbf{B}$ itself comes out in newtons per ampere-meter (as required by the Lorentz force law), or **teslas** (T):¹²

$$1 \text{ T} = 1 \text{ N}/(\text{A} \cdot \text{m}). \quad (5.36)$$

As the starting point for magnetostatics, the Biot–Savart law plays a role analogous to Coulomb’s law in electrostatics. Compare Eq. 5.34 with Eq. 2.6; the $1/r^2$ dependence is common to both. Like Coulomb’s, the Biot–Savart law is empirical: it comes from experiment. The cross product, which might at first seem strange, is just right to account for the observed directions in Fig. 5.2.

Example 5.5. Find the magnetic field a distance s from a long straight wire carrying a steady current I (Fig. 5.17).

Solution. In the diagram, $(d\mathbf{l}' \times \hat{\mathbf{z}})$ points *out* of the page, and has the magnitude

$$dl' \sin \alpha = dl' \cos \theta.$$

Also, $l' = s \tan \theta$, so

$$dl' = \frac{s}{\cos^2 \theta} d\theta,$$

and $s = r \cos \theta$, so

$$\frac{1}{r^2} = \frac{\cos^2 \theta}{s^2}.$$

Thus

$$\begin{aligned} B &= \frac{\mu_0 I}{4\pi} \int_{\theta_1}^{\theta_2} \left(\frac{\cos^2 \theta}{s^2} \right) \left(\frac{s}{\cos^2 \theta} \right) \cos \theta d\theta \\ &= \frac{\mu_0 I}{4\pi s} \int_{\theta_1}^{\theta_2} \cos \theta d\theta = \frac{\mu_0 I}{4\pi s} (\sin \theta_2 - \sin \theta_1). \end{aligned} \quad (5.37)$$

¹¹ This is an exact number, not an empirical constant. It serves (via Eq. 5.40) to define the ampere, and the ampere in turn defines the coulomb.

¹² For some reason, in this one case the cgs unit (the **gauss**) is more commonly used than the SI unit: 1 tesla = 10^4 gauss. The Earth’s magnetic field is about half a gauss; a fairly strong laboratory magnetic field is, say, 1 T (10,000 gauss).

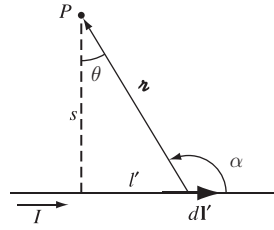


Fig. 5.17

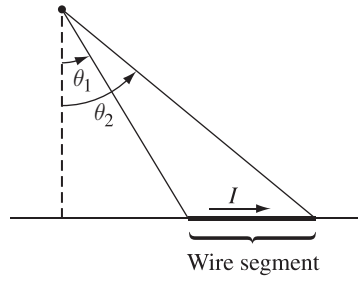


Fig. 5.18

Equation 5.37 gives the field of any straight segment of wire, in terms of the initial and final angles θ_1 and θ_2 (Fig. 5.18). Of course, a finite segment by itself could never support a steady current (where would the charge go when it got to the end?), but it might be a *piece* of some closed circuit, and Eq. 5.37 would then represent its contribution to the total field. In the case of an *infinite* wire, $\theta_1 = -\pi/2$ and $\theta_2 = \pi/2$, so we obtain

$$B = \frac{\mu_0 I}{2\pi s}. \quad (5.38)$$

Notice that the field is inversely proportional to the distance from the wire – just like the electric field of an infinite line charge. In the region *below* the wire, $\mathbf{B}$ points *into* the page, and in general, it “circles around” the wire, in accordance with the right-hand rule (Fig. 5.2):

$$\mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}. \quad (5.39)$$

As an application, let’s find the force of attraction between two long, parallel wires a distance d apart, carrying currents I_1 and I_2 (Fig. 5.19). The field at (2) due to (1) is

$$B = \frac{\mu_0 I_1}{2\pi d},$$

and it points into the page. The Lorentz force law (in the form appropriate to line currents, Eq. 5.17) predicts a force directed toward (1), of magnitude

$$F = I_2 \left(\frac{\mu_0 I_1}{2\pi d} \right) \int dl.$$

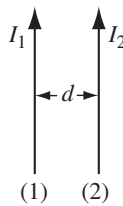


Fig. 5.19

The *total* force, not surprisingly, is infinite, but the force *per unit length* is

$$f = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{d}. \quad (5.40)$$

If the currents are antiparallel (one up, one down), the force is repulsive – consistent again with the qualitative observations in Section 5.1.1.

Example 5.6. Find the magnetic field a distance z above the center of a circular loop of radius R , which carries a steady current I (Fig. 5.20).

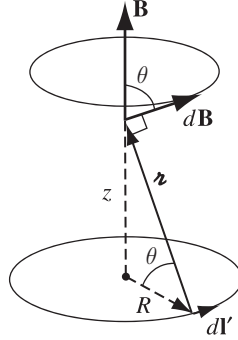


Fig. 5.20

Solution. The field $d\mathbf{B}$ attributable to the segment $d\mathbf{l}'$ points as shown. As we integrate $d\mathbf{l}'$ around the loop, $d\mathbf{B}$ sweeps out a cone. The horizontal components cancel, and the vertical components combine, to give

$$B(z) = \frac{\mu_0}{4\pi} I \int \frac{dl'}{r^2} \cos \theta.$$

(Notice that $d\mathbf{l}'$ and $\mathbf{r}$ are perpendicular, in this case; the factor of $\cos \theta$ projects out the vertical component.) Now, $\cos \theta$ and r^2 are constants, and $\int dl'$ is simply the circumference, $2\pi R$, so

$$B(z) = \frac{\mu_0 I}{4\pi} \left(\frac{\cos \theta}{r^2} \right) 2\pi R = \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + z^2)^{3/2}}. \quad (5.41)$$

For surface and volume currents, the Biot–Savart law becomes

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} da' \quad \text{and} \quad \mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} d\tau', \quad (5.42)$$

respectively. You might be tempted to write down the corresponding formula for a moving point charge, using the “dictionary” (Eq. 5.30):

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{q\mathbf{v} \times \hat{\mathbf{r}}}{r^2}, \quad (5.43)$$

but this is simply *wrong*.¹³ As I mentioned earlier, a point charge does not constitute a steady current, and the Biot–Savart law, which only holds for steady currents, does *not* correctly determine its field.

The superposition principle applies to magnetic fields just as it does to electric fields: if you have a *collection* of source currents, the net field is the (vector) sum of the fields due to each of them taken separately.

Problem 5.8.

- Find the magnetic field at the center of a square loop, which carries a steady current I . Let R be the distance from center to side (Fig. 5.21).
- Find the field at the center of a regular n -sided polygon, carrying a steady current I . Again, let R be the distance from the center to any side.
- Check that your formula reduces to the field at the center of a circular loop, in the limit $n \rightarrow \infty$.

Problem 5.9. Find the magnetic field at point P for each of the steady current configurations shown in Fig. 5.22.

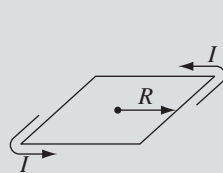
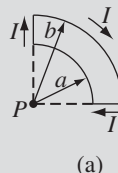
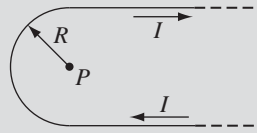


Fig. 5.21



(a)

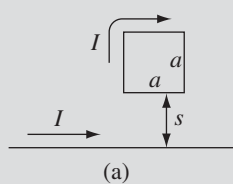


(b)

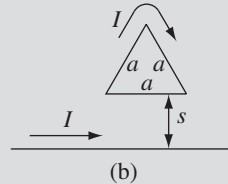
Fig. 5.22

Problem 5.10.

- Find the force on a square loop placed as shown in Fig. 5.23a, near an infinite straight wire. Both the loop and the wire carry a steady current I .
- Find the force on the triangular loop in Fig. 5.23b.



(a)



(b)

Fig. 5.23

¹³ I say this loud and clear to emphasize the point of principle; actually, Eq. 5.43 is *approximately* right for nonrelativistic charges ($v \ll c$), under conditions where retardation can be neglected (see Ex. 10.4).

Problem 5.11. Find the magnetic field at point P on the axis of a tightly wound **solenoid** (helical coil) consisting of n turns per unit length wrapped around a cylindrical tube of radius a and carrying current I (Fig. 5.24). Express your answer in terms of θ_1 and θ_2 (it's tidier that way). Consider the turns to be essentially circular, and use the result of Ex. 5.6. What is the field on the axis of an *infinite* solenoid (infinite in both directions)?

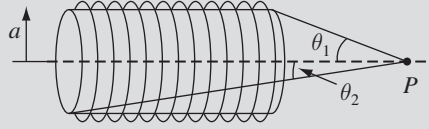


Fig. 5.24

Problem 5.12. Use the result of Ex. 5.6 to calculate the magnetic field at the center of a uniformly charged spherical shell, of radius R and total charge Q , spinning at constant angular velocity ω .

Problem 5.13. Suppose you have two infinite straight-line charges λ , a distance d apart, moving along at a constant speed v (Fig. 5.25). How great would v have to be in order for the magnetic attraction to balance the electrical repulsion? Work out the actual number. Is this a reasonable sort of speed?¹⁴

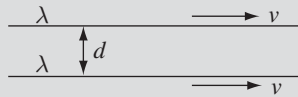


Fig. 5.25

5.3 The Divergence and Curl of $\mathbf{B}$

5.3.1 Straight-Line Currents

The magnetic field of an infinite straight wire is shown in Fig. 5.26 (the current is coming *out* of the page). At a glance, it is clear that this field has a nonzero curl (something you'll never see in an *electrostatic* field); let's calculate it.

According to Eq. 5.38, the integral of $\mathbf{B}$ around a circular path of radius s , centered at the wire, is

$$\oint \mathbf{B} \cdot d\mathbf{l} = \oint \frac{\mu_0 I}{2\pi s} dl = \frac{\mu_0 I}{2\pi s} \oint dl = \mu_0 I.$$

¹⁴ If you've studied special relativity, you may be tempted to look for complexities in this problem that are not really there – λ and v are both measured *in the laboratory frame*, and this is *ordinary electrostatics*.

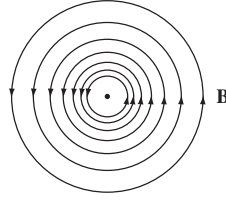


Fig. 5.26

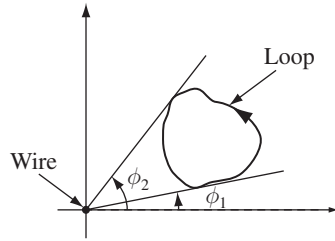


Fig. 5.27

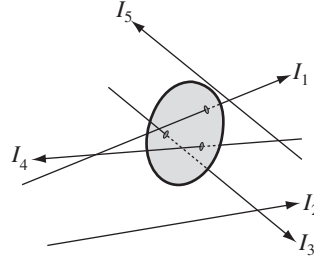


Fig. 5.28

Notice that the answer is independent of s ; that's because B decreases at the same rate as the circumference *increases*. In fact, it doesn't have to be a circle; *any* old loop that encloses the wire would give the same answer. For if we use cylindrical coordinates (s, ϕ, z) , with the current flowing along the z -axis, $\mathbf{B} = (\mu_0 I / 2\pi s) \hat{\phi}$ and $d\mathbf{l} = ds \hat{s} + s d\phi \hat{\phi} + dz \hat{z}$, so

$$\oint \mathbf{B} \cdot d\mathbf{l} = \frac{\mu_0 I}{2\pi} \oint \frac{1}{s} s d\phi = \frac{\mu_0 I}{2\pi} \int_0^{2\pi} d\phi = \mu_0 I.$$

This assumes the loop encircles the wire exactly once; if it went around twice, then ϕ would run from 0 to 4π , and if it didn't enclose the wire at all, then ϕ would go from ϕ_1 to ϕ_2 and back again, with $\int d\phi = 0$ (Fig. 5.27).

Now suppose we have a *bundle* of straight wires. Each wire that passes through our loop contributes $\mu_0 I$, and those outside contribute nothing (Fig. 5.28). The line integral will then be

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}, \quad (5.44)$$

where I_{enc} stands for the total current enclosed by the integration path. If the flow of charge is represented by a volume current density $\mathbf{J}$, the enclosed current is

$$I_{\text{enc}} = \int \mathbf{J} \cdot d\mathbf{a}, \quad (5.45)$$

with the integral taken over any surface bounded by the loop. Applying Stokes' theorem to Eq. 5.44, then,

$$\int (\nabla \times \mathbf{B}) \cdot d\mathbf{a} = \mu_0 \int \mathbf{J} \cdot d\mathbf{a},$$

and hence (since this must hold for *any* loop)

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}. \quad (5.46)$$

With minimal labor, we have actually obtained the general formula for the curl of $\mathbf{B}$. But our derivation is seriously flawed by the restriction to infinite straight-line currents (and combinations thereof). Most current configurations *cannot* be constructed out of infinite straight wires, and we have no right to assume that Eq. 5.46 applies to them. So the next section is devoted to a formal derivation of the divergence and curl of $\mathbf{B}$, starting from the Biot–Savart law itself.

5.3.2 The Divergence and Curl of $\mathbf{B}$

The Biot–Savart law for the general case of a volume current reads

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} d\tau'. \quad (5.47)$$

This formula gives the magnetic field at a point $\mathbf{r} = (x, y, z)$ in terms of an integral over the current distribution $\mathbf{J}(x', y', z')$ (Fig. 5.29). It is best to be absolutely explicit at this stage:

$$\begin{aligned} \mathbf{B} &\text{ is a function of } (x, y, z), \\ \mathbf{J} &\text{ is a function of } (x', y', z'), \\ \mathbf{r} &= (x - x')\hat{\mathbf{x}} + (y - y')\hat{\mathbf{y}} + (z - z')\hat{\mathbf{z}}, \\ d\tau' &= dx' dy' dz'. \end{aligned}$$

The integration is over the *primed* coordinates; the divergence and the curl of $\mathbf{B}$ are with respect to the *unprimed* coordinates.

Applying the divergence to Eq. 5.47, we obtain

$$\nabla \cdot \mathbf{B} = \frac{\mu_0}{4\pi} \int \nabla \cdot \left(\mathbf{J} \times \frac{\hat{\mathbf{r}}}{r^2} \right) d\tau'. \quad (5.48)$$

Invoking product rule number 6,

$$\nabla \cdot \left(\mathbf{J} \times \frac{\hat{\mathbf{r}}}{r^2} \right) = \frac{\hat{\mathbf{r}}}{r^2} \cdot (\nabla \times \mathbf{J}) - \mathbf{J} \cdot \left(\nabla \times \frac{\hat{\mathbf{r}}}{r^2} \right). \quad (5.49)$$

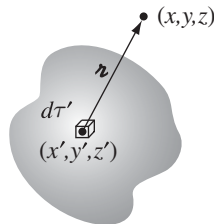


Fig. 5.29

But $\nabla \times \mathbf{J} = \mathbf{0}$, because $\mathbf{J}$ doesn't depend on the unprimed variables, while $\nabla \times (\hat{\mathbf{z}}/r^2) = \mathbf{0}$ (Prob. 1.63b), so

$$\boxed{\nabla \cdot \mathbf{B} = 0.} \quad (5.50)$$

Evidently, the *divergence* of the magnetic field is *zero*.

Applying the curl to Eq. 5.47, we obtain

$$\nabla \times \mathbf{B} = \frac{\mu_0}{4\pi} \int \nabla \times \left(\mathbf{J} \times \frac{\hat{\mathbf{z}}}{r^2} \right) d\tau'. \quad (5.51)$$

Again, our strategy is to expand the integrand, using the appropriate product rule – in this case number 8:

$$\nabla \times \left(\mathbf{J} \times \frac{\hat{\mathbf{z}}}{r^2} \right) = \mathbf{J} \left(\nabla \cdot \frac{\hat{\mathbf{z}}}{r^2} \right) - (\mathbf{J} \cdot \nabla) \frac{\hat{\mathbf{z}}}{r^2}. \quad (5.52)$$

(I have dropped terms involving derivatives of $\mathbf{J}$, because $\mathbf{J}$ does not depend on x, y, z .) The second term on the right integrates to zero, as we'll see in the next paragraph. The first term involves the divergence we were at pains to calculate in Chapter 1 (Eq. 1.100):

$$\nabla \cdot \left(\frac{\hat{\mathbf{z}}}{r^2} \right) = 4\pi\delta^3(\mathbf{r}). \quad (5.53)$$

Thus

$$\nabla \times \mathbf{B} = \frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{r}') 4\pi\delta^3(\mathbf{r} - \mathbf{r}') d\tau' = \mu_0 \mathbf{J}(\mathbf{r}),$$

which confirms that Eq. 5.46 is not restricted to straight-line currents, but holds quite generally in magnetostatics.

To complete the argument, however, we must check that the second term in Eq. 5.52 integrates to zero. Because the derivative acts only on $\hat{\mathbf{z}}/r^2$, we can switch from ∇ to ∇' at the cost of a minus sign:¹⁵

$$-(\mathbf{J} \cdot \nabla) \frac{\hat{\mathbf{z}}}{r^2} = (\mathbf{J} \cdot \nabla') \frac{\hat{\mathbf{z}}}{r^2}. \quad (5.54)$$

The x component, in particular, is

$$(\mathbf{J} \cdot \nabla') \left(\frac{x - x'}{r^3} \right) = \nabla' \cdot \left[\frac{(x - x')}{r^3} \mathbf{J} \right] - \left(\frac{x - x'}{r^3} \right) (\nabla' \cdot \mathbf{J})$$

(using product rule 5). Now, for *steady* currents the divergence of $\mathbf{J}$ is zero (Eq. 5.33), so

$$\left[-(\mathbf{J} \cdot \nabla) \frac{\hat{\mathbf{z}}}{r^2} \right]_x = \nabla' \cdot \left[\frac{(x - x')}{r^3} \mathbf{J} \right],$$

and therefore this contribution to the integral (Eq. 5.51) can be written

$$\int_V \nabla' \cdot \left[\frac{(x - x')}{r^3} \mathbf{J} \right] d\tau' = \oint_S \frac{(x - x')}{r^3} \mathbf{J} \cdot d\mathbf{a}'. \quad (5.55)$$

(The reason for switching from ∇ to ∇' was to permit this use of the divergence theorem.) But what region are we integrating over? Well, it's the

¹⁵ The point here is that r depends only on the *difference* between the coordinates; note that $(\partial/\partial x)f(x - x') = -(\partial/\partial x')f(x - x')$.

volume that appears in the Biot–Savart law (Eq. 5.47) – large enough, that is, to include all the current. You can make it *bigger* than that, if you like; $\mathbf{J} = \mathbf{0}$ out there anyway, so it will add nothing to the integral. The essential point is that *on the boundary* the current is *zero* (all current is safely *inside*) and hence the surface integral (Eq. 5.55) vanishes.¹⁶

5.3.3 Ampère’s Law

The equation for the curl of $\mathbf{B}$,

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}, \quad (5.56)$$

is called **Ampère’s law** (in differential form). It can be converted to integral form by the usual device of applying one of the fundamental theorems – in this case Stokes’ theorem:

$$\int (\nabla \times \mathbf{B}) \cdot d\mathbf{a} = \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int \mathbf{J} \cdot d\mathbf{a}.$$

Now, $\int \mathbf{J} \cdot d\mathbf{a}$ is the total current passing through the surface (Fig. 5.30), which we call I_{enc} (the **current enclosed** by the **Ampèrian loop**). Thus

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}. \quad (5.57)$$

This is the integral version of Ampère’s law; it generalizes Eq. 5.44 to *arbitrary* steady currents. Notice that Eq. 5.57 inherits the sign ambiguity of Stokes’ theorem (Section 1.3.5): Which *way* around the loop am I supposed to go? And which *direction* through the surface corresponds to a “positive” current? The resolution, as always, is the right-hand rule: if the fingers of your right hand indicate the direction of integration around the boundary, then your thumb defines the direction of a positive current.

Just as the Biot–Savart law plays a role in magnetostatics that Coulomb’s law assumed in electrostatics, so Ampère’s plays the part of Gauss’s:

$$\begin{cases} \text{electrostatics :} & \text{Coulomb} \rightarrow \text{Gauss,} \\ \text{magnetostatics :} & \text{Biot–Savart} \rightarrow \text{Ampère.} \end{cases}$$

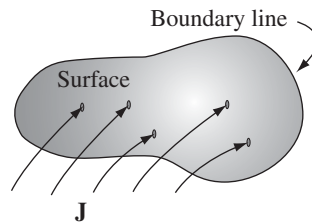


Fig. 5.30

¹⁶ If $\mathbf{J}$ itself extends to infinity (as in the case of an infinite straight wire), the surface integral is still typically zero, though the analysis calls for greater care.

In particular, for currents with appropriate symmetry, Ampère's law in integral form offers a lovely and extraordinarily efficient way of calculating the magnetic field.

Example 5.7. Find the magnetic field a distance s from a long straight wire (Fig. 5.31), carrying a steady current I (the same problem we solved in Ex. 5.5, using the Biot–Savart law).

Solution. We know the direction of $\mathbf{B}$ is “circumferential,” circling around the wire as indicated by the right-hand rule. By symmetry, the magnitude of $\mathbf{B}$ is constant around an Ampèrian loop of radius s , centered on the wire. So Ampère's law gives

$$\oint \mathbf{B} \cdot d\mathbf{l} = B \oint dl = B 2\pi s = \mu_0 I_{\text{enc}} = \mu_0 I,$$

or

$$B = \frac{\mu_0 I}{2\pi s}.$$

This is the same answer we got before (Eq. 5.38), but it was obtained this time with far less effort.

Example 5.8. Find the magnetic field of an infinite uniform surface current $\mathbf{K} = K\hat{\mathbf{x}}$, flowing over the xy plane (Fig. 5.32).

Solution. First of all, what is the *direction* of $\mathbf{B}$? Could it have any x component? *No*: A glance at the Biot–Savart law (Eq. 5.42) reveals that $\mathbf{B}$ is *perpendicular* to $\mathbf{K}$. Could it have a z component? *No again*. You could confirm this by noting that any vertical contribution from a filament at $+y$ is canceled by the corresponding filament at $-y$. But there is a nicer argument: Suppose the field pointed *away* from the plane. By reversing the direction of the current, I could make it point *toward* the plane (in the Biot–Savart law, changing the sign of the current switches the sign of the field). But the z component of $\mathbf{B}$ cannot possibly depend on the *direction* of the current in the xy plane. (Think about it!) So $\mathbf{B}$ can only have a y component, and a quick check with your right hand should convince you that it points to the *left* above the plane and to the *right* below it.

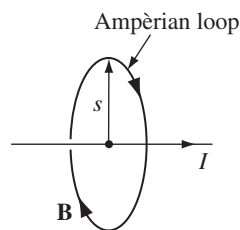


Fig. 5.31

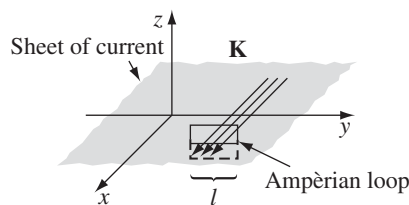


Fig. 5.32

With this in mind, we draw a rectangular Ampèrian loop as shown in Fig. 5.32, parallel to the yz plane and extending an equal distance above and below the surface. Applying Ampère's law,

$$\oint \mathbf{B} \cdot d\mathbf{l} = 2Bl = \mu_0 I_{\text{enc}} = \mu_0 K l,$$

(one Bl comes from the top segment and the other from the bottom), so $B = (\mu_0/2)K$, or, more precisely,

$$\mathbf{B} = \begin{cases} +(\mu_0/2)K\hat{\mathbf{y}} & \text{for } z < 0, \\ -(\mu_0/2)K\hat{\mathbf{y}} & \text{for } z > 0. \end{cases} \quad (5.58)$$

Notice that the field is independent of the distance from the plane, just like the electric field of a uniform surface charge (Ex. 2.5).

Example 5.9. Find the magnetic field of a very long solenoid, consisting of n closely wound turns per unit length on a cylinder of radius R , each carrying a steady current I (Fig. 5.33). [The point of making the windings so close is that one can then pretend each turn is circular. If this troubles you (after all, there is a net current I in the direction of the solenoid's axis, no matter *how* tight the winding), picture instead a sheet of aluminum foil wrapped around the cylinder, carrying the equivalent uniform surface current $K = nI$ (Fig. 5.34). Or make a double winding, going up to one end and then – always in the same sense – going back down again, thereby eliminating the net longitudinal current. But, in truth, this is all unnecessary fastidiousness, for the field inside a solenoid is huge (relatively speaking), and the field of the longitudinal current is at most a tiny refinement.]

Solution. First of all, what is the *direction* of $\mathbf{B}$? Could it have a radial component? *No.* For suppose B_s were *positive*; if we reversed the direction of the current, B_s would then be *negative*. But switching I is physically equivalent to turning the solenoid upside down, and that certainly should not alter the radial field. How about a “circumferential” (azimuthal) component? *No.* For B_ϕ would be constant around an Ampèrian loop concentric with the solenoid (Fig. 5.35), and hence

$$\oint \mathbf{B} \cdot d\mathbf{l} = B_\phi(2\pi s) = \mu_0 I_{\text{enc}} = 0,$$

since the loop encloses no current.

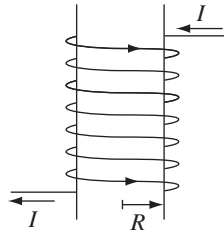


Fig. 5.33

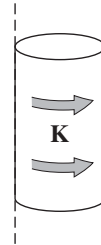


Fig. 5.34

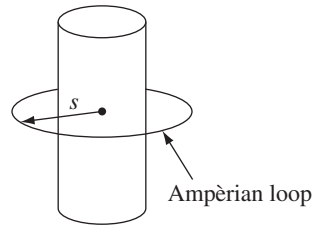


Fig. 5.35

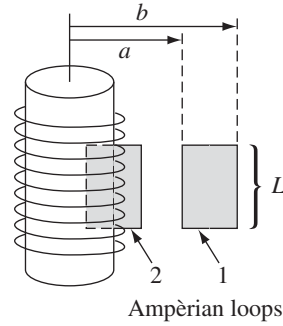


Fig. 5.36

So the magnetic field of an infinite, closely wound solenoid runs *parallel to the axis*. From the right-hand rule, we expect that it points upward inside the solenoid and downward outside. Moreover, it certainly approaches zero as you go very far away. With this in mind, let's apply Ampère's law to the two rectangular loops in Fig. 5.36. Loop 1 lies entirely outside the solenoid, with its sides at distances a and b from the axis:

$$\oint \mathbf{B} \cdot d\mathbf{l} = [B(a) - B(b)]L = \mu_0 I_{\text{enc}} = 0,$$

so

$$B(a) = B(b).$$

Evidently, the *field outside does not depend on the distance from the axis*. But we agreed that it goes to *zero* for large s . It must therefore be zero *everywhere*! (This astonishing result can also be obtained directly from the Biot–Savart law, of course, but it's much more difficult. See Prob. 5.49.)

As for loop 2, which is half inside and half outside, Ampère's law gives

$$\oint \mathbf{B} \cdot d\mathbf{l} = BL = \mu_0 I_{\text{enc}} = \mu_0 nIL,$$

where B is the field inside the solenoid. (The right-hand side of the loop contributes nothing, since $B = 0$ out there.) *Conclusion:*

$$\mathbf{B} = \begin{cases} \mu_0 n I \hat{\mathbf{z}} & \text{inside the solenoid,} \\ \mathbf{0} & \text{outside the solenoid.} \end{cases} \quad (5.59)$$

Notice that the field inside is *uniform* – it doesn't depend on the distance from the axis. In this sense the solenoid is to magnetostatics what the parallel-plate capacitor is to electrostatics: a simple device for producing strong uniform fields.

Like Gauss's law, Ampère's law is always *true* (for steady currents), but it is not always *useful*. Only when the symmetry of the problem enables you to pull B outside the integral $\oint \mathbf{B} \cdot d\mathbf{l}$ can you calculate the magnetic field from Ampère's law. When it *does* work, it's by far the fastest method;

when it doesn't, you have to fall back on the Biot–Savart law. The current configurations that can be handled by Ampère's law are:

- (1) infinite straight lines (prototype: Ex. 5.7),
- (2) infinite planes (prototype: Ex. 5.8),
- (3) infinite solenoids (prototype: Ex. 5.9),
- (4) toroids (prototype: Ex. 5.10).

The last of these is a surprising and elegant application of Ampère's law. As in Exs. 5.8 and 5.9, the hard part is figuring out the *direction* of the field (which we will now have done, once and for all, for each of the four geometries); the actual application of Ampère's law takes only one line.

Example 5.10. A toroidal coil consists of a circular ring, or “donut,” around which a long wire is wrapped (Fig. 5.37). The winding is uniform and tight enough so that each turn can be considered a plane closed loop. The cross-sectional shape of the coil is immaterial. I made it rectangular in Fig. 5.37 for the sake of simplicity, but it could just as well be circular or even some weird asymmetrical form, as in Fig. 5.38, as long as the shape remains the same all the way around the ring. In that case, the *magnetic field of the toroid is circumferential at all points, both inside and outside the coil.*

Proof. According to the Biot–Savart law, the field at $\mathbf{r}$ due to the current element at $\mathbf{r}'$ is

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{\mathbf{I} \times \mathbf{r}}{r^3} dl'.$$

We may as well put $\mathbf{r}$ in the xz plane (Fig. 5.38), so its Cartesian components are $(x, 0, z)$, while the source coordinates are

$$\mathbf{r}' = (s' \cos \phi', s' \sin \phi', z').$$

Then

$$\mathbf{r} = (x - s' \cos \phi', -s' \sin \phi', z - z').$$

Since the current has no ϕ component, $\mathbf{I} = I_s \hat{\mathbf{s}} + I_z \hat{\mathbf{z}}$, or (in Cartesian coordinates)

$$\mathbf{I} = (I_s \cos \phi', I_s \sin \phi', I_z).$$

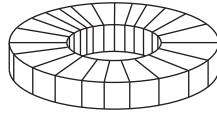


Fig. 5.37

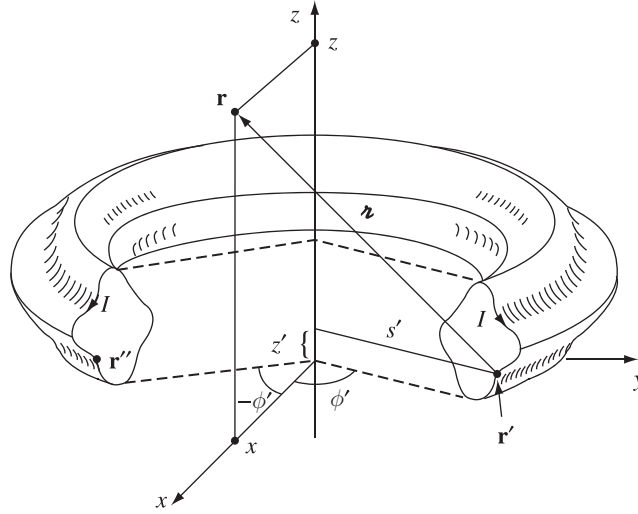


Fig. 5.38

Accordingly,

$$\begin{aligned} \mathbf{I} \times \mathbf{r} &= \begin{bmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ I_s \cos \phi' & I_s \sin \phi' & I_z \\ (x - s' \cos \phi') & (-s' \sin \phi') & (z - z') \end{bmatrix} \\ &= [\sin \phi' (I_s (z - z') + s' I_z)] \hat{\mathbf{x}} + [I_z (x - s' \cos \phi') - I_s \cos \phi' (z - z')] \hat{\mathbf{y}} \\ &\quad + [-I_s x \sin \phi'] \hat{\mathbf{z}}. \end{aligned}$$

But there is a symmetrically located current element at \$\mathbf{r}''\$, with the same \$s'\$, the same \$z\$, the same \$dl'\$, the same \$I_s\$, and the same \$I_z\$, but *negative* \$\phi'\$ (Fig. 5.38). Because \$\sin \phi'\$ changes sign, the \$\hat{\mathbf{x}}\$ and \$\hat{\mathbf{z}}\$ contributions from \$\mathbf{r}'\$ and \$\mathbf{r}''\$ cancel, leaving only a \$\hat{\mathbf{y}}\$ term. Thus the field at \$\mathbf{r}\$ is in the \$\hat{\mathbf{y}}\$ direction, and in general the field points in the \$\hat{\phi}\$ direction. \$\square\$

Now that we know the field is circumferential, determining its magnitude is ridiculously easy. Just apply Ampère's law to a circle of radius \$s\$ about the axis of the toroid:

$$B 2\pi s = \mu_0 I_{\text{enc}},$$

and hence

$$\mathbf{B}(\mathbf{r}) = \begin{cases} \frac{\mu_0 N I}{2\pi s} \hat{\phi} & \text{for points inside the coil,} \\ \mathbf{0} & \text{for points outside the coil,} \end{cases} \quad (5.60)$$

where \$N\$ is the total number of turns.

Problem 5.14. A steady current I flows down a long cylindrical wire of radius a (Fig. 5.39). Find the magnetic field, both inside and outside the wire if:

- the current is uniformly distributed over the surface of the wire;
- the current is distributed in such a way that J is proportional to s , the distance from the axis.

Problem 5.15. A thick slab extending from $z = -a$ to $z = +a$ (and infinite in the x and y directions) carries a uniform volume current $\mathbf{J} = J\hat{\mathbf{x}}$ (Fig. 5.40). Find the magnetic field, as a function of z , both inside and outside the slab.

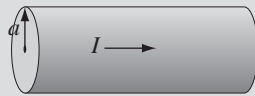


Fig. 5.39

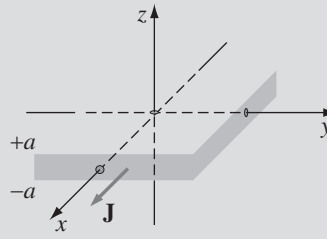


Fig. 5.40

Problem 5.16. Consider the current sheet in Ex. 5.8. A charged particle moving in the xz plane *above* the current sheet (where the field is uniform) executes cyclotron motion. But what if its trajectory crosses the xy plane? Describe the resulting motion, and calculate the average ("drift") velocity.¹⁷

Problem 5.17. Two long coaxial solenoids each carry current I , but in opposite directions, as shown in Fig. 5.41. The inner solenoid (radius a) has n_1 turns per unit length, and the outer one (radius b) has n_2 . Find $\mathbf{B}$ in each of the following three regions: (i) inside the inner solenoid, (ii) between them, and (iii) outside both.

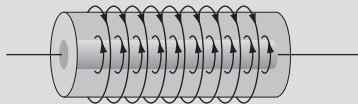


Fig. 5.41

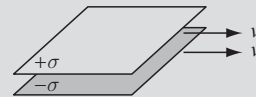


Fig. 5.42

Problem 5.18. A large parallel-plate capacitor with uniform surface charge σ on the upper plate and $-\sigma$ on the lower plate is moving with a constant speed v , as shown in Fig. 5.42.

- Find the magnetic field between the plates and also above and below them.
- Find the magnetic force per unit area on the upper plate, including its direction.
- At what speed v would the magnetic force balance the electrical force?¹⁸

¹⁷ For commentary see H. Essén and A. B. Nordmark, *Eur. Phys. J. D* **70**, 198 (2016).

¹⁸ See footnote to Prob. 5.13.

! **Problem 5.19.** Show that the magnetic field of an infinite solenoid runs parallel to the axis, regardless of the cross-sectional shape of the coil, as long as that shape is constant along the length of the solenoid. What is the magnitude of the field, inside and outside such a coil? Show that the toroid field (Eq. 5.60) reduces to the solenoid field, when the (large) radius of the donut is so much greater than the small radius that a segment can be considered essentially straight.

Problem 5.20. In calculating the current enclosed by an Amperian loop, one must, in general, evaluate an integral of the form

$$I_{\text{enc}} = \int_S \mathbf{J} \cdot d\mathbf{a}.$$

Trouble is, there are infinitely many surfaces that share the same boundary line. Which one are we supposed to use?

5.3.4 Comparison of Magnetostatics and Electrostatics

The divergence and curl of the *electrostatic* field are

$$\begin{cases} \nabla \cdot \mathbf{E} = \rho/\epsilon_0 & (\text{Gauss's law}), \\ \nabla \times \mathbf{E} = \mathbf{0} & (\text{no name}). \end{cases}$$

These are **Maxwell's equations** for electrostatics. Together with the boundary condition $\mathbf{E} \rightarrow \mathbf{0}$ far from all charges,¹⁹ Maxwell's equations determine the field, if the source charge density ρ is given; they contain essentially the same information as Coulomb's law plus the principle of superposition. The divergence and curl of the *magnetostatic* field are

$$\begin{cases} \nabla \cdot \mathbf{B} = 0 & (\text{no name}), \\ \nabla \times \mathbf{B} = \mu_0 \mathbf{J} & (\text{Ampère's law}). \end{cases}$$

These are Maxwell's equations for magnetostatics. Again, together with the boundary condition $\mathbf{B} \rightarrow \mathbf{0}$ far from all currents (or symmetry considerations), Maxwell's equations determine the magnetic field; they are equivalent to the Biot–Savart law (plus superposition). Maxwell's equations and the force law

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

constitute the most elegant formulation of electrostatics and magnetostatics.

Intrinsically, electric forces are enormously stronger than magnetic ones. That's not something obvious from the structure of the theory; it has to do with the sizes of the constants ϵ_0 and μ_0 , and the relatively modest particle speeds typically involved. In general, it is only when both the source charges and the test charge are moving at velocities comparable to the speed

¹⁹ In those artificial problems where the charge (or current) extends to infinity – infinite planes, for example – symmetry considerations can sometimes take the place of boundary conditions.

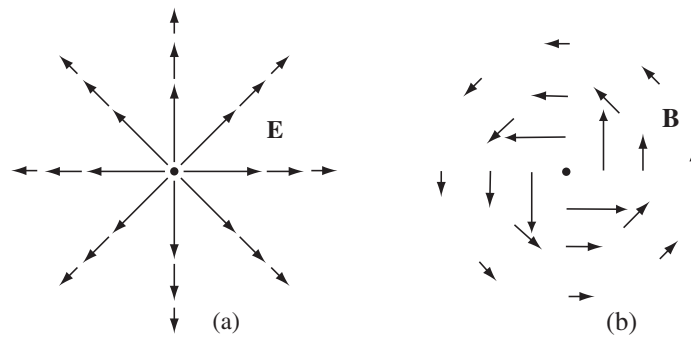


Fig. 5.43

of light that the magnetic force approaches the electric force in strength. (Problems 5.13 and 5.18 illustrate this rule.) How is it, then, that we notice magnetic effects at all? The answer is that both in the production of a magnetic field (Biot–Savart) and in its detection (Lorentz), it is the *current* that matters, and we can compensate for a smallish velocity by pouring prodigious amounts of charge down the wire. Ordinarily, this charge would simultaneously generate so large an *electric* force as to swamp the magnetic one. But if we arrange to keep the wire *neutral*, by embedding in it an equal quantity of opposite charge at rest, the electric field cancels out, leaving the magnetic force to stand alone. It sounds very elaborate, but of course this is precisely what happens in an ordinary current-carrying wire (Prob. 5.21).

The electric field *diverges away from* a (positive) charge; the magnetic field *curls around* a current (Fig. 5.43). There are no stationary point sources for $\mathbf{B}$ – no magnetic analogs to electric charge. This is the physical content of the statement $\nabla \cdot \mathbf{B} = 0$. Coulomb and others believed that magnetism is produced by **magnetic charges** (**magnetic monopoles**, as we would now call them), and in some older books you will still find references to a magnetic version of Coulomb’s law, giving the force of attraction or repulsion between “north” and “south” poles. It was Ampère who first speculated that all magnetic effects are attributable to *electric charges in motion* (currents). As far as we know, Ampère was right, though people still search for magnetic monopoles (they are obviously pretty *rare*, or somebody would have found one by now²⁰), and some recent elementary particle theories actually *require* their existence. In classical electrodynamics, however, $\mathbf{B}$ is divergenceless, and there are no magnetic monopoles. It takes (moving) electric charges to *produce* a magnetic field, and it takes a moving electric charge to “feel” a magnetic field.

In electrostatics we found it illuminating to connect up the field vectors to form **field lines**; the tangent to the field line points in the direction of the

²⁰ An apparent detection (B. Cabrera, *Phys. Rev. Lett.* **48**, 1378 (1982)) has never been reproduced – and not for want of trying. For a delightful brief history of ideas about magnetism, read Chap. 1 in D. C. Mattis, *The Theory of Magnetism*, Harper & Row, New York (1965).

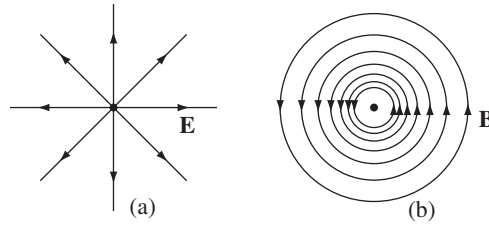


Fig. 5.44

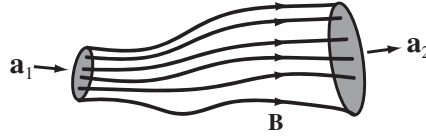


Fig. 5.45

field, and the *density* of field lines (the number per unit cross-sectional area) is proportional to the magnitude of the field. The same device helps us to visualize magnetic fields; thus the *fields* in Fig. 5.43 become the *field lines* in Fig. 5.44. Electric field lines originate on positive charges and terminate on negative ones (either that, or they extend out to infinity). Magnetic field lines do not begin or end anywhere. In the examples we have encountered they form closed loops (or extend to infinity). The direction of the field line still tells you the direction of the field ... but is the density of magnetic field lines proportional to $|\mathbf{B}|$? Yes it is – or rather, we can always choose to make it so.

Proof. Imagine a **flux tube** – a bundle of field lines, sliced off perpendicularly at the two ends (Fig. 5.45). The tube is skinny enough that the field is essentially constant over each end. The surface integral of $\mathbf{B}$ over this tube is

$$\oint_S \mathbf{B} \cdot d\mathbf{a} = B_2 a_2 - B_1 a_1 = \int_V \nabla \cdot \mathbf{B} d\tau = 0.$$

Thus the flux of $\mathbf{B}$ is constant along a given flux tube. If at the “start” (end 1) we have drawn a number of field lines proportional to the flux ($n = \kappa B_1 a_1$, or $B_1 = n/(\kappa a_1)$) then B is proportional to the number of field lines per unit area all along the tube. Doing the same for every flux tube (with the same proportionality factor κ), the density of field lines represents the strength of the field everywhere. $\square$

But there’s a dirty little secret about magnetic field lines that you’re not supposed to talk about in public. In many simple problems (the infinite wire, the circular current ring, the solenoid, the toroidal coil, ...) the field lines form *closed loops*. Indeed, you will often read that magnetic fields *always* form closed loops. Unfortunately, this is the *exception*, not the rule, and it depends in each case on a special symmetry of the source. Consider, for

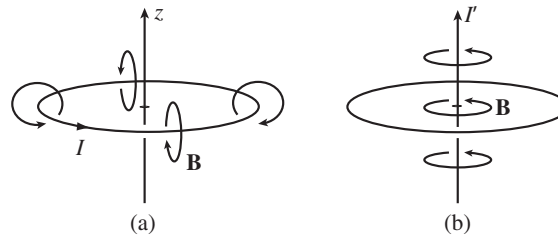


Fig. 5.46

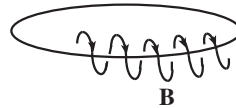


Fig. 5.47

instance, a circular ring carrying a steady current I : the field lines form closed oval-shaped loops that circle the wire (Fig. 5.46a).

But suppose we now introduce a long straight wire along the z -axis, with an adjustable current I' . This will generate an azimuthal field (Fig. 5.46b) that, combined with the field of the ring, yields a pattern of *helical* field lines that wind around the ring like a slinky (Fig. 5.47). Except for certain very special values of the currents, the “helix” will not close on itself after circling the ring. Indeed, it will wind around and around the ring, weaving a donut-shaped sheath – all from a single field line (see Prob. 5.68). So much for the notion that the density of field lines represents the magnitude of the field! And this is not an unusual case; it’s the *typical* behavior of magnetic field lines.²¹ What’s gone wrong, and can it be fixed?

Let’s go back to the flux tube in Fig. 5.45. We start with a bunch of points on the “entering” patch a_1 , construct the field lines that issue from them, and track those lines out to a_2 . These are the field lines inside the flux tube ... but if we continue to follow one of those lines out past a_2 , it wanders around (outside the tube) and eventually, perhaps, finds itself back at a_1 . If it happens to hit one of our starting points, great: it won’t change the total number. But what if it lands *between* the dots? When it enters the flux tube we will suddenly have *too many* field lines to represent the strength of the field. What is to be done? Either we abandon the idea that the density of field lines represents the magnitude of the field (in which case we sacrifice the main point of the field-line picture) or else we must stop the errant line at the gate, and identify its continuation as one of the lines we already drew. As long as just as many field lines exit a_1 as entered it, that won’t violate $\nabla \cdot \mathbf{B} = 0$... but they are *discontinuous* at the

²¹ See J. Slepian, *Am. J. Phys.* **19**, 87 (1951); M. Lieberherr, *Am. J. Phys.* **78**, 1117 (2010); L. Zilberti, *IEEE Magn. Lett.* **8**, 1306005 (2017). See also Prob. 5.67.

boundary – they lose their identity, and hop over to an existing field line. (The field line in Fig. 5.47 makes one circuit of the loop, and then jumps over to rejoin its tail.)

This complication vitiates the utility of field lines for any but the very simplest examples (which is why we don't like to talk about it). I guess the moral of the story is not to take magnetic²² field lines too literally. They work within finite neighborhoods, but not globally. Faraday, who introduced the concept, thought of field lines as real physical entities – invisible wires (or spider webs) that exist out there in space (to this day that is how most electrical engineers think of them). If you really want to sustain this picture, you're going to have to live with magnetic field lines that are discontinuous at places that are entirely arbitrary. I prefer to say that field lines are not physically real; they can sometimes be useful as visual aids, but don't trust them with serious money. At any rate, in Maxwell's formulation, field lines play no role: the "real" physical quantities are the *fields* – *vectors* at each point – and field *lines* are nothing but cartoons.

Problem 5.21.

- Find the density ρ of mobile charges in a piece of copper, assuming each atom contributes one free electron. [Look up the necessary physical constants.]
- Calculate the average electron velocity in a copper wire 1 mm in diameter, carrying a current of 1 A. [Note: This is literally a *snail's* pace. How, then, can you carry on a long distance telephone conversation?]
- What is the force of attraction between two such wires, 1 cm apart?
- If you could somehow remove the stationary positive charges, what would the electrical repulsion force be? How many times greater than the magnetic force is it?

Problem 5.22. Is Ampère's law consistent with the general rule (Eq. 1.46) that divergence-of-curl is always zero? Show that Ampère's law *cannot* be valid, in general, outside magnetostatics. Is there any such "defect" in the other three Maxwell equations?

Problem 5.23. Suppose there *did* exist magnetic monopoles. How would you modify Maxwell's equations and the force law to accommodate them? If you think there are several plausible options, list them, and suggest how you might decide experimentally which one is right.

Problem 5.24. Why doesn't the "dirty little secret" apply to *electrostatics*?

²² Electrostatic field lines do not suffer from this defect. See Prob. 5.24.

5.4 Magnetic Vector Potential

5.4.1 The Vector Potential

Just as $\nabla \times \mathbf{E} = \mathbf{0}$ permitted us to introduce a scalar potential (V) in electrostatics,

$$\mathbf{E} = -\nabla V,$$

so $\nabla \cdot \mathbf{B} = 0$ invites the introduction of a *vector* potential $\mathbf{A}$ in magnetostatics:

$$\boxed{\mathbf{B} = \nabla \times \mathbf{A}.} \quad (5.61)$$

The former is authorized by Theorem 1 (of Section 1.6.2), the latter by Theorem 2 (the proof of Theorem 2 is developed in Prob. 5.33). The potential formulation automatically takes care of $\nabla \cdot \mathbf{B} = 0$ (since the divergence of a curl is *always* zero); there remains Ampère's law:

$$\nabla \times \mathbf{B} = \nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J}. \quad (5.62)$$

Now, the electric potential had a built-in ambiguity: you can add to V any function whose gradient is zero (which is to say, any *constant*), without altering the *physical* quantity $\mathbf{E}$. Likewise, you can add to $\mathbf{A}$ any function whose *curl* vanishes (which is to say, the *gradient of any scalar*), with no effect on $\mathbf{B}$. We can exploit this freedom to eliminate the divergence of $\mathbf{A}$:

$$\boxed{\nabla \cdot \mathbf{A} = 0.} \quad (5.63)$$

To prove that this is always possible, suppose that our original potential, $\mathbf{A}_0$, is *not* divergenceless. If we add to it the gradient of λ ($\mathbf{A} = \mathbf{A}_0 + \nabla \lambda$), the new divergence is

$$\nabla \cdot \mathbf{A} = \nabla \cdot \mathbf{A}_0 + \nabla^2 \lambda.$$

We can accommodate Eq. 5.63, then, if a function λ can be found that satisfies

$$\nabla^2 \lambda = -\nabla \cdot \mathbf{A}_0.$$

But this is mathematically identical to Poisson's equation (2.24),

$$\nabla^2 V = -\frac{\rho}{\epsilon_0},$$

with $\nabla \cdot \mathbf{A}_0$ in place of ρ/ϵ_0 as the "source." And we *know* how to solve Poisson's equation – that's what electrostatics is all about ("given the charge distribution, find the potential"). In particular, if ρ goes to zero at infinity, the solution is Eq. 2.29:

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{\rho}{r} d\tau',$$

and by the same token, if $\nabla \cdot \mathbf{A}_0$ goes to zero at infinity, then

$$\lambda = \frac{1}{4\pi} \int \frac{\nabla \cdot \mathbf{A}_0}{r} d\tau'.$$

If $\nabla \cdot \mathbf{A}_0$ does *not* go to zero at infinity, we'll have to use other means to discover the appropriate λ , just as we get the electric potential by other means when the charge distribution extends to infinity. But the essential point remains: *it is always possible to make the vector potential divergenceless*. To put it the other way around: the definition $\mathbf{B} = \nabla \times \mathbf{A}$ specifies the *curl* of $\mathbf{A}$, but it doesn't say anything about the *divergence* – we are at liberty to pick that as we see fit, and zero is ordinarily the simplest choice.

With this condition on $\mathbf{A}$, Ampère's law (Eq. 5.62) becomes

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}. \quad (5.64)$$

This *again* is nothing but Poisson's equation – or rather, it is *three* Poisson equations, one for each Cartesian²³ component. Assuming $\mathbf{J}$ goes to zero at infinity, we can read off the solution:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{r} d\tau'. \quad (5.65)$$

For line and surface currents,

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I}}{r} dl' = \frac{\mu_0 I}{4\pi} \int \frac{1}{r} dl', \quad \mathbf{A} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{r} da'. \quad (5.66)$$

(If the current does *not* go to zero at infinity, we have to find other ways to get $\mathbf{A}$; some of these are explored in Ex. 5.12 and in the problems at the end of the section.)

It must be said that $\mathbf{A}$ is not as *useful* as V . For one thing, it's still a *vector*, and although Eqs. 5.65 and 5.66 are somewhat easier to work with than the Biot–Savart law, you still have to fuss with components. It would be nice if we could get away with a *scalar* potential

$$\mathbf{B} = -\nabla U, \quad (5.67)$$

but this is incompatible with Ampère's law, since the curl of a gradient is always zero. (A **magnetostatic scalar potential** *can* be used, if you stick scrupulously to simply connected, current-free regions, but as a theoretical tool, it is of limited interest. See Prob. 5.31.) Moreover, since magnetic forces do no work, $\mathbf{A}$ does not admit a simple physical interpretation in terms of potential energy per unit charge. (In some contexts it can be

²³ In Cartesian coordinates, $\nabla^2 \mathbf{A} = (\nabla^2 A_x)\hat{\mathbf{x}} + (\nabla^2 A_y)\hat{\mathbf{y}} + (\nabla^2 A_z)\hat{\mathbf{z}}$, so Eq. 5.64 reduces to $\nabla^2 A_x = -\mu_0 J_x$, $\nabla^2 A_y = -\mu_0 J_y$, and $\nabla^2 A_z = -\mu_0 J_z$. In curvilinear coordinates the unit vectors *themselves* are functions of position, and must be differentiated, so it is *not* the case, for example, that $\nabla^2 A_r = -\mu_0 J_r$. Remember that even if you plan to *evaluate* integrals such as 5.65 using curvilinear coordinates, you must first express $\mathbf{J}$ in terms of its *Cartesian* components (see Section 1.4.1).

interpreted as *momentum* per unit charge.²⁴) Nevertheless, the vector potential has substantial theoretical importance, as we shall see in Chapter 10.

Example 5.11. A spherical shell of radius R , carrying a uniform surface charge σ , is set spinning at angular velocity ω . Find the vector potential it produces at point $\mathbf{r}$ (Fig. 5.48).

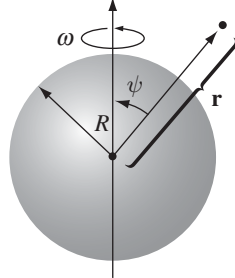


Fig. 5.48

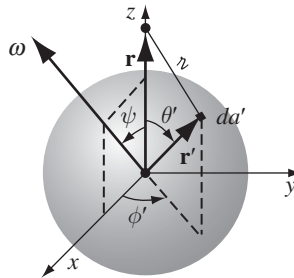


Fig. 5.49

Solution. It might seem natural to set the polar axis along ω , but in fact the integration is easier if we let $\mathbf{r}$ lie on the z -axis, so that ω is tilted at an angle ψ . We might as well orient the x -axis so that ω lies in the xz plane, as shown in Fig. 5.49. According to Eq. 5.66,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}(\mathbf{r}')}{z} da',$$

where $\mathbf{K} = \sigma \mathbf{v}$, $z = \sqrt{R^2 + r^2 - 2Rr \cos \theta'}$, and $da' = R^2 \sin \theta' d\theta' d\phi'$. Now the velocity of a point $\mathbf{r}'$ in a rotating rigid body is given by $\omega \times \mathbf{r}'$; in this case,

$$\begin{aligned} \mathbf{v} = \omega \times \mathbf{r}' &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \omega \sin \psi & 0 & \omega \cos \psi \\ R \sin \theta' \cos \phi' & R \sin \theta' \sin \phi' & R \cos \theta' \end{vmatrix} \\ &= R\omega [-(\cos \psi \sin \theta' \sin \phi')\hat{\mathbf{x}} + (\cos \psi \sin \theta' \cos \phi' - \sin \psi \cos \theta')\hat{\mathbf{y}} \\ &\quad + (\sin \psi \sin \theta' \sin \phi')\hat{\mathbf{z}}]. \end{aligned}$$

Notice that each of these terms, save one, involves either $\sin \phi'$ or $\cos \phi'$. Since

$$\int_0^{2\pi} \sin \phi' d\phi' = \int_0^{2\pi} \cos \phi' d\phi' = 0,$$

such terms contribute nothing. There remains

$$\mathbf{A}(\mathbf{r}) = -\frac{\mu_0 R^3 \sigma \omega \sin \psi}{2} \left(\int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2Rr \cos \theta'}} d\theta' \right) \hat{\mathbf{y}}.$$

²⁴ M. D. Semon and J. R. Taylor, *Am. J. Phys.* **64**, 1361 (1996).

Letting $u \equiv \cos \theta'$, the integral becomes

$$\begin{aligned} \int_{-1}^{+1} \frac{u}{\sqrt{R^2 + r^2 - 2Rru}} du &= -\frac{(R^2 + r^2 + Rru)}{3R^2r^2} \sqrt{R^2 + r^2 - 2Rru} \Big|_{-1}^{+1} \\ &= -\frac{1}{3R^2r^2} [(R^2 + r^2 + Rr)|R - r| \\ &\quad - (R^2 + r^2 - Rr)(R + r)]. \end{aligned}$$

If the point $\mathbf{r}$ lies *inside* the sphere, then $R > r$, and this expression reduces to $(2r/3R^2)$; if $\mathbf{r}$ lies *outside* the sphere, so that $R < r$, it reduces to $(2R/3r^2)$. Noting that $(\boldsymbol{\omega} \times \mathbf{r}) = -\omega r \sin \psi \hat{\mathbf{y}}$, we have, finally,

$$\mathbf{A}(\mathbf{r}) = \begin{cases} \frac{\mu_0 R \sigma}{3} (\boldsymbol{\omega} \times \mathbf{r}) & \text{for points inside the sphere,} \\ \frac{\mu_0 R^4 \sigma}{3r^3} (\boldsymbol{\omega} \times \mathbf{r}) & \text{for points outside the sphere.} \end{cases} \quad (5.68)$$

Having evaluated the integral, I revert to the “natural” coordinates of Fig. 5.48, in which $\boldsymbol{\omega}$ coincides with the z -axis and the point $\mathbf{r}$ is at (r, θ, ϕ) :

$$\mathbf{A}(r, \theta, \phi) = \begin{cases} \frac{\mu_0 R \omega \sigma}{3} r \sin \theta \hat{\boldsymbol{\phi}} & (r \leq R), \\ \frac{\mu_0 R^4 \omega \sigma}{3} \frac{\sin \theta}{r^2} \hat{\boldsymbol{\phi}} & (r \geq R). \end{cases} \quad (5.69)$$

Curiously, the *field* inside this spherical shell is *uniform*:

$$\mathbf{B} = \nabla \times \mathbf{A} = \frac{2\mu_0 R \omega \sigma}{3} (\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}}) = \frac{2}{3} \mu_0 \sigma R \omega \hat{\mathbf{z}} = \frac{2}{3} \mu_0 \sigma R \boldsymbol{\omega}. \quad (5.70)$$

Example 5.12. Find the vector potential of an infinite solenoid with n turns per unit length, radius R , and current I .

Solution. This time we cannot use Eq. 5.66, since the current itself extends to infinity. But here’s a cute method that does the job. Notice that

$$\oint \mathbf{A} \cdot d\mathbf{l} = \int (\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \int \mathbf{B} \cdot d\mathbf{a} = \Phi, \quad (5.71)$$

where Φ is the flux of $\mathbf{B}$ through the loop in question. This is reminiscent of Ampère’s law in integral form (Eq. 5.57),

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}.$$

In fact, it’s the same equation, with $\mathbf{B} \rightarrow \mathbf{A}$ and $\mu_0 I_{\text{enc}} \rightarrow \Phi$. If symmetry permits, we can determine $\mathbf{A}$ from Φ in the same way we got $\mathbf{B}$ from I_{enc} , in Section 5.3.3. The present problem (with a uniform longitudinal magnetic field $\mu_0 n I$ inside the solenoid and no field outside) is analogous to the Ampère’s law problem of a fat

wire carrying a uniformly distributed current. The vector potential is “circumferential” (it mimics the magnetic field in the analog); using a circular “Ampèrian loop” at radius s *inside* the solenoid, we have

$$\oint \mathbf{A} \cdot d\mathbf{l} = A(2\pi s) = \int \mathbf{B} \cdot d\mathbf{a} = \mu_0 n I (\pi s^2),$$

so

$$\mathbf{A} = \frac{\mu_0 n I}{2} s \hat{\phi} \quad \text{for } s \leq R. \quad (5.72)$$

For an Ampèrian loop *outside* the solenoid, the flux is

$$\int \mathbf{B} \cdot d\mathbf{a} = \mu_0 n I (\pi R^2),$$

since the field only extends out to R . Thus

$$\mathbf{A} = \frac{\mu_0 n I}{2} \frac{R^2}{s} \hat{\phi} \quad \text{for } s \geq R. \quad (5.73)$$

If you have any doubts about this answer, *check* it: Does $\nabla \times \mathbf{A} = \mathbf{B}$? Does $\nabla \cdot \mathbf{A} = 0$? If so, we’re done.²⁵

Typically, the direction of $\mathbf{A}$ mimics the direction of the current. For instance, both were azimuthal in Exs. 5.11 and 5.12. Indeed, if all the current flows in *one* direction, then Eq. 5.65 suggests that $\mathbf{A}$ *must* point that way too. Thus the potential of a finite segment of straight wire (Prob. 5.25) is in the direction of the current. Of course, if the current extends to infinity you can’t use Eq. 5.65 in the first place (see Probs. 5.28 and 5.29). Moreover, you can always add an arbitrary constant vector to $\mathbf{A}$ – this is analogous to changing the reference point for V , and it won’t affect the divergence or curl of $\mathbf{A}$, which is all that matters (in Eq. 5.65 we have chosen the constant so that $\mathbf{A}$ goes to zero at infinity). In principle you could even use a vector potential that is not divergenceless, in which case all bets are off. Despite these caveats, the essential point remains: *ordinarily* the direction of $\mathbf{A}$ will match the direction of the current.

Problem 5.25. Find the magnetic vector potential of a finite segment of straight wire carrying a current I . [Put the wire on the z -axis, from z_1 to z_2 , and use Eq. 5.66.] Check that your answer is consistent with Eq. 5.37.

Problem 5.26.

- (a) What current density would produce the vector potential $\mathbf{A} = k\hat{\phi}$ (where k is a constant), in cylindrical coordinates?

²⁵ Notice that $\mathbf{A}$ is nonzero outside the solenoid, even though $\mathbf{B}$ ($= \nabla \times \mathbf{A}$) vanishes there. In classical electrodynamics only $\mathbf{B}$ can exert a magnetic force on a charged particle (via the Lorentz force law). In the *quantum* theory, by striking contrast, the magnetic influence on a charge is expressed in terms of $\mathbf{A}$, not $\mathbf{B}$, and an electron passing nearby but *outside* the solenoid experiences a change in the phase of its wave function, which can be detected in the famous **Aharonov–Bohm effect**. (See, for instance, D. J. Griffiths and D. F. Schroeter, *Introduction to Quantum Mechanics*, 3rd ed., Cambridge University Press (2018), Sect. 4.5.)

- (b) Consider an azimuthally symmetric magnetic field; it points in the z direction, and its magnitude is a function only of s . Check that

$$\mathbf{A} = A(s)\hat{\phi}, \quad \text{where } A(s) = \frac{1}{s} \int_0^s B(s')s' ds',$$

by calculating its divergence and curl. (This generalizes Ex. 5.12.)

Problem 5.27. If $\mathbf{B}$ is *uniform*, show that $\mathbf{A}(\mathbf{r}) = -\frac{1}{2}(\mathbf{r} \times \mathbf{B})$ works. That is, check that $\nabla \cdot \mathbf{A} = 0$ and $\nabla \times \mathbf{A} = \mathbf{B}$. Is this result unique, or are there other functions with the same divergence and curl?

Problem 5.28.

- (a) By whatever means you can think of (short of looking it up), find the vector potential a distance s from an infinite straight wire carrying a current I . Check that $\nabla \cdot \mathbf{A} = 0$ and $\nabla \times \mathbf{A} = \mathbf{B}$.
- (b) Find the magnetic potential *inside* the wire, if it has radius R and the current is uniformly distributed.

Problem 5.29. Find the vector potential above and below the plane surface current in Ex. 5.8.

Problem 5.30.

- (a) Check that Eq. 5.65 is consistent with Eq. 5.63, by applying the *divergence*.
- (b) Check that Eq. 5.65 is consistent with Eq. 5.47, by applying the *curl*.
- (c) Check that Eq. 5.65 is consistent with Eq. 5.64, by applying the *Laplacian*.

Problem 5.31. Suppose you want to define a magnetic scalar potential U (Eq. 5.67) in the vicinity of a current-carrying wire. First of all, you must stay away from the wire itself (there $\nabla \times \mathbf{B} \neq \mathbf{0}$); but that's not enough. Show, by applying Ampère's law to a path that starts at $\mathbf{a}$ and circles the wire, returning to $\mathbf{b}$ (Fig. 5.50), that the scalar potential cannot be single valued (that is, $U(\mathbf{a}) \neq U(\mathbf{b})$, even if they represent the same physical point). As an example, find the scalar potential for an infinite straight wire. (To avoid a multivalued potential, you must restrict yourself to simply connected regions that remain on one side or the other of every wire, never allowing you to go all the way around.)

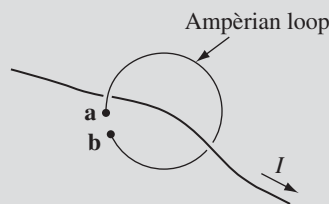


Fig. 5.50

Problem 5.32. Use the results of Ex. 5.11 to find the magnetic field inside a solid sphere, of uniform charge density ρ and radius R , that is rotating at a constant angular velocity ω .

Problem 5.33.

- (a) Complete the proof of Theorem 2, Section 1.6.2. That is, show that any divergenceless vector field $\mathbf{F}$ can be written as the curl of a vector potential $\mathbf{A}$. What you have to do is find A_x , A_y , and A_z such that (i) $\partial A_z/\partial y - \partial A_y/\partial z = F_x$, (ii) $\partial A_x/\partial z - \partial A_z/\partial x = F_y$, and (iii) $\partial A_y/\partial x - \partial A_x/\partial y = F_z$. Here's one way to do it: Pick $A_x = 0$, and solve (ii) and (iii) for A_y and A_z . Note that the "constants of integration" are themselves functions of y and z – they're constant only with respect to x . Now plug these expressions into (i), and use the fact that $\nabla \cdot \mathbf{F} = 0$ to obtain

$$A_y = \int_0^x F_z(x', y, z) dx', \quad A_z = \int_0^y F_x(0, y', z) dy' - \int_0^x F_y(x', y, z) dx'.$$

- (b) By direct differentiation, check that the $\mathbf{A}$ you obtained in part (a) satisfies $\nabla \times \mathbf{A} = \mathbf{F}$. Is $\mathbf{A}$ divergenceless? [This was a very asymmetrical construction, and it would be surprising if it were – although we know that there *exists* a vector whose curl is $\mathbf{F}$ and whose divergence is zero.]
- (c) As an example, let $\mathbf{F} = y\hat{\mathbf{x}} + z\hat{\mathbf{y}} + x\hat{\mathbf{z}}$. Calculate $\mathbf{A}$, and confirm that $\nabla \times \mathbf{A} = \mathbf{F}$. (For further discussion, see Prob. 5.56.)

5.4.2 Boundary Conditions

In Chapter 2, I drew a triangular diagram to summarize the relations among the three fundamental quantities of electrostatics: the charge density ρ , the electric field $\mathbf{E}$, and the potential V . A similar figure can be constructed for magnetostatics (Fig. 5.51), relating the current density $\mathbf{J}$, the field $\mathbf{B}$, and the potential $\mathbf{A}$. There is one "missing link" in the diagram: the equation for $\mathbf{A}$ in terms of $\mathbf{B}$. It's unlikely you would ever need such a formula, but in case you are interested, see Probs. 5.55 and 5.56.

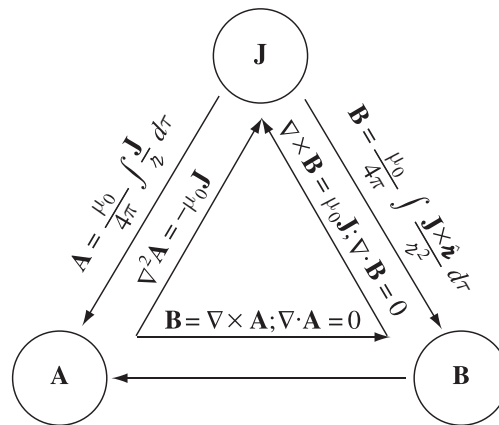


Fig. 5.51

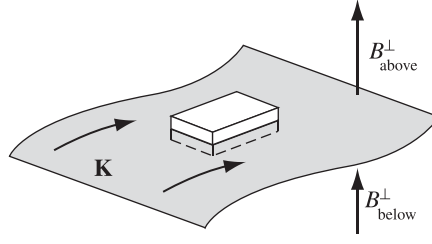


Fig. 5.52

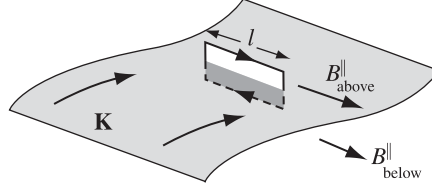


Fig. 5.53

Just as the electric field suffers a discontinuity at a surface *charge*, so the magnetic field is discontinuous at a surface *current*. Only this time it is the *tangential* component that changes. For if we apply Eq. 5.50, in integral form,

$$\oint \mathbf{B} \cdot d\mathbf{a} = 0,$$

to a wafer-thin pillbox straddling the surface (Fig. 5.52), we get

$$B_{\text{above}}^{\perp} = B_{\text{below}}^{\perp}. \quad (5.74)$$

As for the tangential components, an Ampèrian loop running perpendicular to the current (Fig. 5.53) yields

$$\oint \mathbf{B} \cdot d\mathbf{l} = (B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel})l = \mu_0 I_{\text{enc}} = \mu_0 K l,$$

or

$$B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel} = \mu_0 K. \quad (5.75)$$

Thus the component of $\mathbf{B}$ that is parallel to the surface but perpendicular to the current is discontinuous in the amount $\mu_0 K$. A similar Ampèrian loop running *parallel* to the current reveals that the *parallel* component is *continuous*. These results can be summarized in a single formula:

$$\mathbf{B}_{\text{above}} - \mathbf{B}_{\text{below}} = \mu_0 (\mathbf{K} \times \hat{\mathbf{n}}), \quad (5.76)$$

where $\hat{\mathbf{n}}$ is a unit vector perpendicular to the surface, pointing “upward.”

Like the scalar potential in electrostatics, the vector potential is continuous across any boundary:

$$\mathbf{A}_{\text{above}} = \mathbf{A}_{\text{below}}, \quad (5.77)$$

for $\nabla \cdot \mathbf{A} = 0$ guarantees²⁶ that the *normal* component is continuous, and $\nabla \times \mathbf{A} = \mathbf{B}$, in the form

$$\oint \mathbf{A} \cdot d\mathbf{l} = \int \mathbf{B} \cdot d\mathbf{a} = \Phi,$$

means that the tangential components are continuous (the flux through an Ampèrian loop of vanishing thickness is zero). But the *derivative* of $\mathbf{A}$ inherits the discontinuity of $\mathbf{B}$:

$$\frac{\partial \mathbf{A}_{\text{above}}}{\partial n} - \frac{\partial \mathbf{A}_{\text{below}}}{\partial n} = -\mu_0 \mathbf{K}. \quad (5.78)$$

Problem 5.34.

- (a) Check Eq. 5.76 for the configuration in Ex. 5.9.
- (b) Check Eqs. 5.77 and 5.78 for the configuration in Ex. 5.11.

Problem 5.35. Prove Eq. 5.78, using Eqs. 5.63, 5.76, and 5.77. [Suggestion: I'd set up Cartesian coordinates at the surface, with z perpendicular to the surface and x parallel to the current.]

5.4.3 Multipole Expansion of the Vector Potential

If you want an approximate formula for the vector potential of a localized current distribution, valid at distant points, a multipole expansion is in order. Remember: the idea of a multipole expansion is to write the potential as a power series in $1/r$, where r is the distance to the point in question (Fig. 5.54); if r is sufficiently large, the series will be dominated by the lowest nonvanishing contribution, and the higher terms can be ignored. As we found in Eq. 3.94,

$$\frac{1}{z} = \frac{1}{\sqrt{r^2 + (r')^2 - 2rr' \cos \alpha}} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{r'}{r}\right)^n P_n(\cos \alpha), \quad (5.79)$$

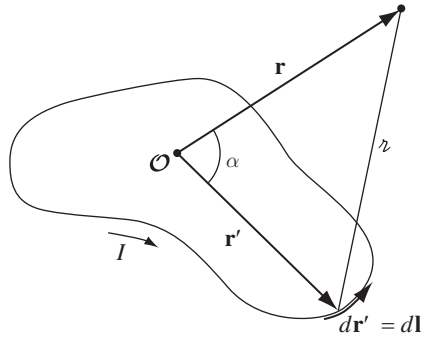


Fig. 5.54

²⁶ Note that Eqs. 5.77 and 5.78 presuppose that $\mathbf{A}$ is divergenceless.

where α is the angle between $\mathbf{r}$ and $\mathbf{r}'$. Accordingly, the vector potential of a current loop can be written

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \oint \frac{1}{r} d\mathbf{l}' = \frac{\mu_0 I}{4\pi} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \oint (r')^n P_n(\cos \alpha) d\mathbf{l}', \quad (5.80)$$

or, more explicitly,

$$\begin{aligned} \mathbf{A}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} & \left[\frac{1}{r} \oint d\mathbf{l}' + \frac{1}{r^2} \oint r' \cos \alpha d\mathbf{l}' \right. \\ & \left. + \frac{1}{r^3} \oint (r')^2 \left(\frac{3}{2} \cos^2 \alpha - \frac{1}{2} \right) d\mathbf{l}' + \dots \right]. \end{aligned} \quad (5.81)$$

As in the multipole expansion of V , we call the first term (which goes like $1/r$) the **monopole** term, the second (which goes like $1/r^2$) **dipole**, the third **quadrupole**, and so on.

Now, the *magnetic monopole term is always zero*, for the integral is just the total vector displacement around a closed loop:

$$\oint d\mathbf{l}' = \mathbf{0}. \quad (5.82)$$

This reflects the fact that there are no magnetic monopoles in nature (an assumption contained in Maxwell's equation $\nabla \cdot \mathbf{B} = 0$, on which the entire theory of vector potential is predicated).

In the absence of any monopole contribution, the dominant term is the dipole (except in the rare case where it, too, vanishes):

$$\mathbf{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0 I}{4\pi r^2} \oint r' \cos \alpha d\mathbf{l}' = \frac{\mu_0 I}{4\pi r^2} \oint (\hat{\mathbf{r}} \cdot \mathbf{r}') d\mathbf{l}'. \quad (5.83)$$

This integral can be rewritten in a more illuminating way if we invoke Eq. 1.108, with $\mathbf{c} = \hat{\mathbf{r}}$:

$$\oint (\hat{\mathbf{r}} \cdot \mathbf{r}') d\mathbf{l}' = -\hat{\mathbf{r}} \times \int d\mathbf{a}'. \quad (5.84)$$

Then

$$\boxed{\mathbf{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \hat{\mathbf{r}}}{r^2}}, \quad (5.85)$$

where $\mathbf{m}$ is the **magnetic dipole moment**:

$$\boxed{\mathbf{m} \equiv I \int d\mathbf{a} = I \mathbf{a}}. \quad (5.86)$$

Here $\mathbf{a}$ is the “vector area” of the loop (Prob. 1.62); if the loop is *flat*, $\mathbf{a}$ is the ordinary area enclosed, with the direction assigned by the usual right-hand rule (fingers in the direction of the current).

Example 5.13. Find the magnetic dipole moment of the “bookend-shaped” loop shown in Fig. 5.55. All sides have length w , and it carries a current I .

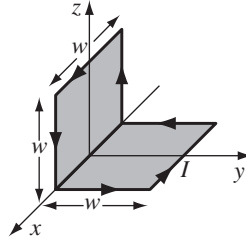


Fig. 5.55

Solution. This wire could be considered the superposition of two plane square loops (Fig. 5.56). The “extra” sides (AB) cancel when the two are put together, since the currents flow in opposite directions. The net magnetic dipole moment is

$$\mathbf{m} = Iw^2\hat{\mathbf{y}} + Iw^2\hat{\mathbf{z}};$$

its magnitude is $\sqrt{2}Iw^2$, and it points along the 45° line $z = y$.

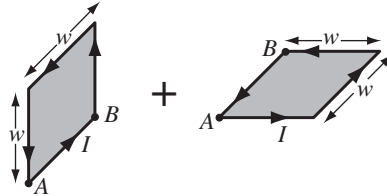


Fig. 5.56

It is clear from Eq. 5.86 that the magnetic dipole moment is independent of the choice of origin. You may remember that the *electric* dipole moment is independent of the origin only when the total charge vanishes (Section 3.4.3). Since the *magnetic* monopole moment is *always* zero, it is not too surprising that the magnetic dipole moment is always independent of origin.

Although the dipole term *dominates* the multipole expansion (unless $\mathbf{m} = \mathbf{0}$) and thus offers a good approximation to the true potential, it is not ordinarily the *exact* potential; there will be quadrupole, octopole, and higher contributions. You might ask, is it possible to devise a current distribution whose potential is “pure” dipole – for which Eq. 5.85 is *exact*? Well, yes and no: like the electrical analog, it can be done, but the model is a bit contrived. To begin with, you must take an *infinitesimally small* loop at the origin, but then, in order to keep the dipole moment finite, you have to crank the current up to infinity, with the product $m = Ia$ held fixed.

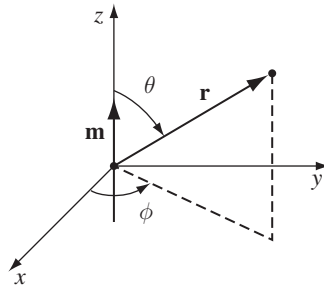


Fig. 5.57

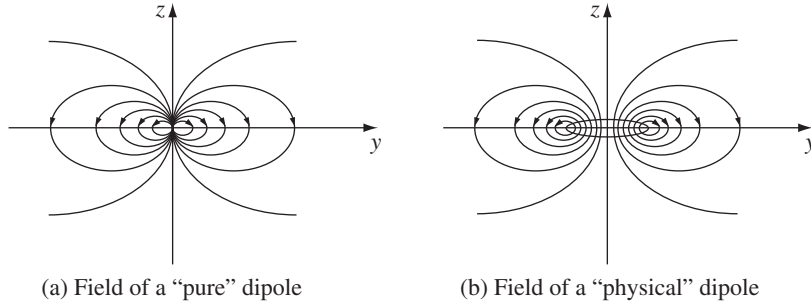


Fig. 5.58

In practice, the dipole potential is a suitable *approximation* whenever the distance r greatly exceeds the size of the loop.

The magnetic *field* of a (perfect) dipole is easiest to calculate if we put $\mathbf{m}$ at the origin and let it point in the z direction (Fig. 5.57). According to Eq. 5.85, the potential at point (r, θ, ϕ) is

$$\mathbf{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{m \sin \theta}{r^2} \hat{\phi}, \quad (5.87)$$

and hence

$$\mathbf{B}_{\text{dip}}(\mathbf{r}) = \nabla \times \mathbf{A} = \frac{\mu_0 m}{4\pi r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\theta}). \quad (5.88)$$

Surprisingly, this is *identical* in structure to the field of an *electric* dipole (Eq. 3.103)! (Up close, however, the field of a *physical* magnetic dipole – a small current loop – looks quite different from the field of a physical electric dipole – plus and minus charges a short distance apart. Compare Fig. 5.58 with Fig. 3.37.)

- **Problem 5.36.** Show that the magnetic field of a dipole can be written in coordinate-free form:

$$\mathbf{B}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{1}{r^3} [3(\mathbf{m} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}]. \quad (5.89)$$

Problem 5.37. A circular loop of wire, with radius R , lies in the xy plane (centered at the origin) and carries a current I running counterclockwise as viewed from the positive z -axis.

- What is its magnetic dipole moment?
- What is the (approximate) magnetic field at points far from the origin?
- Show that, for points on the z -axis, your answer is consistent with the *exact* field (Ex. 5.6), when $z \gg R$.

Problem 5.38. Find the exact magnetic field a distance z above the center of a square loop of side w , carrying a current I . Verify that it reduces to the field of a dipole, with the appropriate dipole moment, when $z \gg w$.

Problem 5.39.

- A phonograph record of radius R , carrying a uniform surface charge σ , is rotating at constant angular velocity ω . Find its magnetic dipole moment.
- Find the magnetic dipole moment of the spinning spherical shell in Ex. 5.11. Show that for points $r > R$ the potential is that of a perfect dipole.

Problem 5.40. I worked out the multipole expansion for the vector potential of a *line* current because that's the most common type, and in some respects the easiest to handle. For a *volume* current $\mathbf{J}$:

- write down the multipole expansion, analogous to Eq. 5.80;
- write down the monopole potential, and prove that it vanishes;
- using Eqs. 1.107 and 5.86, show that the dipole moment can be written

$$\mathbf{m} = \frac{1}{2} \int (\mathbf{r} \times \mathbf{J}) d\tau. \quad (5.90)$$

More Problems on Chapter 5

Problem 5.41. Analyze the motion of a particle (charge q , mass m) in the magnetic field of a long straight wire carrying a steady current I .²⁷

- Is its kinetic energy conserved?
- Find the force on the particle, in cylindrical coordinates, with I along the z -axis.
- Obtain the equations of motion – that is, equations for $\ddot{s}$, $\ddot{\phi}$, and $\ddot{z}$. The latter two can be integrated once, yielding equations for $\dot{\phi}$ and $\dot{z}$ (as functions of s). The equation for $\ddot{s}$ is not so easy; instead, use the fact that the speed is a constant to get the formula for $\dot{s}$.
- Suppose $\dot{z}$ is constant. Describe the motion.

²⁷ For a numerical study of this configuration see Prob. 5.66.

Problem 5.42. Imagine two infinite straight steady currents, I_1 and I_2 , in arbitrary relative orientations. The distance of closest approach is d , and if you slide one over (without changing its direction) so that they intersect, they would make an angle α . Calculate the magnetic force of attraction/repulsion between the currents (in the original configuration).²⁸ [Answer: $\frac{1}{2}\mu_0 I_1 I_2 \cot \alpha$.]

Problem 5.43. It may have occurred to you that since parallel currents attract, the current within a single wire should contract into a tiny concentrated stream along the axis. Yet in practice the current typically distributes itself quite uniformly over the wire. How do you account for this? If the positive charges (density ρ_+) are “nailed down,” and the negative charges (density ρ_-) move at speed v (and none of these depends on the distance from the axis), show that $\rho_- = -\rho_+ \gamma^2$, where $\gamma \equiv 1/\sqrt{1 - (v/c)^2}$ and $c^2 = 1/\mu_0 \epsilon_0$. If the wire as a whole is neutral, where is the compensating charge located?²⁹ [Notice that for typical velocities (see Prob. 5.21), the two charge densities are essentially unchanged by the current, since $\gamma \approx 1$. In **plasmas**, however, where the positive charges are *also* free to move, this so-called **pinch effect** can be very significant.]

Problem 5.44. A current I flows to the right through a rectangular bar of conducting material, in the presence of a uniform magnetic field $\mathbf{B}$ pointing out of the page (Fig. 5.59).

- If the moving charges are *positive*, in which direction are they deflected by the magnetic field? This deflection results in an accumulation of charge on the upper and lower surfaces of the bar, which in turn produces an electric force to counteract the magnetic one. Equilibrium occurs when the two exactly cancel. (This is known as the **Hall effect**.)
- Find the resulting potential difference (the **Hall voltage**) between the top and bottom of the bar, in terms of B , v (the speed of the charges), and the relevant dimensions of the bar.³⁰
- How would your analysis change if the moving charges were *negative*? [The Hall effect is the classic way of determining the sign of the mobile charge carriers in a material.]

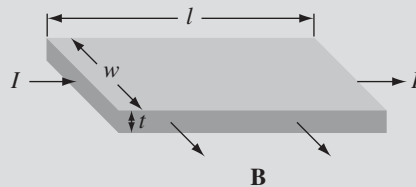


Fig. 5.59

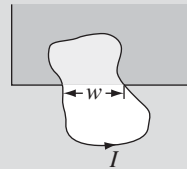


Fig. 5.60

Problem 5.45. A plane wire loop of irregular shape is situated so that part of it is in a uniform magnetic field $\mathbf{B}$ (in Fig. 5.60 the field occupies the shaded region, and points perpendicular to the plane of the loop). The loop carries a current I . Show that the net magnetic force on the

²⁸ P.-M. Binder, F. Cross, and J. K. Silva, *Eur. J. Phys.* **37**, 045206 (2016).

²⁹ For further discussion, see D. C. Gabuzda, *Am. J. Phys.* **61**, 360 (1993).

³⁰ The potential *within* the bar makes an interesting boundary-value problem. See M. J. Moelter, J. Evans, G. Elliot, and M. Jackson, *Am. J. Phys.* **66**, 668 (1998).

loop is $F = IBw$, where w is the chord subtended. Generalize this result to the case where the magnetic field region itself has an irregular shape. What is the direction of the force?

- ! **Problem 5.46.** A circularly symmetrical magnetic field (B depends only on the distance from the axis), pointing perpendicular to the page, occupies the shaded region in Fig. 5.61.

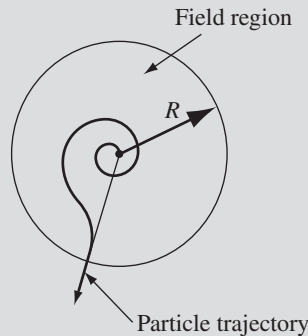


Fig. 5.61

- (a) If the total flux ($\int \mathbf{B} \cdot d\mathbf{a}$) is zero, show that a charged particle that starts out at the center (with velocity $\mathbf{v}$, in the plane) will emerge from the field region on a *radial* path (provided it escapes at all). On the reverse trajectory, a particle fired toward the center from outside will hit its target (if it has sufficient energy), though it may follow a weird route getting there. [Hint: Calculate the total angular momentum acquired by the particle, using the Lorentz force law.]
- (b) Alternatively, if the speed is insufficient ($v < v_{\text{esc}}$), a particle fired out from the center will reach a critical radius s_c (still within the field region) and return to the center – or a particle fired radially inward from the outside will come to a critical radius and then fly back out, without ever making it to the center. Show that the critical radius is given by

$$s_c = -\frac{q}{2\pi m v} \Phi(s_c),$$

where m is the mass of the particle, q is its charge, and v is its speed. (If there are multiple critical radii, a third possibility is for the particle to be trapped between them.)

- (c) Consider the case of a field that is *linear* in s/R (that is, $B = B_0[(s/R) - a]$). First determine the constant a , then find the critical radii and the “escape velocity” v_{esc} .³¹
- (d) Suppose the field is *uniform* (B_0) out to radius a , and then reverses direction ($-B_0$) out to R . Find a , s_c , and v_{esc} , and sketch the trajectory for (i) v slightly less than v_{esc} and (ii) v slightly greater than v_{esc} .

Problem 5.47. Calculate the magnetic force of attraction between the northern and southern hemispheres of a spinning charged spherical shell (Ex. 5.11).

[Answer: $(\pi/4)\mu_0\sigma^2\omega^2 R^4$.]

³¹ For more on this curious problem, see J. Franklin and K. Cole Newton, *Am. J. Phys.* **84**, 263 (2016).

- ! **Problem 5.48.** Consider the motion of a particle with mass m and electric charge q_e in the field of a (hypothetical) stationary magnetic *monopole* q_m at the origin:

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{q_m}{r^2} \hat{\mathbf{r}}.$$

- (a) Find the acceleration of q_e , expressing your answer in terms of q , q_m , m , $\mathbf{r}$ (the position of the particle), and $\mathbf{v}$ (its velocity).
 (b) Show that the speed $v = |\mathbf{v}|$ is a constant of the motion.
 (c) Show that the vector quantity

$$\mathbf{Q} \equiv m(\mathbf{r} \times \mathbf{v}) - \frac{\mu_0 q_e q_m}{4\pi} \hat{\mathbf{r}}$$

is a constant of the motion. [Hint: Differentiate it with respect to time, and prove – using the equation of motion from (a) – that the derivative is zero.] For the physical interpretation of $\mathbf{Q}$, see Prob. 8.21.

- (d) Choosing spherical coordinates (r, θ, ϕ) , with the polar (z) axis along $\mathbf{Q}$:
- (i) calculate $\mathbf{Q} \cdot \hat{\phi}$, and show that θ is a constant of the motion (so q_e moves on the surface of a cone – something Poincaré discovered in 1896);³²
 - (ii) calculate $\mathbf{Q} \cdot \hat{\mathbf{r}}$, and show that the magnitude of $\mathbf{Q}$ is

$$Q = \frac{\mu_0}{4\pi} \left| \frac{q_e q_m}{\cos \theta} \right|;$$

- (iii) calculate $\mathbf{Q} \cdot \hat{\theta}$, show that

$$\frac{d\phi}{dt} = \frac{k}{r^2},$$

and determine the constant k .

- (e) By expressing v^2 in spherical coordinates, obtain the equation for the trajectory, in the form

$$\frac{dr}{d\phi} = f(r)$$

(that is: determine the function $f(r)$).

- (f) Solve this equation for $r(\phi)$.

- ! **Problem 5.49.** Use the Biot–Savart law (most conveniently in the form of Eq. 5.42 appropriate to surface currents) to find the field inside and outside an infinitely long solenoid of radius R , with n turns per unit length, carrying a steady current I .

Problem 5.50. The magnetic field on the axis of a circular current loop (Eq. 5.41) is far from uniform (it falls off sharply with increasing z). You can produce a more nearly uniform field by using *two* such loops a distance d apart (Fig. 5.62).

³² In point of fact, the charge follows a *geodesic* on the cone. The original paper is H. Poincaré, *Comptes rendus de l'Académie des Sciences* **123**, 530 (1896); for a more modern treatment, see B. Rossi and S. Olbert, *Introduction to the Physics of Space*, McGraw-Hill, New York (1970). For extensive history see K. T. McDonald, <http://kirkmcd.princeton.edu/examples/birkeland.pdf>.

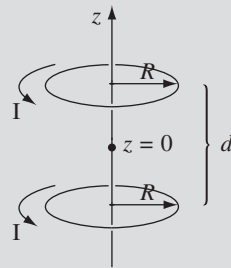


Fig. 5.62

- (a) Find the field (B) as a function of z , and show that $\partial B/\partial z$ is zero at the point midway between them ($z = 0$).
- (b) If you pick d just right, the *second* derivative of B will *also* vanish at the midpoint. This arrangement is known as a **Helmholtz coil**; it's a convenient way of producing locally uniform fields in the laboratory. Determine d such that $\partial^2 B/\partial z^2 = 0$ at the midpoint, and find the resulting magnetic field at the center. [Answer: $8\mu_0 I/5\sqrt{5}R$.]

Problem 5.51. Use Eq. 5.41 to obtain the magnetic field on the axis of the rotating disk in Prob. 5.39a. Show that the dipole field (Eq. 5.88), with the dipole moment you found in Prob. 5.39, is a good approximation if $z \gg R$.

Problem 5.52. Suppose you wanted to find the field of a circular loop (Ex. 5.6) at a point $\mathbf{r}$ that is *not* directly above the center (Fig. 5.63). You might as well choose your axes so that $\mathbf{r}$ lies in the yz plane at $(0, y, z)$. The source point is $(R \cos \phi', R \sin \phi', 0)$, and ϕ' runs from 0 to 2π . Set up the integrals³³ from which you could calculate B_x , B_y , and B_z , and evaluate B_x explicitly.

Problem 5.53. Magnetostatics treats the “source current” (the one that sets up the field) and the “recipient current” (the one that experiences the force) so asymmetrically that it is by no means obvious that the magnetic force between two current loops is consistent with Newton’s third law. Show, starting with the Biot–Savart law (Eq. 5.34) and the Lorentz force law (Eq. 5.16), that the force on loop 2 due to loop 1 (Fig. 5.64) can be written as

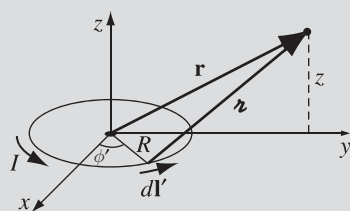


Fig. 5.63

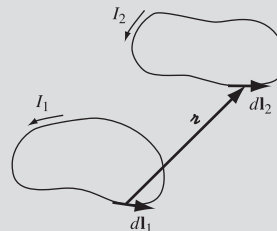


Fig. 5.64

³³ These are **elliptic integrals** – see R. H. Good, *Eur. J. Phys.* **22**, 119 (2001). For a numerical study see Prob. 5.68.

$$\mathbf{F}_2 = -\frac{\mu_0}{4\pi} I_1 I_2 \oint \oint \frac{\hat{\mathbf{z}}}{r^2} d\mathbf{l}_1 \cdot d\mathbf{l}_2. \quad (5.91)$$

In this form, it is clear that $\mathbf{F}_2 = -\mathbf{F}_1$, since $\hat{\mathbf{z}}$ changes direction when the roles of 1 and 2 are interchanged. (If you seem to be getting an “extra” term, it will help to note that $d\mathbf{l}_2 \cdot \hat{\mathbf{z}} = d\mathbf{z}$.)

Problem 5.54. Consider a *plane* loop of wire that carries a steady current I ; we want to calculate the magnetic field at a point in the plane. We might as well take that point to be the origin (it could be inside or outside the loop). The shape of the wire is given, in polar coordinates, by a specified function $r(\theta)$ (Fig. 5.65).

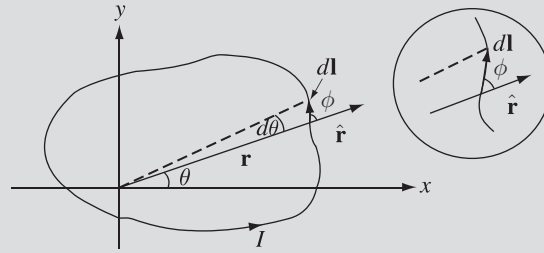


Fig. 5.65

- (a) Show that the magnitude of the field is³⁴

$$B = \frac{\mu_0 I}{4\pi} \oint \frac{d\theta}{r}. \quad (5.92)$$

[Hint: Start with the Biot–Savart law; note that $\mathbf{z} = -\mathbf{r}$, and $d\mathbf{l} \times \hat{\mathbf{r}}$ points perpendicular to the plane; show that $|d\mathbf{l} \times \hat{\mathbf{r}}| = dl \sin \phi = r d\theta$.]

- (b) Test this formula by calculating the field at the center of a circular loop.
(c) The **lituus spiral** is defined by

$$r(\theta) = \frac{a}{\sqrt{\theta}} \quad (0 < \theta \leq 2\pi)$$

(for some constant a). Sketch this figure, and complete the loop with a straight segment along the x -axis. What is the magnetic field at the origin?

- (d) For a conic section with focus at the origin,

$$r(\theta) = \frac{p}{1 + e \cos \theta},$$

where p is the **semilatus rectum** (the y -intercept) and e is the **eccentricity** ($e = 0$ for a circle, $0 < e < 1$ for an ellipse, $e = 1$ for a parabola). Show that the field (at the origin) is

$$B = \frac{\mu_0 I}{2p},$$

regardless of the eccentricity.³⁵

³⁴ J. A. Miranda, *Am. J. Phys.* **68**, 254 (2000).

³⁵ C. Christodoulides, *Am. J. Phys.* **77**, 1195 (2009).

Problem 5.55.

(a) One way to fill in the “missing link” in Fig. 5.51 is to exploit the analogy between the defining equations for $\mathbf{A}$ (viz. $\nabla \cdot \mathbf{A} = 0$, $\nabla \times \mathbf{A} = \mathbf{B}$) and Maxwell’s equations for $\mathbf{B}$ (viz. $\nabla \cdot \mathbf{B} = 0$, $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$). Evidently, $\mathbf{A}$ depends on $\mathbf{B}$ in exactly the same way that $\mathbf{B}$ depends on $\mu_0 \mathbf{J}$ (to wit, the Biot–Savart law). Use this observation to write down the formula for $\mathbf{A}$ in terms of $\mathbf{B}$.

(b) The electrical analog to your result in (a) is

$$V(\mathbf{r}) = -\frac{1}{4\pi} \int \frac{\mathbf{E}(\mathbf{r}') \cdot \hat{\mathbf{z}}}{r^2} d\tau'.$$

Derive it, by exploiting the appropriate analogy.

! **Problem 5.56.** Another way to fill in the “missing link” in Fig. 5.51 is to look for a magnetostatic analog to Eq. 2.21. The obvious candidate would be

$$\mathbf{A}(\mathbf{r}) = \int_O^{\mathbf{r}} (\mathbf{B} \times d\mathbf{l}).$$

(a) Test this formula for the simplest possible case – uniform $\mathbf{B}$ (use the origin as your reference point). Is the result consistent with Prob. 5.27? You could cure this problem by throwing in a factor of $\frac{1}{2}$, but the flaw in this equation runs deeper.

(b) Show that $\int (\mathbf{B} \times d\mathbf{l})$ is *not* independent of path, by calculating $\oint (\mathbf{B} \times d\mathbf{l})$ around the rectangular loop in Fig. 5.66.

As far as I know,³⁶ the best one can do along these lines is the pair of equations

$$(i) \quad V(\mathbf{r}) = -\mathbf{r} \cdot \int_0^1 \mathbf{E}(\lambda \mathbf{r}) d\lambda,$$

$$(ii) \quad \mathbf{A}(\mathbf{r}) = -\mathbf{r} \times \int_0^1 \lambda \mathbf{B}(\lambda \mathbf{r}) d\lambda.$$

[Equation (i) amounts to selecting a *radial* path for the integral in Eq. 2.21, while (ii) constitutes a more “symmetrical” solution to Prob. 5.33.]

(c) Use (ii) to find the vector potential for *uniform* $\mathbf{B}$.

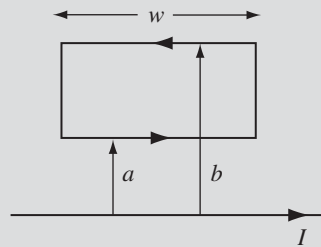


Fig. 5.66

³⁶ R. L. Bishop and S. I. Goldberg, *Tensor Analysis on Manifolds*, Macmillan, New York (1968), Sect. 4.5. This is known as the **Poincaré gauge** (see, for instance, J. D. Jackson, *Am. J. Phys.* **70**, 917 (2002)).

- (d) Use (ii) to find the vector potential of an infinite straight wire carrying a steady current I . Does (ii) automatically satisfy $\nabla \cdot \mathbf{A} = 0$? [Answer: $(\mu_0 I / 2\pi s)(z\hat{s} - s\hat{z})$.]

Problem 5.57.

- (a) Construct the scalar potential $U(\mathbf{r})$ (Eq. 5.67) for an ideal magnetic dipole $\mathbf{m}$.
 (b) Construct a scalar potential for the spinning spherical shell (Ex. 5.11). [Hint: For $r > R$ this is a pure dipole field, as you can see by comparing Eqs. 5.69 and 5.87.]
 (c) Try doing the same for the interior of a *solid* spinning sphere. [Hint: If you solved Prob. 5.32, you already know the *field*; set it equal to $-\nabla U$, and solve for U . What's the trouble?]

Problem 5.58. Just as $\nabla \cdot \mathbf{B} = 0$ allows us to express $\mathbf{B}$ as the curl of a vector potential ($\mathbf{B} = \nabla \times \mathbf{A}$), so $\nabla \cdot \mathbf{A} = 0$ permits us to write $\mathbf{A}$ itself as the curl of a “higher” potential: $\mathbf{A} = \nabla \times \mathbf{W}$. (And this hierarchy can be extended ad infinitum.)

- (a) Find the general formula for $\mathbf{W}$ (as an integral over $\mathbf{B}$), which holds when $\mathbf{B} \rightarrow \mathbf{0}$ at ∞ .
 (b) Determine $\mathbf{W}$ for the case of a *uniform* magnetic field $\mathbf{B}$. [Hint: See Prob. 5.27.]
 (c) Find $\mathbf{W}$ inside and outside an infinite solenoid. [Hint: See Ex. 5.12.]

Problem 5.59. Prove the following uniqueness theorem: If the current density $\mathbf{J}$ is specified throughout a volume $\mathcal{V}$, and *either* the potential $\mathbf{A}$ *or* the magnetic field $\mathbf{B}$ is specified on the surface $\mathcal{S}$ bounding $\mathcal{V}$, then the magnetic field itself is uniquely determined throughout $\mathcal{V}$. [Hint: First use the divergence theorem to show that

$$\int \{(\nabla \times \mathbf{U}) \cdot (\nabla \times \mathbf{V}) - \mathbf{U} \cdot [\nabla \times (\nabla \times \mathbf{V})]\} d\tau = \oint [\mathbf{U} \times (\nabla \times \mathbf{V})] \cdot d\mathbf{a}$$

for arbitrary vector functions $\mathbf{U}$ and $\mathbf{V}$.]

Problem 5.60. A magnetic dipole $\mathbf{m} = -m_0\hat{z}$ is situated at the origin, in an otherwise uniform magnetic field $\mathbf{B} = B_0\hat{z}$. Show that there exists a spherical surface, centered at the origin, through which no magnetic field lines pass. Find the radius of this sphere, and sketch the field lines, inside and out.

Problem 5.61. A thin uniform donut, carrying charge Q and mass M , rotates about its axis as shown in Fig. 5.67.

- (a) Find the ratio of its magnetic dipole moment to its angular momentum. This is called the **gyromagnetic ratio** (or **magnetomechanical ratio**).

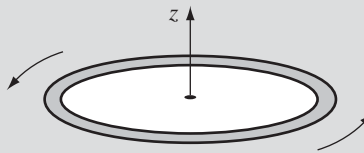


Fig. 5.67

- (b) What is the gyromagnetic ratio for a uniform spinning sphere? [This requires no new calculation; simply decompose the sphere into infinitesimal rings, and apply the result of part (a).]
- (c) According to quantum mechanics, the angular momentum of a spinning electron is $\frac{1}{2}\hbar$, where $\hbar$ is Planck's constant. What, then, is the electron's magnetic dipole moment, in $\mathbf{A} \cdot \text{m}^2$? [This semiclassical value is actually off by a factor of almost exactly 2. Dirac's relativistic electron theory got the 2 right, and Feynman, Schwinger, and Tomonaga later calculated tiny further corrections. The determination of the electron's magnetic dipole moment remains the finest achievement of quantum electrodynamics, and exhibits perhaps the most stunningly precise agreement between theory and experiment in all of physics. Incidentally, the quantity $(e\hbar/2m)$, where e is the charge of the electron and m is its mass, is called the **Bohr magneton**.]

• **Problem 5.62.**

- (a) Prove that the average magnetic field, over a sphere of radius R , due to steady currents inside the sphere, is

$$\mathbf{B}_{\text{ave}} = \frac{\mu_0}{4\pi} \frac{2\mathbf{m}}{R^3}, \quad (5.93)$$

where $\mathbf{m}$ is the total dipole moment of the sphere. Contrast the electrostatic result, Eq. 3.105. [This is tough, so I'll give you a start:

$$\mathbf{B}_{\text{ave}} = \frac{1}{\frac{4}{3}\pi R^3} \int \mathbf{B} d\tau.$$

Write $\mathbf{B}$ as $(\nabla \times \mathbf{A})$, and apply Prob. 1.61b. Now put in Eq. 5.65, and do the surface integral first, showing that

$$\int \frac{1}{r} d\mathbf{a} = \frac{4}{3}\pi \mathbf{r}'$$

(see Fig. 5.68). Use Eq. 5.90, if you like.]

- (b) Show that the average magnetic field due to steady currents *outside* the sphere is the same as the field they produce at the center.

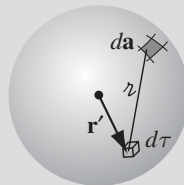


Fig. 5.68

Problem 5.63. A uniformly charged solid sphere of radius R carries a total charge Q , and is set spinning with angular velocity ω about the z -axis.

- (a) What is the magnetic dipole moment of the sphere?
- (b) Find the average magnetic field within the sphere (see Prob. 5.62).

- (c) Find the approximate vector potential at a point (r, θ) , where $r \gg R$.
- (d) Find the *exact* potential at a point (r, θ) outside the sphere, and check that it is consistent with (c). [Hint: Refer to Ex. 5.11.]
- (e) Find the magnetic field at a point (r, θ) *inside* the sphere (Prob. 5.32), and check that it is consistent with (b).

Problem 5.64. Using Eq. 5.88, calculate the average magnetic field of a dipole over a sphere of radius R centered at the origin. Do the angular integrals first. Compare your answer with the general theorem in Prob. 5.62. Explain the discrepancy, and indicate how Eq. 5.89 can be corrected to resolve the ambiguity at $r = 0$. (If you get stuck, refer to Prob. 3.52.) Evidently, the *true* field of a magnetic dipole is³⁷

$$\mathbf{B}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{1}{r^3} [3(\mathbf{m} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}] + \frac{2\mu_0}{3} \mathbf{m} \delta^3(\mathbf{r}). \quad (5.94)$$

Compare the electrostatic analog, Eq. 3.106.

Problem 5.65. A thin glass rod of radius R and length L carries a uniform surface charge σ . It is set spinning about its axis, at an angular velocity ω . Find the magnetic field at a distance $s \gg R$ from the axis, in the xy plane (Fig. 5.69). [Hint: Treat it as a stack of magnetic dipoles. Answer: $\mu_0 \omega \sigma L R^3 / 4 [s^2 + (L/2)^2]^{3/2}$.]

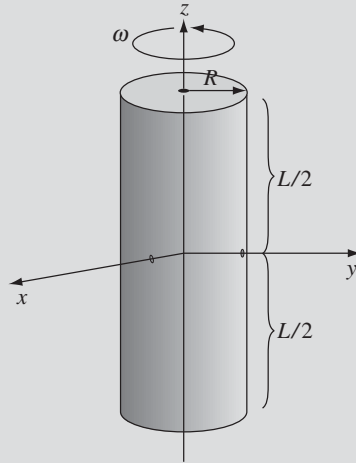


Fig. 5.69

Problem 5.66. As we have seen in Ex. 5.2 and Probs. 5.16, 5.46, and 5.48, the Lorentz force law can produce bizarre particle trajectories, even in relatively simple magnetic fields. Analytically soluble cases, though, are few and far between, and to explore further we must resort to numerical methods. An algorithm particularly well suited to motion in magnetostatic fields is **magnetic velocity Verlet**,³⁸ which can be implemented in Mathematica using the following code:

³⁷ The delta-function term is responsible for the **hyperfine splitting** in atomic spectra – see, for example, D. J. Griffiths, *Am. J. Phys.* **50**, 698 (1982).

³⁸ A. Chambliss and J. Franklin, *Am. J. Phys.* **88**, 1075 (2020).

```

MVVerlet[x0_, v0_, m_, q_, dt_, n_, BF_] :=
Module[{X, V, alpha, x, v, d, C, j},
  X = Table[0., {i, 1, n}];
  X[[1]] = x0;
  V = Table[0., {i, 1, n}];
  V[[1]] = v0;
  x = x0;
  v = v0;
  alpha = q dt / (2m);
  For[j = 2, j <= n, j = j + 1,
    d = v + alpha Cross[v, BF[x]];
    x = x + d dt;
    C = BF[x];
    v = (d + alpha Cross[d, C]
      + alpha^2 C d.C)/(1 + alpha^2 C.C);
    X[[j]] = x;
    V[[j]] = v;
  ];
  Return[{X, V}];
]

```

and

```

ShowTrajectory[pts_] := Show[Graphics3D [Table[Line
  [{pts[[jj]], pts[[jj + 1]]}, {jj, 1, Length[pts] - 1}]]]

```

MVVerlet takes as input parameters

x_0 = $\{x(0), y(0), z(0)\}$ (the initial position),
 v_0 = $\{v_x(0), v_y(0), v_z(0)\}$ (the initial velocity),
 m (the mass of the particle – typically set to 1),
 q (the charge of the particle – typically set to 1),
 dt (the time step – small),
 n (the number of iterations – large).

First you specify the magnetic field:

```
BF[{x_, y_, z_}] := {Bx(x, y, z), By(x, y, z), Bz(x, y, z)};
```

then type

```
{X, V} = MVVerlet[x0, v0, m, q, dt, n, BF];
```

(inserting your chosen parameters), and finally plot the trajectory by invoking

```
ShowTrajectory[X]
```

(a) Test it, for a *uniform* magnetic field (you might start with $dt = 0.005$, $n = 5000$, $x_0 = \{1, 0, 0\}$, $v_0 = \{0, 1, 0.1\}$, $BF[\{x, y, z\}] = \{0, 0, 1\}$, but to get a feel for how it works, try varying the parameters).

(b) Check that the speed is constant, using

```

vsquared = Table[V[[jj]].V[[jj]], {jj, 1, Length[V]}];
ListPlot[vsquared]

```

- (c) Now apply it to the case of a long straight steady current (Prob. 5.41). First reproduce the helical motion predicted in Prob. 5.41d. You might as well set $\mu_0 I/(2\pi) = 1$, $\mathbf{x}_0 = \{1, 0, 0\}$, $\mathbf{v}_0 = \{0, .1, v_z\}$, but you have to pick v_z just right . . .
- (d) Try other initial conditions to explore the rich variety of motions this field admits.³⁹ Here's a particularly striking one: $\mathbf{x}_0 = \{1, 0, 0\}$, $\mathbf{v}_0 = \{.05, .05, 0\}$.



Problem 5.67. Imagine an infinite wire running parallel to the y -axis, at $x = a$ and $z = 0$, and a second wire running parallel to the z -axis, at $x = -a$ and $y = 0$. Each wire carries a steady current I . In addition there is a constant magnetic field $\mathbf{B}_0$.⁴⁰

- (a) What is the net magnetic field, $\mathbf{B}(x, y, z)$? (Leave it as the sum of three terms.)
- (b) Use the method of Prob. 2.66 to draw the magnetic field line starting at $(0.4, 0, 0)$. Begin with $t_{\max} = 5$; then try 75 and 150. Note that this is just a snippet of a *single* field line! [Set $a = 1$, $\mu_0 I/2\pi = 1$, $\mathbf{B}_0 = -(0.1, 0.1, 0.1)$.]



Problem 5.68. In Ex. 5.6 we used the Biot–Savart law to calculate the magnetic field on the axis of a circular steady current loop of radius R . For points *off* the axis (Prob. 5.52) the field is more complicated,⁴¹ in cylindrical coordinates,

$$B_z(s, \phi, z) = \frac{\mu_0 I}{2\pi} \frac{1}{\sqrt{(R+s)^2 + z^2}} \left[\frac{R^2 - s^2 - z^2}{(R-s)^2 + z^2} E(k) + K(k) \right],$$

$$B_s(s, \phi, z) = \frac{\mu_0 I}{2\pi} \frac{z}{s} \frac{1}{\sqrt{(R+s)^2 + z^2}} \left[\frac{R^2 + s^2 + z^2}{(R-s)^2 + z^2} E(k) - K(k) \right],$$

where $K(k)$ is the complete elliptic integral of the first kind (`EllipticK[k]`, in Mathematica), $E(k)$ is the complete elliptic integral of the second kind (`EllipticE[k]`), and

$$k \equiv \frac{4Rs}{(R+s)^2 + z^2}.$$

- (a) Using the method of Prob. 2.66, plot the field line starting at $(3R/4)\hat{\mathbf{y}}$. (Set $\mu_0 I/2\pi = 1$ and $R = 1$; let t run from 0 to 10.)
- (b) Now introduce a current $0.1I$ along the z -axis, and plot the field line for the combined field. This time let t run to 200. Note that this is a *single* field line; eventually it would fill in the entire “toroidal” surface.



Problem 5.69. In Prob. 5.48 you studied the motion of an electric charge q_e in the field of a stationary magnetic monopole q_m , and found that it moves on the surface of a cone of opening angle θ (measured from the axis), with the monopole at the vertex. In spherical coordinates (r, θ, ϕ) ,

³⁹ J. Franklin, D. J. Griffiths, and N. Mann, *Am. J. Phys.* **90**, 513 (2022).

⁴⁰ M. Hosada et al., *Phys. Rev. E* **80**, 067202 (2009); M. Lieberherr, footnote 21.

⁴¹ W. R. Smythe, *Static and Dynamic Electricity*, 3rd ed., Hemisphere, New York (1989), p. 291.

$$r(\phi) = \frac{A}{\cos(\phi \sin \theta)}, \quad \frac{d\phi}{dt} = \frac{v \cos^2(\phi \sin \theta)}{A \sin \theta}, \quad \text{where } A = \frac{\mu_0 |q_e q_m| \tan \theta}{4\pi m v}$$

(m is the mass of the particle and v is its speed).

- (a) Plot the trajectory, using Mathematica's `ParametricPlot3D`:

```
ParametricPlot3D[{x[phi], y[phi], z[phi]}, {phi, -phi0, phi0}]
```

(use $A = 1$, $\sin \theta = 0.05$, and $\phi_0 = 9\pi$).

- (b) Match the plot in (a) using magnetic velocity Verlet (Prob. 5.66). You can set $m = 1$, $q_e = 1$, and $v = 1$, but you'll need to figure out the initial position and velocity.
- (c) Notice how the charge approaches the monopole and then bounces back as the field lines converge. Of course, magnetic monopoles are (presumably) mythical, but the same "mirror" effect⁴² occurs in the dipole field of the Earth, causing electrons to bounce back and forth between the north and south (magnetic) poles and producing the "whistlers" that used to irritate (and mystify) amateur radio operators. Use the method of part (b) and the field of a magnetic dipole ($\mathbf{m} = m\hat{\mathbf{z}}$) to explore this phenomenon. Let $\mu_0 m / 4\pi = 100$, and start with, say, $\mathbf{x}_0 = \{1., 0., 0.\}$, $\mathbf{v}_0 = \{0., 1.5, -3.\}$. You'll discover a curious twist to the story: the electron retrace its path, but gradually circles around the globe.⁴³

⁴² See A. Chambliss and J. Franklin, footnote 38.

⁴³ This is an example of **Störmer's problem**, the study of charged particle motion in the magnetic field of the Earth. C. Störmer, *The Polar Aurora*, Clarendon Press, Oxford (1955); Rossi and Olbert, footnote 32.